Exploratory Data Analysis Assignments in Python

Name: Jayacharan R
Registration number: 23BDS0065
Course: BCSE331P - Exploratory Data Analysis
Semester: Fall 2026-27

Video Link

https://drive.google.com/drive/folders/1f1yuMfQSepzRKk1ZTfraCkUKdmYQTEp9?usp=sharing

Github Link

https://github.com/JayacharanR/EDA-Assignment

Contents

Assignment	Topic	Pages
1	Merging and Data Transformation	4-11
2	Data Import, Transformation and Basic Statistical Analysis	12-33
3	Data Cleaning and Univariate Analysis	34-74
4	Bivariate Analysis	75-90
5	Multivariate Analysis	91-101
6	Time Series Analysis	102-110
7	One, Two and Multiple Variable Analysis	111-129
8	K-Means and Hierarchical Clustering	130-145
9	Principal Component Analysis	146-152

Setup

Required packages: `numpy`, `pandas`, `matplotlib`, `seaborn`, `scipy`, `scikit-learn`, `statsmodels`, and `adjustText`. If needed, uncomment and run the installation line below once. Excel imports additionally require `openpyxl`.

Python code

```
# %pip install numpy pandas matplotlib seaborn scipy scikit-learn statsmodels adjustText openpyxl
from pathlib import Path
from io import StringIO
import base64
import gzip
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from IPython.display import display
from scipy.spatial.distance import pdist
from scipy.cluster.hierarchy import linkage, dendrogram, cut_tree, set_link_color_palette
from sklearn.cluster import KMeans
from sklearn.decomposition import PCA
from statsmodels.tsa.seasonal import seasonal_decompose
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
from statsmodels.tsa.statespace.sarimax import SARIMAX
from adjustText import adjust_text

sns.set_theme(style="whitegrid", context="notebook")
plt.rcParams.update({"figure.figsize": (8, 4.5), "figure.dpi": 110})
pd.set_option("display.max_rows", 40)
pd.set_option("display.max_columns", 30)
```

Output

```
Embedded horse-colic dataset: 110 rows × 28 columns
```

Python code

```
clinical_columns = ["rectal_temp", "pulse", "respiratory_rate", "packed_cell_volume", "total_protein"]


def show(label, value):
    print(label)
    display(value)


def finish(title, xlabel=None, ylabel=None):
    ax = plt.gca()
    ax.set_title(title)
```

```
    if xlabel is not None:
        ax.set_xlabel(xlabel)
    if ylabel is not None:
        ax.set_ylabel(ylabel)
    plt.tight_layout()
    plt.show()


def first_mode(values):
    counts = values.dropna().value_counts(sort=False)
    return counts.idxmax() if len(counts) else np.nan


def describe_distribution(values, label):
    values = values.dropna()
    skew = values.skew()
    shape = "right-skewed" if skew > 0.5 else "left-skewed" if skew < -0.5 else "approximately symmetric"
    print(f"{label}: n={len(values)}, median={values.median():.1f}, "
          f"middle 50%={values.quantile(.25):.1f}-{values.quantile(.75):.1f}; {shape} (skewness={skew:.2f}).")


def prepare_clusters(frame):
    numeric = frame.select_dtypes(include="number").drop(columns="hospital_number", errors="ignore")
    numeric = numeric.dropna(axis=1, how="all")
    numeric = numeric.fillna(numeric.mean())
    numeric = numeric.loc[:, numeric.std(ddof=1).gt(0)]
    return numeric, (numeric - numeric.mean()) / numeric.std(ddof=1)
```

Assignment 1 - Merging and Data Transformation

This assignment creates product data frames, combines them by rows and columns, and demonstrates key-based joins. It also applies min-max, Z-score, and decimal scaling transformations to a vector beginning with the last four digits of my registration number.

Questions

1. Dataframe Operations

Write an R program to create three data frames named `dataframe1`, `dataframe2`, and `dataframe3`. `dataframe1` and `dataframe3` should contain `ProductID`, `NameofProd`, and `Price`; `dataframe2` should contain `ProductID` and `Company_Location`. Insert a minimum of 10 records into each data frame, then:

1. Append the records of `dataframe1` and `dataframe3`.

2. Concatenate the data frames wherever applicable.

3. Merge `dataframe1` and `dataframe2` using `ProductID`.

4. Perform inner, left, right, and full outer joins between `dataframe1` and `dataframe2`.

2. Data Transformations

Create a numeric vector using your first name as its object name. Include 10 numeric values, starting with the last four digits of registration number **23BDS0065**, followed by nine integers of your choice. Normalize it to **[-3, +3]**, **[-1, +1]**, and **[0, 1]**; calculate Z-scores; and apply decimal scaling normalization.

1.1 Create the product data frames

Python code

```
dataframe1 = pd.DataFrame({
    "ProductID": range(101, 111),
    "NameofProd": ["ThinkPad", "Logitech", "Keychron", "UltraSharp", "LaserJet", "ScanSnap", "Soundbar", "Brio", "Netgear", "IronWolf"],
    "Price": [68999, 1499, 5499, 21999, 16499, 12499, 7999, 4599, 3499, 8999]
})
dataframe2 = pd.DataFrame({
    "ProductID": [101, 102, 113, 124, 205, 116, 107, 108, 109, 110],
    "Company_Location": ["Delhi", "Mumbai", "Chennai", "Bangalore", "Hyderabad", "Pune", "Kolkata", "Jaipur", "Lucknow", "Ahmedabad"]
})
dataframe3 = pd.DataFrame({
    "ProductID": range(111, 121),
    "NameofProd": ["iPad", "Fitbit", "SanDisk", "FireCuda", "Epson", "Lumix", "BlueYeti", "AirPods", "Anker", "CyberPower"],
    "Price": [42999, 18999, 1299, 9999, 38999, 52999, 8499, 14999, 2499, 7499]
})
for i, frame in enumerate([dataframe1, dataframe2, dataframe3], 1):
    show(f"Data frame {i}", frame)
```

Output

```
Data frame 1
```

	ProductID	NameofProd	Price
0	101	ThinkPad	68999
1	102	Logitech	1499
2	103	Keychron	5499
3	104	UltraSharp	21999
4	105	LaserJet	16499
5	106	ScanSnap	12499
6	107	Soundbar	7999
7	108	Brio	4599
8	109	Netgear	3499
9	110	IronWolf	8999

```
Data frame 2
```

	ProductID	Company_Location
0	101	Delhi
1	102	Mumbai
2	113	Chennai
3	124	Bangalore
4	205	Hyderabad
5	116	Pune
6	107	Kolkata
7	108	Jaipur
8	109	Lucknow
9	110	Ahmedabad

```
Data frame 3
```

	ProductID	NameofProd	Price
0	111	iPad	42999
1	112	Fitbit	18999
2	113	SanDisk	1299
3	114	FireCuda	9999
4	115	Epson	38999
5	116	Lumix	52999
6	117	BlueYeti	8499
7	118	AirPods	14999
8	119	Anker	2499
9	120	CyberPower	7499

Column concatenation below aligns rows by position, as in the source `cbind()` call. Product IDs are matched explicitly in the merges and joins.

1.2 Append, concatenate, merge, and join

Python code

```
append_df = pd.concat([dataframe1, dataframe3], ignore_index=True)
show("Appended data frame", append_df)
concat_df = pd.concat([dataframe1, dataframe2[["Company_Location"]]], axis=1)
show("Column concatenation", concat_df)
merged_df = dataframe1.merge(dataframe2, on="ProductID", sort=True)
show("Merged data frame", merged_df)
inner_join_df = dataframe1.merge(dataframe2, on="ProductID", how="inner", sort=True)
left_join_df = dataframe1.merge(dataframe2, on="ProductID", how="left", sort=True)
right_join_df = dataframe1.merge(dataframe2, on="ProductID", how="right", sort=True)
full_join_df = dataframe1.merge(dataframe2, on="ProductID", how="outer", sort=True)
for label, frame in [("Inner join", inner_join_df), ("Left join", left_join_df),
                     ("Right join", right_join_df), ("Full outer join", full_join_df)]:
    show(label, frame)
```

Output

```
Appended data frame
```

	ProductID	NameofProd	Price
0	101	ThinkPad	68999
1	102	Logitech	1499

	ProductID	NameofProd	Price
2	103	Keychron	5499
3	104	UltraSharp	21999
4	105	LaserJet	16499
5	106	ScanSnap	12499
6	107	Soundbar	7999
7	108	Brio	4599
8	109	Netgear	3499
9	110	IronWolf	8999
10	111	iPad	42999
11	112	Fitbit	18999
12	113	SanDisk	1299
13	114	FireCuda	9999
14	115	Epson	38999
15	116	Lumix	52999
16	117	BlueYeti	8499
17	118	AirPods	14999
18	119	Anker	2499
19	120	CyberPower	7499

`Column concatenation`

	ProductID	NameofProd	Price	Company_Location
0	101	ThinkPad	68999	Delhi
1	102	Logitech	1499	Mumbai
2	103	Keychron	5499	Chennai
3	104	UltraSharp	21999	Bangalore
4	105	LaserJet	16499	Hyderabad
5	106	ScanSnap	12499	Pune

	ProductID	NameofProd	Price	Company_Location
6	107	Soundbar	7999	Kolkata
7	108	Brio	4599	Jaipur
8	109	Netgear	3499	Lucknow
9	110	IronWolf	8999	Ahmedabad

`Merged data frame`

	ProductID	NameofProd	Price	Company_Location
0	101	ThinkPad	68999	Delhi
1	102	Logitech	1499	Mumbai
2	107	Soundbar	7999	Kolkata
3	108	Brio	4599	Jaipur
4	109	Netgear	3499	Lucknow
5	110	IronWolf	8999	Ahmedabad

`Inner join`

	ProductID	NameofProd	Price	Company_Location
0	101	ThinkPad	68999	Delhi
1	102	Logitech	1499	Mumbai
2	107	Soundbar	7999	Kolkata
3	108	Brio	4599	Jaipur
4	109	Netgear	3499	Lucknow
5	110	IronWolf	8999	Ahmedabad

`Left join`

	ProductID	NameofProd	Price	Company_Location
0	101	ThinkPad	68999	Delhi
1	102	Logitech	1499	Mumbai
2	103	Keychron	5499	NaN

	ProductID	NameofProd	Price	Company_Location
3	104	UltraSharp	21999	NaN
4	105	LaserJet	16499	NaN
5	106	ScanSnap	12499	NaN
6	107	Soundbar	7999	Kolkata
7	108	Brio	4599	Jaipur
8	109	Netgear	3499	Lucknow
9	110	IronWolf	8999	Ahmedabad

`Right join`

	ProductID	NameofProd	Price	Company_Location
0	101	ThinkPad	68999.0	Delhi
1	102	Logitech	1499.0	Mumbai
2	107	Soundbar	7999.0	Kolkata
3	108	Brio	4599.0	Jaipur
4	109	Netgear	3499.0	Lucknow
5	110	IronWolf	8999.0	Ahmedabad
6	113	NaN	NaN	Chennai
7	116	NaN	NaN	Pune
8	124	NaN	NaN	Bangalore
9	205	NaN	NaN	Hyderabad

`Full outer join`

	ProductID	NameofProd	Price	Company_Location
0	101	ThinkPad	68999.0	Delhi
1	102	Logitech	1499.0	Mumbai
2	103	Keychron	5499.0	NaN
3	104	UltraSharp	21999.0	NaN
4	105	LaserJet	16499.0	NaN

	ProductID	NameofProd	Price	Company_Location
5	106	ScanSnap	12499.0	NaN
6	107	Soundbar	7999.0	Kolkata
7	108	Brio	4599.0	Jaipur
8	109	Netgear	3499.0	Lucknow
9	110	IronWolf	8999.0	Ahmedabad
10	113	NaN	NaN	Chennai
11	116	NaN	NaN	Pune
12	124	NaN	NaN	Bangalore
13	205	NaN	NaN	Hyderabad

1.3 Normalize the numeric vector

Python code

```
Jayacharan = np.array([65, 25, 40, 55, 70, 85, 100, 115, 130, 145], dtype=float)
Fname = Jayacharan
minmax_norm = (Jayacharan - Jayacharan.min()) / np.ptp(Jayacharan)
norm_neg3_pos3 = minmax_norm * 6 - 3
norm_neg1_pos1 = minmax_norm * 2 - 1
z_score = (Jayacharan - Jayacharan.mean()) / Jayacharan.std(ddof=1)
j = len(str(int(np.abs(Jayacharan).max())))
decimal_scaling = Jayacharan / 10**j
display(pd.DataFrame({"Original": Jayacharan, "[-3, 3]": norm_neg3_pos3,
                      "[-1, 1]": norm_neg1_pos1, "[0, 1]": minmax_norm,
                      "Z-score": z_score, "Decimal scaling": decimal_scaling}))
```

Output

	Original	[-3, 3]	[-1, 1]	[0, 1]	Z-score	Decimal scaling
0	65.0	-1.00	-0.333333	0.333333	-0.458682	0.065
1	25.0	-3.00	-1.000000	0.000000	-1.477977	0.025
2	40.0	-2.25	-0.750000	0.125000	-1.095741	0.040
3	55.0	-1.50	-0.500000	0.250000	-0.713506	0.055
4	70.0	-0.75	-0.250000	0.375000	-0.331271	0.070

	Original	[-3, 3]	[-1, 1]	[0, 1]	Z-score	Decimal scaling
5	85.0	0.00	0.000000	0.500000	0.050965	0.085
6	100.0	0.75	0.250000	0.625000	0.433200	0.100
7	115.0	1.50	0.500000	0.750000	0.815436	0.115
8	130.0	2.25	0.750000	0.875000	1.197671	0.130
9	145.0	3.00	1.000000	1.000000	1.579906	0.145

Assignment 2 - Data Import, Transformation and Basic Statistical Analysis

This assignment imports the horse-colic data, inspects its structure, demonstrates several missing-value treatments, and removes duplicates. It also converts categories, renames a column, derives a ratio, and computes descriptive statistics.

Questions

Write an R program to import the given dataset and perform preliminary data exploration.

1. Import the given dataset (CSV/Excel) into R. (Dataset share through email)
2. Display the first 10 records of the dataset.
3. Display the last 10 records of the dataset.
4. Find the total number of rows and columns.
5. Display the names of all variables (columns).
6. Display the data type (structure) of each variable.
7. Display the summary statistics of the dataset.
8. Determine the number of missing values in each column.
9. Display the total number of missing values in the dataset.
10. Display the number of missing values in each column.
11. Identify the rows containing at least one missing value.
12. Remove all rows containing missing values and display the cleaned dataset.
13. Replace missing values in a selected numerical column using the mean of that column.
14. Replace missing values in another numerical column using the median.
15. Replace missing values in a categorical column using the mode.
16. Perform forward filling to replace missing values with the previous non-missing value. (Load the package zoo, and use the function locf())
17. Perform backward filling to replace missing values with the next non-missing value.
18. Replace remaining missing values with a constant value (0 or "Unknown").
19. Remove duplicate records from the dataset.

20. Convert appropriate variables into factors.

21. Rename one or more column names.

22. Create a new derived column using existing variables.

23. Display the summary statistics of the dataset using summary().

24. Find the minimum and maximum values.

25. Calculate the range of a numerical variable.

26. Calculate the variance.

27. Calculate the standard deviation.

28. Calculate the interquartile range (IQR).

Note: Don’t display all the record, if more than 10 records from your dataframe are displayed, then you can use head command

Q1-Q7 Import and explore

Python code

```
data = load_horse()
show("First 10 records", data.head(10))
show("Last 10 records", data.tail(10))
print(f"Rows: {len(data)}, columns: {data.shape[1]}")
print("Columns:", data.columns.tolist())
data.info()
show("Summary statistics", data.describe(include="all"))
```

Output

```
First 10 records
```

Table columns 1 of 3

	surgery	age	hospital_number	rectal_temp	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time	pain
0	no	adult	530101	38.5	66.0	28.0	cool	reduced	NaN	more_3_sec	extreme_pain
1	yes	adult	534817	39.2	88.0	20.0	NaN	NaN	pale_cyanotic	less_3_sec	mild_pain
2	no	adult	530334	38.3	40.0	24.0	normal	normal	pale_pink	less_3_sec	mild_pain
3	yes	young	5290409	39.1	164.0	84.0	cold	normal	dark_cyanotic	more_3_sec	depressed
4	no	adult	530255	37.3	104.0	35.0	NaN	NaN	dark_cyanotic	more_3_sec	NaN

	surgery	age	hospital_number	rectal_temp	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time	pain
5	no	adult	528355	NaN	NaN	NaN	warm	normal	pale_pink	less_3_sec	depressed
6	yes	adult	526802	37.9	48.0	16.0	normal	normal	normal_pink	less_3_sec	mild_pain
7	yes	adult	529607	NaN	60.0	NaN	cool	NaN	NaN	less_3_sec	NaN
8	no	adult	530051	NaN	80.0	36.0	cool	absent	pale_pink	less_3_sec	severe_pain
9	no	young	5299629	38.3	90.0	NaN	normal	NaN	normal_pink	less_3_sec	extreme_pain

Table columns 2 of 3

	peristalsis	abdominal_distention	nasogastric_tube	nasogastric_reflux	nasogastric_reflux_ph	rectal_exam_feces	abdomen	packed_cell_volume	total_protein
0	absent	severe	NaN	NaN	NaN	decreased	distend_large	45.0	8.4
1	absent	slight	NaN	NaN	NaN	absent	other	50.0	85.0
2	hypomotile	none	NaN	NaN	NaN	normal	normal	33.0	6.7
3	absent	severe	none	less_1_liter	5.0	decreased	NaN	48.0	7.2
4	NaN	NaN	NaN	NaN	NaN	NaN	NaN	74.0	7.4
5	hypomotile	slight	slight	none	NaN	decreased	firm	NaN	NaN
6	hypomotile	moderate	none	none	NaN	decreased	distend_large	37.0	7.0
7	absent	slight	slight	none	NaN	decreased	distend_small	44.0	8.3
8	absent	severe	slight	none	NaN	decreased	distend_large	38.0	6.2
9	hypomotile	none	slight	none	NaN	decreased	NaN	40.0	6.2

Table columns 3 of 3

	abdomo_appearance	abdomo_protein	outcome	surgical_lesion	lesion_1	lesion_2	lesion_3	cp_data
0	NaN	NaN	died	no	11300.0	0.0	0.0	no
1	cloudy	2.0	euthanized	no	2208.0	0.0	0.0	no
2	NaN	NaN	lived	no	0.0	0.0	0.0	yes
3	serosanguious	5.3	died	yes	2208.0	0.0	0.0	yes
4	NaN	NaN	died	no	4300.0	0.0	0.0	no
5	NaN	NaN	lived	no	0.0	0.0	0.0	no

	abdomo_appearance	abdomo_protein	outcome	surgical_lesion	lesion_1	lesion_2	lesion_3	cp_data
6	NaN	NaN	lived	yes	3124.0	0.0	0.0	no
7	NaN	NaN	died	yes	2208.0	0.0	0.0	no
8	NaN	NaN	euthanized	yes	3205.0	0.0	0.0	no
9	clear	2.2	lived	no	0.0	0.0	0.0	yes

Last 10 records

Table columns 1 of 3

	surgery	age	hospital_number	rectal_temp	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time	pain
100	yes	adult	535364	38.2	40.0	16.0	cool	reduced	normal_pink	less_3_sec	alert
101	yes	adult	534899	NaN	60.0	20.0	cold	reduced	pale_cyanotic	more_3_sec	extreme_pain
102	no	adult	534938	38.3	40.0	16.0	cool	NaN	normal_pink	less_3_sec	depressed
103	yes	young	5292489	38.0	140.0	68.0	normal	normal	normal_pink	less_3_sec	mild_pain
104	yes	adult	527563	37.8	52.0	24.0	normal	reduced	pale_pink	less_3_sec	severe_pain
105	yes	adult	530381	NaN	70.0	36.0	normal	NaN	pale_pink	more_3_sec	depressed
106	yes	adult	528668	38.3	52.0	96.0	NaN	reduced	pale_pink	less_3_sec	NaN
107	no	adult	529764	37.3	50.0	32.0	normal	normal	pale_pink	less_3_sec	alert
108	yes	adult	533696	38.7	60.0	32.0	cold	reduced	bright_pink	more_3_sec	severe_pain
109	yes	young	5297379	38.4	84.0	40.0	cool	reduced	bright_pink	less_3_sec	mild_pain

Table columns 2 of 3

	peristalsis	abdominal_distention	nasogastric_tube	nasogastric_reflux	nasogastric_reflux_ph	rectal_exam_feces	abdomen	packed_cell_volume	total_protein
100	hypomotile	NaN	NaN	NaN	NaN	normal	normal	34.0	66.0
101	absent	NaN	NaN	none	NaN	absent	distend_large	NaN	NaN
102	NaN	NaN	NaN	NaN	NaN	NaN	NaN	37.0	57.0
103	hypomotile	slight	NaN	NaN	NaN	increased	normal	39.0	5.3
104	absent	none	slight	more_1_liter	5.7	increased	distend_large	48.0	6.6
105	hypomotile	slight	slight	NaN	NaN	absent	distend_large	36.0	7.3

	peristalsis	abdominal_distention	nasogastric_tube	nasogastric_reflux	nasogastric_reflux_ph	rectal_exam_feces	abdomen	packed_cell_volume	total_protein
106	NaN	NaN	none	none	NaN	normal	NaN	43.0	6.1
107	hypomotile	slight	NaN	NaN	NaN	normal	NaN	44.0	7.0
108	absent	severe	NaN	NaN	NaN	absent	distend_large	53.0	64.0
109	hypomotile	moderate	none	none	NaN	NaN	NaN	36.0	6.6

Table columns 3 of 3

	abdomo_appearance	abdomo_protein	outcome	surgical_lesion	lesion_1	lesion_2	lesion_3	cp_data
100	NaN	NaN	lived	no	0.0	0.0	0.0	no
101	NaN	NaN	euthanized	yes	3205.0	0.0	0.0	no
102	NaN	NaN	lived	no	0.0	0.0	0.0	no
103	NaN	NaN	lived	yes	7111.0	0.0	0.0	no
104	clear	3.7	died	yes	5400.0	0.0	0.0	no
105	NaN	NaN	lived	yes	31110.0	0.0	0.0	no
106	NaN	NaN	lived	yes	3209.0	0.0	0.0	yes
107	NaN	NaN	lived	no	0.0	0.0	0.0	no
108	serosanguious	2.0	euthanized	yes	3205.0	0.0	0.0	no
109	cloudy	2.0	NaN	NaN	NaN	NaN	NaN	NaN

```
Rows: 110, columns: 28
Columns: ['surgery', 'age', 'hospital_number', 'rectal_temp', 'pulse', 'respiratory_rate', 'temp_of_extremities', 'peripheral_pulse', 'mucous_membrane',
    'capillary_refill_time', 'pain', 'peristalsis', 'abdominal_distention', 'nasogastric_tube', 'nasogastric_reflux', 'nasogastric_reflux_ph', 'rectal_exam_feces',
    'abdomen', 'packed_cell_volume', 'total_protein', 'abdomo_appearance', 'abdomo_protein', 'outcome', 'surgical_lesion', 'lesion_1', 'lesion_2', 'lesion_3',
    'cp_data']
<class 'pandas.DataFrame'>
RangeIndex: 110 entries, 0 to 109
Data columns (total 28 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   surgery               110 non-null    str
 1   age                   110 non-null    str
 2   hospital_number       110 non-null    int64
 3   rectal_temp           85 non-null     float64
 4   pulse                 101 non-null    float64
 5   respiratory_rate      87 non-null     float64
 6   temp_of_extremities   90 non-null     str
```

```
 7   peripheral_pulse       83 non-null     str
 8   mucous_membrane        94 non-null     str
 9   capillary_refill_time  99 non-null     str
 10  pain                   90 non-null     str
 11  peristalsis            91 non-null     str
 12  abdominal_distention   86 non-null     str
 13  nasogastric_tube       73 non-null     str
 14  nasogastric_reflux     72 non-null     str
 15  nasogastric_reflux_ph  21 non-null     float64
 16  rectal_exam_feces      80 non-null     str
 17  abdomen                74 non-null     str
 18  packed_cell_volume     101 non-null    float64
 19  total_protein          99 non-null     float64
 20  abdomo_appearance      49 non-null     str
 21  abdomo_protein         38 non-null     float64
 22  outcome                109 non-null    str
 23  surgical_lesion        109 non-null    str
 24  lesion_1               109 non-null    float64
 25  lesion_2               109 non-null    float64
 26  lesion_3               109 non-null    float64
 27  cp_data                109 non-null    str
dtypes: float64(10), int64(1), str(17)
memory usage: 24.2 KB
Summary statistics
```

Table columns 1 of 3

	surgery	age	hospital_number	rectal_temp	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time	pain
count	110	110	1.100000e+02	85.000000	101.000000	87.000000	90	83	94	99	90
unique	2	2	NaN	NaN	NaN	NaN	4	4	6	3	5
top	yes	adult	NaN	NaN	NaN	NaN	cool	reduced	normal_pink	less_3_sec	mild_pain
freq	59	97	NaN	NaN	NaN	NaN	40	42	26	64	26
mean	NaN	NaN	1.179631e+06	38.149412	75.544554	31.954023	NaN	NaN	NaN	NaN	NaN
std	NaN	NaN	1.640871e+06	0.783251	30.672308	18.842621	NaN	NaN	NaN	NaN	NaN
min	NaN	NaN	5.216810e+05	35.400000	38.000000	10.000000	NaN	NaN	NaN	NaN	NaN
25%	NaN	NaN	5.293520e+05	37.700000	52.000000	20.000000	NaN	NaN	NaN	NaN	NaN
50%	NaN	NaN	5.301290e+05	38.100000	64.000000	28.000000	NaN	NaN	NaN	NaN	NaN
75%	NaN	NaN	5.347552e+05	38.600000	96.000000	36.000000	NaN	NaN	NaN	NaN	NaN
max	NaN	NaN	5.301219e+06	40.300000	164.000000	96.000000	NaN	NaN	NaN	NaN	NaN

Table columns 2 of 3

	peristalsis	abdominal_distention	nasogastric_tube	nasogastric_reflux	nasogastric_reflux_ph	rectal_exam_feces	abdomen	packed_cell_volume	total_protein
count	91	86	73	72	21.00000	80	74	101.000000	99.000000
unique	4	4	3	3	NaN	4	5	NaN	NaN
top	hypomotile	slight	slight	none	NaN	absent	distend_large	NaN	NaN
freq	50	24	32	43	NaN	32	34	NaN	NaN
mean	NaN	NaN	NaN	NaN	4.87619	NaN	NaN	47.336634	21.077778
std	NaN	NaN	NaN	NaN	2.02803	NaN	NaN	10.038204	26.761927
min	NaN	NaN	NaN	NaN	1.00000	NaN	NaN	24.000000	4.500000
25%	NaN	NaN	NaN	NaN	4.00000	NaN	NaN	40.000000	6.500000
50%	NaN	NaN	NaN	NaN	5.50000	NaN	NaN	46.000000	7.500000
75%	NaN	NaN	NaN	NaN	7.00000	NaN	NaN	53.000000	8.650000
max	NaN	NaN	NaN	NaN	7.20000	NaN	NaN	74.000000	86.000000

Table columns 3 of 3

	abdomo_appearance	abdomo_protein	outcome	surgical_lesion	lesion_1	lesion_2	lesion_3	cp_data
count	49	38.000000	109	109	109.000000	109.000000	109.0	109
unique	3	NaN	3	2	NaN	NaN	NaN	2
top	cloudy	NaN	lived	yes	NaN	NaN	NaN	no
freq	21	NaN	64	66	NaN	NaN	NaN	71
mean	NaN	3.184211	NaN	NaN	3114.715596	12.844037	0.0	NaN
std	NaN	1.787574	NaN	NaN	3505.769296	134.095680	0.0	NaN
min	NaN	0.100000	NaN	NaN	0.000000	0.000000	0.0	NaN
25%	NaN	2.000000	NaN	NaN	2112.000000	0.000000	0.0	NaN
50%	NaN	2.450000	NaN	NaN	3111.000000	0.000000	0.0	NaN
75%	NaN	4.400000	NaN	NaN	3207.000000	0.000000	0.0	NaN
max	NaN	7.400000	NaN	NaN	31110.000000	1400.000000	0.0	NaN

Q8-Q12 Inspect and remove missing records

Python code

```
missing_counts = data.isna().sum()
show("Q8 and Q10: Missing values in each column", missing_counts)
print("Q9: Total missing values:", int(missing_counts.sum()))
missing_rows = data.loc[data.isna().any(axis=1)]
show("Q11: Rows containing missing values (first 10)", missing_rows.head(10))
data_clean = data.dropna().copy()
show("Q12: Complete records (first 10)", data_clean.head(10))
print("Complete record count:", len(data_clean))
```

Output

```
Q8 and Q10: Missing values in each column

surgery                   0
age                       0
hospital_number           0
rectal_temp              25
pulse                     9
respiratory_rate         23
temp_of_extremities      20
peripheral_pulse         27
mucous_membrane          16
capillary_refill_time    11
pain                     20
peristalsis              19
abdominal_distention     24
nasogastric_tube         37
nasogastric_reflux       38
nasogastric_reflux_ph    89
rectal_exam_feces        30
abdomen                  36
packed_cell_volume        9
total_protein            11
abdomo_appearance        61
abdomo_protein           72
outcome                   1
surgical_lesion           1
lesion_1                  1
lesion_2                  1
lesion_3                  1
cp_data                   1
dtype: int64

Q9: Total missing values: 583
Q11: Rows containing missing values (first 10)
```

Table columns 1 of 3

	surgery	age	hospital_number	rectal_temp	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time	pain
0	no	adult	530101	38.5	66.0	28.0	cool	reduced	NaN	more_3_sec	extreme_pain
1	yes	adult	534817	39.2	88.0	20.0	NaN	NaN	pale_cyanotic	less_3_sec	mild_pain
2	no	adult	530334	38.3	40.0	24.0	normal	normal	pale_pink	less_3_sec	mild_pain
3	yes	young	5290409	39.1	164.0	84.0	cold	normal	dark_cyanotic	more_3_sec	depressed
4	no	adult	530255	37.3	104.0	35.0	NaN	NaN	dark_cyanotic	more_3_sec	NaN
5	no	adult	528355	NaN	NaN	NaN	warm	normal	pale_pink	less_3_sec	depressed
6	yes	adult	526802	37.9	48.0	16.0	normal	normal	normal_pink	less_3_sec	mild_pain
7	yes	adult	529607	NaN	60.0	NaN	cool	NaN	NaN	less_3_sec	NaN
8	no	adult	530051	NaN	80.0	36.0	cool	absent	pale_pink	less_3_sec	severe_pain
9	no	young	5299629	38.3	90.0	NaN	normal	NaN	normal_pink	less_3_sec	extreme_pain

Table columns 2 of 3

	peristalsis	abdominal_distention	nasogastric_tube	nasogastric_reflux	nasogastric_reflux_ph	rectal_exam_feces	abdomen	packed_cell_volume	total_protein
0	absent	severe	NaN	NaN	NaN	decreased	distend_large	45.0	8.4
1	absent	slight	NaN	NaN	NaN	absent	other	50.0	85.0
2	hypomotile	none	NaN	NaN	NaN	normal	normal	33.0	6.7
3	absent	severe	none	less_1_liter	5.0	decreased	NaN	48.0	7.2
4	NaN	NaN	NaN	NaN	NaN	NaN	NaN	74.0	7.4
5	hypomotile	slight	slight	none	NaN	decreased	firm	NaN	NaN
6	hypomotile	moderate	none	none	NaN	decreased	distend_large	37.0	7.0
7	absent	slight	slight	none	NaN	decreased	distend_small	44.0	8.3
8	absent	severe	slight	none	NaN	decreased	distend_large	38.0	6.2
9	hypomotile	none	slight	none	NaN	decreased	NaN	40.0	6.2

Table columns 3 of 3

	abdomo_appearance	abdomo_protein	outcome	surgical_lesion	lesion_1	lesion_2	lesion_3	cp_data
0	NaN	NaN	died	no	11300.0	0.0	0.0	no

	abdomo_appearance	abdomo_protein	outcome	surgical_lesion	lesion_1	lesion_2	lesion_3	cp_data
1	cloudy	2.0	euthanized	no	2208.0	0.0	0.0	no
2	NaN	NaN	lived	no	0.0	0.0	0.0	yes
3	serosanguious	5.3	died	yes	2208.0	0.0	0.0	yes
4	NaN	NaN	died	no	4300.0	0.0	0.0	no
5	NaN	NaN	lived	no	0.0	0.0	0.0	no
6	NaN	NaN	lived	yes	3124.0	0.0	0.0	no
7	NaN	NaN	died	yes	2208.0	0.0	0.0	no
8	NaN	NaN	euthanized	yes	3205.0	0.0	0.0	no
9	clear	2.2	lived	no	0.0	0.0	0.0	yes

```
Q12: Complete records (first 10)
```

Table columns 1 of 3

	surgery	age	hospital_number	rectal_temp	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time	pain
10	yes	adult	528548	38.1	66.0	12.0	cool	reduced	bright_red	less_3_sec	mild_pain
54	no	adult	529461	40.3	114.0	36.0	cool	reduced	normal_pink	more_3_sec	depressed
64	yes	adult	529667	39.0	64.0	36.0	cool	normal	pale_cyanotic	more_3_sec	mild_pain
91	no	adult	529461	40.3	114.0	36.0	cool	reduced	normal_pink	more_3_sec	depressed
104	yes	adult	527563	37.8	52.0	24.0	normal	reduced	pale_pink	less_3_sec	severe_pain

Table columns 2 of 3

	peristalsis	abdominal_distention	nasogastric_tube	nasogastric_reflux	nasogastric_reflux_ph	rectal_exam_feces	abdomen	packed_cell_volume	total_protein
10	hypomotile	none	slight	none	3.0	increased	distend_large	44.0	6.0
54	hypomotile	moderate	slight	none	7.0	normal	distend_large	57.0	8.1
64	hypomotile	slight	none	less_1_liter	7.0	absent	distend_large	44.0	7.5
91	hypomotile	moderate	slight	none	7.0	normal	distend_large	57.0	8.1
104	absent	none	slight	more_1_liter	5.7	increased	distend_large	48.0	6.6

Table columns 3 of 3

	abdomo_appearance	abdomo_protein	outcome	surgical_lesion	lesion_1	lesion_2	lesion_3	cp_data
10	cloudy	3.6	lived	yes	2124.0	0.0	0.0	yes
54	serosanguious	4.5	euthanized	yes	7400.0	0.0	0.0	yes
64	serosanguious	5.0	lived	yes	2113.0	0.0	0.0	yes
91	serosanguious	4.5	died	yes	3205.0	0.0	0.0	yes
104	clear	3.7	died	yes	5400.0	0.0	0.0	no

```
Complete record count: 5
```

Q13-Q18 Missing-value replacement methods

Python code

```
data_changed = data.copy()
data_changed["rectal_temp"] = data_changed["rectal_temp"].fillna(data_changed["rectal_temp"].mean())
data_changed["pulse"] = data_changed["pulse"].fillna(data_changed["pulse"].median())
data_changed["temp_of_extremities"] = data_changed["temp_of_extremities"].fillna(first_mode(data_changed["temp_of_extremities"]))
show("Q13-Q15: Mean, median, and mode imputation", data_changed[["rectal_temp", "pulse", "temp_of_extremities"]].head(10))
data_f = data.ffill()
data_b = data.bfill()
show("Q16: Forward fill", data_f.head(10))
show("Q17: Backward fill", data_b.head(10))
data_const = data.copy()
num_cols = data_const.select_dtypes(include="number").columns
cat_cols = data_const.select_dtypes(exclude="number").columns
data_const[num_cols] = data_const[num_cols].fillna(0)
data_const[cat_cols] = data_const[cat_cols].fillna("Unknown")
show("Q18: Constant replacement", data_const.head(10))
```

Output

```
Q13-Q15: Mean, median, and mode imputation
```

	rectal_temp	pulse	temp_of_extremities
0	38.500000	66.0	cool
1	39.200000	88.0	cool
2	38.300000	40.0	normal
3	39.100000	164.0	cold
4	37.300000	104.0	cool

	rectal_temp	pulse	temp_of_extremities
5	38.149412	64.0	warm
6	37.900000	48.0	normal
7	38.149412	60.0	cool
8	38.149412	80.0	cool
9	38.300000	90.0	normal

Q16: Forward fill

Table columns 1 of 3

	surgery	age	hospital_number	rectal_temp	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time	pain
0	no	adult	530101	38.5	66.0	28.0	cool	reduced	NaN	more_3_sec	extreme_pain
1	yes	adult	534817	39.2	88.0	20.0	cool	reduced	pale_cyanotic	less_3_sec	mild_pain
2	no	adult	530334	38.3	40.0	24.0	normal	normal	pale_pink	less_3_sec	mild_pain
3	yes	young	5290409	39.1	164.0	84.0	cold	normal	dark_cyanotic	more_3_sec	depressed
4	no	adult	530255	37.3	104.0	35.0	cold	normal	dark_cyanotic	more_3_sec	depressed
5	no	adult	528355	37.3	104.0	35.0	warm	normal	pale_pink	less_3_sec	depressed
6	yes	adult	526802	37.9	48.0	16.0	normal	normal	normal_pink	less_3_sec	mild_pain
7	yes	adult	529607	37.9	60.0	16.0	cool	normal	normal_pink	less_3_sec	mild_pain
8	no	adult	530051	37.9	80.0	36.0	cool	absent	pale_pink	less_3_sec	severe_pain
9	no	young	5299629	38.3	90.0	36.0	normal	absent	normal_pink	less_3_sec	extreme_pain

Table columns 2 of 3

	peristalsis	abdominal_distention	nasogastric_tube	nasogastric_reflux	nasogastric_reflux_ph	rectal_exam_feces	abdomen	packed_cell_volume	total_protein
0	absent	severe	NaN	NaN	NaN	decreased	distend_large	45.0	8.4
1	absent	slight	NaN	NaN	NaN	absent	other	50.0	85.0
2	hypomotile	none	NaN	NaN	NaN	normal	normal	33.0	6.7
3	absent	severe	none	less_1_liter	5.0	decreased	normal	48.0	7.2

	peristalsis	abdominal_distention	nasogastric_tube	nasogastric_reflux	nasogastric_reflux_ph	rectal_exam_feces	abdomen	packed_cell_volume	total_protein
4	absent	severe	none	less_1_liter	5.0	decreased	normal	74.0	7.4
5	hypomotile	slight	slight	none	5.0	decreased	firm	74.0	7.4
6	hypomotile	moderate	none	none	5.0	decreased	distend_large	37.0	7.0
7	absent	slight	slight	none	5.0	decreased	distend_small	44.0	8.3
8	absent	severe	slight	none	5.0	decreased	distend_large	38.0	6.2
9	hypomotile	none	slight	none	5.0	decreased	distend_large	40.0	6.2

Table columns 3 of 3

	abdomo_appearance	abdomo_protein	outcome	surgical_lesion	lesion_1	lesion_2	lesion_3	cp_data
0	NaN	NaN	died	no	11300.0	0.0	0.0	no
1	cloudy	2.0	euthanized	no	2208.0	0.0	0.0	no
2	cloudy	2.0	lived	no	0.0	0.0	0.0	yes
3	serosanguious	5.3	died	yes	2208.0	0.0	0.0	yes
4	serosanguious	5.3	died	no	4300.0	0.0	0.0	no
5	serosanguious	5.3	lived	no	0.0	0.0	0.0	no
6	serosanguious	5.3	lived	yes	3124.0	0.0	0.0	no
7	serosanguious	5.3	died	yes	2208.0	0.0	0.0	no
8	serosanguious	5.3	euthanized	yes	3205.0	0.0	0.0	no
9	clear	2.2	lived	no	0.0	0.0	0.0	yes

```
Q17: Backward fill
```

Table columns 1 of 3

	surgery	age	hospital_number	rectal_temp	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time	pain
0	no	adult	530101	38.5	66.0	28.0	cool	reduced	pale_cyanotic	more_3_sec	extreme_pain
1	yes	adult	534817	39.2	88.0	20.0	normal	normal	pale_cyanotic	less_3_sec	mild_pain
2	no	adult	530334	38.3	40.0	24.0	normal	normal	pale_pink	less_3_sec	mild_pain
3	yes	young	5290409	39.1	164.0	84.0	cold	normal	dark_cyanotic	more_3_sec	depressed

	surgery	age	hospital_number	rectal_temp	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time	pain
4	no	adult	530255	37.3	104.0	35.0	warm	normal	dark_cyanotic	more_3_sec	depressed
5	no	adult	528355	37.9	48.0	16.0	warm	normal	pale_pink	less_3_sec	depressed
6	yes	adult	526802	37.9	48.0	16.0	normal	normal	normal_pink	less_3_sec	mild_pain
7	yes	adult	529607	38.3	60.0	36.0	cool	absent	pale_pink	less_3_sec	severe_pain
8	no	adult	530051	38.3	80.0	36.0	cool	absent	pale_pink	less_3_sec	severe_pain
9	no	young	5299629	38.3	90.0	12.0	normal	reduced	normal_pink	less_3_sec	extreme_pain

Table columns 2 of 3

	peristalsis	abdominal_distention	nasogastric_tube	nasogastric_reflux	nasogastric_reflux_ph	rectal_exam_feces	abdomen	packed_cell_volume	total_protein
0	absent	severe	none	less_1_liter	5.0	decreased	distend_large	45.0	8.4
1	absent	slight	none	less_1_liter	5.0	absent	other	50.0	85.0
2	hypomotile	none	none	less_1_liter	5.0	normal	normal	33.0	6.7
3	absent	severe	none	less_1_liter	5.0	decreased	firm	48.0	7.2
4	hypomotile	slight	slight	none	3.0	decreased	firm	74.0	7.4
5	hypomotile	slight	slight	none	3.0	decreased	firm	37.0	7.0
6	hypomotile	moderate	none	none	3.0	decreased	distend_large	37.0	7.0
7	absent	slight	slight	none	3.0	decreased	distend_small	44.0	8.3
8	absent	severe	slight	none	3.0	decreased	distend_large	38.0	6.2
9	hypomotile	none	slight	none	3.0	decreased	distend_large	40.0	6.2

Table columns 3 of 3

	abdomo_appearance	abdomo_protein	outcome	surgical_lesion	lesion_1	lesion_2	lesion_3	cp_data
0	cloudy	2.0	died	no	11300.0	0.0	0.0	no
1	cloudy	2.0	euthanized	no	2208.0	0.0	0.0	no
2	serosanguious	5.3	lived	no	0.0	0.0	0.0	yes
3	serosanguious	5.3	died	yes	2208.0	0.0	0.0	yes
4	clear	2.2	died	no	4300.0	0.0	0.0	no

	abdomo_appearance	abdomo_protein	outcome	surgical_lesion	lesion_1	lesion_2	lesion_3	cp_data
5	clear	2.2	lived	no	0.0	0.0	0.0	no
6	clear	2.2	lived	yes	3124.0	0.0	0.0	no
7	clear	2.2	died	yes	2208.0	0.0	0.0	no
8	clear	2.2	euthanized	yes	3205.0	0.0	0.0	no
9	clear	2.2	lived	no	0.0	0.0	0.0	yes

`Q18: Constant replacement`

Table columns 1 of 3

	surgery	age	hospital_number	rectal_temp	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time	pain
0	no	adult	530101	38.5	66.0	28.0	cool	reduced	Unknown	more_3_sec	extreme_pain
1	yes	adult	534817	39.2	88.0	20.0	Unknown	Unknown	pale_cyanotic	less_3_sec	mild_pain
2	no	adult	530334	38.3	40.0	24.0	normal	normal	pale_pink	less_3_sec	mild_pain
3	yes	young	5290409	39.1	164.0	84.0	cold	normal	dark_cyanotic	more_3_sec	depressed
4	no	adult	530255	37.3	104.0	35.0	Unknown	Unknown	dark_cyanotic	more_3_sec	Unknown
5	no	adult	528355	0.0	0.0	0.0	warm	normal	pale_pink	less_3_sec	depressed
6	yes	adult	526802	37.9	48.0	16.0	normal	normal	normal_pink	less_3_sec	mild_pain
7	yes	adult	529607	0.0	60.0	0.0	cool	Unknown	Unknown	less_3_sec	Unknown
8	no	adult	530051	0.0	80.0	36.0	cool	absent	pale_pink	less_3_sec	severe_pain
9	no	young	5299629	38.3	90.0	0.0	normal	Unknown	normal_pink	less_3_sec	extreme_pain

Table columns 2 of 3

	peristalsis	abdominal_distention	nasogastric_tube	nasogastric_reflux	nasogastric_reflux_ph	rectal_exam_feces	abdomen	packed_cell_volume	total_protein
0	absent	severe	Unknown	Unknown	0.0	decreased	distend_large	45.0	8.4
1	absent	slight	Unknown	Unknown	0.0	absent	other	50.0	85.0
2	hypomotile	none	Unknown	Unknown	0.0	normal	normal	33.0	6.7
3	absent	severe	none	less_1_liter	5.0	decreased	Unknown	48.0	7.2
4	Unknown	Unknown	Unknown	Unknown	0.0	Unknown	Unknown	74.0	7.4

	peristalsis	abdominal_distention	nasogastric_tube	nasogastric_reflux	nasogastric_reflux_ph	rectal_exam_feces	abdomen	packed_cell_volume	total_protein
5	hypomotile	slight	slight	none	0.0	decreased	firm	0.0	0.0
6	hypomotile	moderate	none	none	0.0	decreased	distend_large	37.0	7.0
7	absent	slight	slight	none	0.0	decreased	distend_small	44.0	8.3
8	absent	severe	slight	none	0.0	decreased	distend_large	38.0	6.2
9	hypomotile	none	slight	none	0.0	decreased	Unknown	40.0	6.2

Table columns 3 of 3

	abdomo_appearance	abdomo_protein	outcome	surgical_lesion	lesion_1	lesion_2	lesion_3	cp_data
0	Unknown	0.0	died	no	11300.0	0.0	0.0	no
1	cloudy	2.0	euthanized	no	2208.0	0.0	0.0	no
2	Unknown	0.0	lived	no	0.0	0.0	0.0	yes
3	serosanguious	5.3	died	yes	2208.0	0.0	0.0	yes
4	Unknown	0.0	died	no	4300.0	0.0	0.0	no
5	Unknown	0.0	lived	no	0.0	0.0	0.0	no
6	Unknown	0.0	lived	yes	3124.0	0.0	0.0	no
7	Unknown	0.0	died	yes	2208.0	0.0	0.0	no
8	Unknown	0.0	euthanized	yes	3205.0	0.0	0.0	no
9	clear	2.2	lived	no	0.0	0.0	0.0	yes

Q19-Q23 Deduplicate and transform

Python code

```
data_unique = data.drop_duplicates()
show("Q19: Unique records", data_unique.head(10))
print("Duplicates removed:", len(data) - len(data_unique))
data[["surgery", "age", "outcome"]] = data[["surgery", "age", "outcome"]].astype("category")
show("Q20: Categorical data types", data[["surgery", "age", "outcome"]].dtypes)
data = data.rename(columns={"rectal_temp": "body_temperature"})
data["pulse_respiratory_ratio"] = data["pulse"] / data["respiratory_rate"].replace(0, np.nan)
show("Q21-Q22: Renamed and derived columns", data.head(10))
show("Q23: Updated summary", data.describe(include="all"))
```

Output

```
Q19: Unique records
```

Table columns 1 of 3

	surgery	age	hospital_number	rectal_temp	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time	pain
0	no	adult	530101	38.5	66.0	28.0	cool	reduced	NaN	more_3_sec	extreme_pain
1	yes	adult	534817	39.2	88.0	20.0	NaN	NaN	pale_cyanotic	less_3_sec	mild_pain
2	no	adult	530334	38.3	40.0	24.0	normal	normal	pale_pink	less_3_sec	mild_pain
3	yes	young	5290409	39.1	164.0	84.0	cold	normal	dark_cyanotic	more_3_sec	depressed
4	no	adult	530255	37.3	104.0	35.0	NaN	NaN	dark_cyanotic	more_3_sec	NaN
5	no	adult	528355	NaN	NaN	NaN	warm	normal	pale_pink	less_3_sec	depressed
6	yes	adult	526802	37.9	48.0	16.0	normal	normal	normal_pink	less_3_sec	mild_pain
7	yes	adult	529607	NaN	60.0	NaN	cool	NaN	NaN	less_3_sec	NaN
8	no	adult	530051	NaN	80.0	36.0	cool	absent	pale_pink	less_3_sec	severe_pain
9	no	young	5299629	38.3	90.0	NaN	normal	NaN	normal_pink	less_3_sec	extreme_pain

Table columns 2 of 3

	peristalsis	abdominal_distention	nasogastric_tube	nasogastric_reflux	nasogastric_reflux_ph	rectal_exam_feces	abdomen	packed_cell_volume	total_protein
0	absent	severe	NaN	NaN	NaN	decreased	distend_large	45.0	8.4
1	absent	slight	NaN	NaN	NaN	absent	other	50.0	85.0
2	hypomotile	none	NaN	NaN	NaN	normal	normal	33.0	6.7
3	absent	severe	none	less_1_liter	5.0	decreased	NaN	48.0	7.2
4	NaN	NaN	NaN	NaN	NaN	NaN	NaN	74.0	7.4
5	hypomotile	slight	slight	none	NaN	decreased	firm	NaN	NaN
6	hypomotile	moderate	none	none	NaN	decreased	distend_large	37.0	7.0
7	absent	slight	slight	none	NaN	decreased	distend_small	44.0	8.3
8	absent	severe	slight	none	NaN	decreased	distend_large	38.0	6.2
9	hypomotile	none	slight	none	NaN	decreased	NaN	40.0	6.2

Table columns 3 of 3

	abdomo_appearance	abdomo_protein	outcome	surgical_lesion	lesion_1	lesion_2	lesion_3	cp_data
0	NaN	NaN	died	no	11300.0	0.0	0.0	no
1	cloudy	2.0	euthanized	no	2208.0	0.0	0.0	no
2	NaN	NaN	lived	no	0.0	0.0	0.0	yes
3	serosanguious	5.3	died	yes	2208.0	0.0	0.0	yes
4	NaN	NaN	died	no	4300.0	0.0	0.0	no
5	NaN	NaN	lived	no	0.0	0.0	0.0	no
6	NaN	NaN	lived	yes	3124.0	0.0	0.0	no
7	NaN	NaN	died	yes	2208.0	0.0	0.0	no
8	NaN	NaN	euthanized	yes	3205.0	0.0	0.0	no
9	clear	2.2	lived	no	0.0	0.0	0.0	yes

```
Duplicates removed: 0
Q20: Categorical data types

surgery    category
age        category
outcome    category
dtype: object

Q21-Q22: Renamed and derived columns
```

Table columns 1 of 3

	surgery	age	hospital_number	body_temperature	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time	pain
0	no	adult	530101	38.5	66.0	28.0	cool	reduced	NaN	more_3_sec	extreme_pain
1	yes	adult	534817	39.2	88.0	20.0	NaN	NaN	pale_cyanotic	less_3_sec	mild_pain
2	no	adult	530334	38.3	40.0	24.0	normal	normal	pale_pink	less_3_sec	mild_pain
3	yes	young	5290409	39.1	164.0	84.0	cold	normal	dark_cyanotic	more_3_sec	depressed
4	no	adult	530255	37.3	104.0	35.0	NaN	NaN	dark_cyanotic	more_3_sec	NaN
5	no	adult	528355	NaN	NaN	NaN	warm	normal	pale_pink	less_3_sec	depressed
6	yes	adult	526802	37.9	48.0	16.0	normal	normal	normal_pink	less_3_sec	mild_pain
7	yes	adult	529607	NaN	60.0	NaN	cool	NaN	NaN	less_3_sec	NaN

	surgery	age	hospital_number	body_temperature	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time	pain
8	no	adult	530051	NaN	80.0	36.0	cool	absent	pale_pink	less_3_sec	severe_pain
9	no	young	5299629	38.3	90.0	NaN	normal	NaN	normal_pink	less_3_sec	extreme_pain

Table columns 2 of 3

	peristalsis	abdominal_distention	nasogastric_tube	nasogastric_reflux	nasogastric_reflux_ph	rectal_exam_feces	abdomen	packed_cell_volume	total_protein
0	absent	severe	NaN	NaN	NaN	decreased	distend_large	45.0	8.4
1	absent	slight	NaN	NaN	NaN	absent	other	50.0	85.0
2	hypomotile	none	NaN	NaN	NaN	normal	normal	33.0	6.7
3	absent	severe	none	less_1_liter	5.0	decreased	NaN	48.0	7.2
4	NaN	NaN	NaN	NaN	NaN	NaN	NaN	74.0	7.4
5	hypomotile	slight	slight	none	NaN	decreased	firm	NaN	NaN
6	hypomotile	moderate	none	none	NaN	decreased	distend_large	37.0	7.0
7	absent	slight	slight	none	NaN	decreased	distend_small	44.0	8.3
8	absent	severe	slight	none	NaN	decreased	distend_large	38.0	6.2
9	hypomotile	none	slight	none	NaN	decreased	NaN	40.0	6.2

Table columns 3 of 3

	abdomo_appearance	abdomo_protein	outcome	surgical_lesion	lesion_1	lesion_2	lesion_3	cp_data	pulse_respiratory_ratio
0	NaN	NaN	died	no	11300.0	0.0	0.0	no	2.357143
1	cloudy	2.0	euthanized	no	2208.0	0.0	0.0	no	4.400000
2	NaN	NaN	lived	no	0.0	0.0	0.0	yes	1.666667
3	serosanguious	5.3	died	yes	2208.0	0.0	0.0	yes	1.952381
4	NaN	NaN	died	no	4300.0	0.0	0.0	no	2.971429
5	NaN	NaN	lived	no	0.0	0.0	0.0	no	NaN
6	NaN	NaN	lived	yes	3124.0	0.0	0.0	no	3.000000
7	NaN	NaN	died	yes	2208.0	0.0	0.0	no	NaN

	abdomo_appearance	abdomo_protein	outcome	surgical_lesion	lesion_1	lesion_2	lesion_3	cp_data	pulse_respiratory_ratio
8	NaN	NaN	euthanized	yes	3205.0	0.0	0.0	no	2.222222
9	clear	2.2	lived	no	0.0	0.0	0.0	yes	NaN

`Q23: Updated summary`

Table columns 1 of 3

	surgery	age	hospital_number	body_temperature	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time
count	110	110	1.100000e+02	85.000000	101.000000	87.000000	90	83	94	99
unique	2	2	NaN	NaN	NaN	NaN	4	4	6	3
top	yes	adult	NaN	NaN	NaN	NaN	cool	reduced	normal_pink	less_3_sec
freq	59	97	NaN	NaN	NaN	NaN	40	42	26	64
mean	NaN	NaN	1.179631e+06	38.149412	75.544554	31.954023	NaN	NaN	NaN	NaN
std	NaN	NaN	1.640871e+06	0.783251	30.672308	18.842621	NaN	NaN	NaN	NaN
min	NaN	NaN	5.216810e+05	35.400000	38.000000	10.000000	NaN	NaN	NaN	NaN
25%	NaN	NaN	5.293520e+05	37.700000	52.000000	20.000000	NaN	NaN	NaN	NaN
50%	NaN	NaN	5.301290e+05	38.100000	64.000000	28.000000	NaN	NaN	NaN	NaN
75%	NaN	NaN	5.347552e+05	38.600000	96.000000	36.000000	NaN	NaN	NaN	NaN
max	NaN	NaN	5.301219e+06	40.300000	164.000000	96.000000	NaN	NaN	NaN	NaN

Table columns 2 of 3

	pain	peristalsis	abdominal_distention	nasogastric_tube	nasogastric_reflux	nasogastric_reflux_ph	rectal_exam_feces	abdomen	packed_cell_volume	total_protein
count	90	91	86	73	72	21.00000	80	74	101.000000	99.000000
unique	5	4	4	3	3	NaN	4	5	NaN	NaN
top	mild_pain	hypomotile	slight	slight	none	NaN	absent	distend_large	NaN	NaN
freq	26	50	24	32	43	NaN	32	34	NaN	NaN
mean	NaN	NaN	NaN	NaN	NaN	4.87619	NaN	NaN	47.336634	21.077778
std	NaN	NaN	NaN	NaN	NaN	2.02803	NaN	NaN	10.038204	26.761927

	pain	peristalsis	abdominal_distention	nasogastric_tube	nasogastric_reflux	nasogastric_reflux_ph	rectal_exam_feces	abdomen	packed_cell_volume	total_protein
min	NaN	NaN	NaN	NaN	NaN	1.00000	NaN	NaN	24.000000	4.500000
25%	NaN	NaN	NaN	NaN	NaN	4.00000	NaN	NaN	40.000000	6.500000
50%	NaN	NaN	NaN	NaN	NaN	5.50000	NaN	NaN	46.000000	7.500000
75%	NaN	NaN	NaN	NaN	NaN	7.00000	NaN	NaN	53.000000	8.650000
max	NaN	NaN	NaN	NaN	NaN	7.20000	NaN	NaN	74.000000	86.000000

Table columns 3 of 3

	abdomo_appearance	abdomo_protein	outcome	surgical_lesion	lesion_1	lesion_2	lesion_3	cp_data	pulse_respiratory_ratio
count	49	38.000000	109	109	109.000000	109.000000	109.0	109	85.000000
unique	3	NaN	3	2	NaN	NaN	NaN	2	NaN
top	cloudy	NaN	lived	yes	NaN	NaN	NaN	no	NaN
freq	21	NaN	64	66	NaN	NaN	NaN	71	NaN
mean	NaN	3.184211	NaN	NaN	3114.715596	12.844037	0.0	NaN	2.851969
std	NaN	1.787574	NaN	NaN	3505.769296	134.095680	0.0	NaN	1.582264
min	NaN	0.100000	NaN	NaN	0.000000	0.000000	0.0	NaN	0.541667
25%	NaN	2.000000	NaN	NaN	2112.000000	0.000000	0.0	NaN	1.875000
50%	NaN	2.450000	NaN	NaN	3111.000000	0.000000	0.0	NaN	2.400000
75%	NaN	4.400000	NaN	NaN	3207.000000	0.000000	0.0	NaN	3.250000
max	NaN	7.400000	NaN	NaN	31110.000000	1400.000000	0.0	NaN	9.000000

Q24-Q28 Numerical statistics

Python code

```
pulse = data["pulse"].dropna()
display(pd.Series({"Minimum": pulse.min(), "Maximum": pulse.max(),
                   "Range (maximum - minimum)": pulse.max() - pulse.min(),
                   "Variance": pulse.var(ddof=1), "Standard deviation": pulse.std(ddof=1),
                   "IQR": pulse.quantile(.75) - pulse.quantile(.25)}))
print("Range endpoints:", (pulse.min(), pulse.max()))
```

Output

```
Minimum                      38.000000
Maximum                     164.000000
Range (maximum - minimum)   126.000000
Variance                    940.790495
Standard deviation           30.672308
IQR                          44.000000
dtype: float64

Range endpoints: (np.float64(38.0), np.float64(164.0))
```

Assignment 3 - Data Cleaning and Univariate Analysis

This assignment checks missing values, duplicates, data types, and pulse outliers, then compares alternative cleaning methods. It visualizes numeric distributions and category frequencies with histograms, density curves, violin plots, bar charts, pie charts, and facets.

Questions

Load dplyr, tidyr, ggplot2, stringr, forcats into R

1. Import a dataset into R. (Dataset shared through e-mail)

2. Display the first and last six records of the dataset

3. Identify the total number of missing values in the dataset.

4. Display all records containing at least one missing value.

5. Remove columns having more than 70 percentage of missing values.

6. Replace missing numerical values with the mean.

7. Replace missing numerical values with the median.

8. Replace missing categorical values with the mode.

9. Identify duplicate rows if any.

10. Remove the duplicate rows if any.

11. Identify columns stored with incorrect data types.

12. Convert character variables into factors if any.

13. Convert text values into lowercase/ uppercase.

14. Detect outliers using boxplots.

15. Identify outliers using the interquartile range method.

16. Remove extreme outliers from the dataset.

17. Create a histogram for a numerical variable and interpret its distribution.

18. Add a density curve to the histogram.

19. Create a violin plot

20. Create a dot plot for a small numerical dataset.

21. Create a frequency table for a categorical variable.

22. Calculate the proportion and percentage of each category.

23. Create a bar chart showing the frequency of each category.

24. Create a horizontal bar chart for a variable with several categories.

25. Create a percentage bar chart instead of a frequency bar chart.

26. Create a pie chart for a categorical variable with a small number of categories.

27. Create a lollipop chart for categorical frequencies.

28. Create a dot chart

29. Create a density plot using ggplot2.

30. Use facet_wrap() to display the distribution of one variable across different groups while interpreting each distribution separately.

Q1-Q5 Import and inspect missing data

Python code

```
data = load_horse()
show("First six records", data.head(6))
show("Last six records", data.tail(6))
print("Total missing values:", int(data.isna().sum().sum()))
print("All records with at least one missing value:")
print(data.loc[data.isna().any(axis=1)].to_string(index=False))
data_col_filtered = data.loc[:, data.isna().mean().le(.70)].copy()
print("Columns retained:", data_col_filtered.columns.tolist())
print("Columns removed:", data.columns.difference(data_col_filtered.columns).tolist())
```

Output

```
First six records
```

Table columns 1 of 3

	surgery	age	hospital_number	rectal_temp	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time	pain
0	no	adult	530101	38.5	66.0	28.0	cool	reduced	NaN	more_3_sec	extreme_pain
1	yes	adult	534817	39.2	88.0	20.0	NaN	NaN	pale_cyanotic	less_3_sec	mild_pain
2	no	adult	530334	38.3	40.0	24.0	normal	normal	pale_pink	less_3_sec	mild_pain

	surgery	age	hospital_number	rectal_temp	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time	pain
3	yes	young	5290409	39.1	164.0	84.0	cold	normal	dark_cyanotic	more_3_sec	depressed
4	no	adult	530255	37.3	104.0	35.0	NaN	NaN	dark_cyanotic	more_3_sec	NaN
5	no	adult	528355	NaN	NaN	NaN	warm	normal	pale_pink	less_3_sec	depressed

Table columns 2 of 3

	peristalsis	abdominal_distention	nasogastric_tube	nasogastric_reflux	nasogastric_reflux_ph	rectal_exam_feces	abdomen	packed_cell_volume	total_protein
0	absent	severe	NaN	NaN	NaN	decreased	distend_large	45.0	8.4
1	absent	slight	NaN	NaN	NaN	absent	other	50.0	85.0
2	hypomotile	none	NaN	NaN	NaN	normal	normal	33.0	6.7
3	absent	severe	none	less_1_liter	5.0	decreased	NaN	48.0	7.2
4	NaN	NaN	NaN	NaN	NaN	NaN	NaN	74.0	7.4
5	hypomotile	slight	slight	none	NaN	decreased	firm	NaN	NaN

Table columns 3 of 3

	abdomo_appearance	abdomo_protein	outcome	surgical_lesion	lesion_1	lesion_2	lesion_3	cp_data
0	NaN	NaN	died	no	11300.0	0.0	0.0	no
1	cloudy	2.0	euthanized	no	2208.0	0.0	0.0	no
2	NaN	NaN	lived	no	0.0	0.0	0.0	yes
3	serosanguious	5.3	died	yes	2208.0	0.0	0.0	yes
4	NaN	NaN	died	no	4300.0	0.0	0.0	no
5	NaN	NaN	lived	no	0.0	0.0	0.0	no

```
Last six records
```

Table columns 1 of 3

	surgery	age	hospital_number	rectal_temp	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time	pain
104	yes	adult	527563	37.8	52.0	24.0	normal	reduced	pale_pink	less_3_sec	severe_pain
105	yes	adult	530381	NaN	70.0	36.0	normal	NaN	pale_pink	more_3_sec	depressed

	surgery	age	hospital_number	rectal_temp	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time	pain
106	yes	adult	528668	38.3	52.0	96.0	NaN	reduced	pale_pink	less_3_sec	NaN
107	no	adult	529764	37.3	50.0	32.0	normal	normal	pale_pink	less_3_sec	alert
108	yes	adult	533696	38.7	60.0	32.0	cold	reduced	bright_pink	more_3_sec	severe_pain
109	yes	young	5297379	38.4	84.0	40.0	cool	reduced	bright_pink	less_3_sec	mild_pain

Table columns 2 of 3

	peristalsis	abdominal_distention	nasogastric_tube	nasogastric_reflux	nasogastric_reflux_ph	rectal_exam_feces	abdomen	packed_cell_volume	total_protein
104	absent	none	slight	more_1_liter	5.7	increased	distend_large	48.0	6.6
105	hypomotile	slight	slight	NaN	NaN	absent	distend_large	36.0	7.3
106	NaN	NaN	none	none	NaN	normal	NaN	43.0	6.1
107	hypomotile	slight	NaN	NaN	NaN	normal	NaN	44.0	7.0
108	absent	severe	NaN	NaN	NaN	absent	distend_large	53.0	64.0
109	hypomotile	moderate	none	none	NaN	NaN	NaN	36.0	6.6

Table columns 3 of 3

	abdomo_appearance	abdomo_protein	outcome	surgical_lesion	lesion_1	lesion_2	lesion_3	cp_data
104	clear	3.7	died	yes	5400.0	0.0	0.0	no
105	NaN	NaN	lived	yes	31110.0	0.0	0.0	no
106	NaN	NaN	lived	yes	3209.0	0.0	0.0	yes
107	NaN	NaN	lived	no	0.0	0.0	0.0	no
108	serosanguious	2.0	euthanized	yes	3205.0	0.0	0.0	no
109	cloudy	2.0	NaN	NaN	NaN	NaN	NaN	NaN

```
Total missing values: 583
```

All records with at least one missing value

Table columns 1 of 3

Row	surgery	age	hospital_number	rectal_temp	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time	pain
0	no	adult	530101	38.5	66.0	28.0	cool	reduced	NaN	more_3_sec	extreme_pain
1	yes	adult	534817	39.2	88.0	20.0	NaN	NaN	pale_cyanotic	less_3_sec	mild_pain
2	no	adult	530334	38.3	40.0	24.0	normal	normal	pale_pink	less_3_sec	mild_pain
3	yes	young	5290409	39.1	164.0	84.0	cold	normal	dark_cyanotic	more_3_sec	depressed
4	no	adult	530255	37.3	104.0	35.0	NaN	NaN	dark_cyanotic	more_3_sec	NaN
5	no	adult	528355	NaN	NaN	NaN	warm	normal	pale_pink	less_3_sec	depressed
6	yes	adult	526802	37.9	48.0	16.0	normal	normal	normal_pink	less_3_sec	mild_pain
7	yes	adult	529607	NaN	60.0	NaN	cool	NaN	NaN	less_3_sec	NaN
8	no	adult	530051	NaN	80.0	36.0	cool	absent	pale_pink	less_3_sec	severe_pain
9	no	young	5299629	38.3	90.0	NaN	normal	NaN	normal_pink	less_3_sec	extreme_pain
11	no	adult	527927	39.1	72.0	52.0	warm	NaN	bright_pink	less_3_sec	depressed
12	yes	adult	528031	37.2	42.0	12.0	warm	normal	normal_pink	less_3_sec	mild_pain
13	no	young	5291329	38.0	92.0	28.0	normal	normal	bright_pink	less_3_sec	alert
14	yes	adult	534917	38.2	76.0	28.0	cool	normal	normal_pink	less_3_sec	mild_pain
15	yes	adult	530233	37.6	96.0	48.0	cool	normal	pale_cyanotic	less_3_sec	extreme_pain
16	yes	young	5301219	NaN	128.0	36.0	cool	reduced	pale_cyanotic	more_3_sec	severe_pain
17	no	adult	526639	37.5	48.0	24.0	NaN	NaN	NaN	NaN	NaN
18	yes	adult	5290481	37.6	64.0	21.0	normal	normal	bright_pink	less_3_sec	depressed
19	no	adult	532110	39.4	110.0	35.0	cold	reduced	dark_cyanotic	NaN	NaN
20	yes	adult	530157	39.9	72.0	60.0	normal	normal	bright_red	more_3_sec	extreme_pain
21	no	adult	529340	38.4	48.0	16.0	normal	NaN	normal_pink	less_3_sec	alert
22	yes	adult	521681	38.6	42.0	34.0	warm	normal	pale_cyanotic	NaN	depressed
23	yes	young	534998	38.3	130.0	60.0	NaN	reduced	NaN	less_3_sec	depressed
24	yes	adult	533692	38.1	60.0	12.0	cool	reduced	pale_pink	less_3_sec	NaN
25	no	adult	529518	37.8	60.0	42.0	NaN	NaN	NaN	less_3_sec	NaN
26	yes	adult	530526	38.3	72.0	30.0	cold	reduced	pale_pink	more_3_sec	mild_pain

Row	surgery	age	hospital_number	rectal_temp	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time	pain
27	yes	adult	528653	37.8	48.0	12.0	cool	normal	normal_pink	less_3_sec	NaN
28	yes	adult	5279442	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
29	no	adult	535415	37.7	48.0	NaN	warm	normal	normal_pink	less_3_sec	alert
30	no	adult	529475	37.7	96.0	30.0	cool	reduced	pale_cyanotic	more_3_sec	extreme_pain
31	no	adult	530242	37.2	108.0	12.0	cool	reduced	pale_cyanotic	more_3_sec	depressed
32	yes	adult	529427	37.2	60.0	NaN	warm	normal	normal_pink	less_3_sec	mild_pain
33	yes	adult	529663	38.2	64.0	28.0	normal	normal	normal_pink	less_3_sec	mild_pain
34	yes	adult	529796	NaN	100.0	30.0	cool	reduced	pale_cyanotic	more_3_sec	extreme_pain
35	no	adult	528812	NaN	104.0	24.0	cold	reduced	pale_pink	more_3_sec	severe_pain
36	no	adult	529493	38.3	112.0	16.0	NaN	reduced	bright_red	more_3_sec	NaN
37	yes	adult	533847	37.8	72.0	NaN	NaN	reduced	NaN	less_3_sec	extreme_pain
38	no	adult	528996	38.6	52.0	NaN	normal	normal	normal_pink	less_3_sec	mild_pain
39	yes	young	5277409	39.2	146.0	96.0	NaN	NaN	NaN	NaN	NaN
40	yes	adult	529498	NaN	88.0	NaN	cool	reduced	dark_cyanotic	more_3_sec	extreme_pain
41	no	young	5288249	39.0	150.0	72.0	NaN	NaN	NaN	NaN	NaN
42	no	adult	530301	38.0	60.0	12.0	cool	normal	pale_pink	less_3_sec	mild_pain
43	yes	adult	534069	NaN	120.0	NaN	cool	absent	pale_cyanotic	less_3_sec	severe_pain
44	yes	adult	535407	35.4	140.0	24.0	cool	reduced	pale_cyanotic	more_3_sec	severe_pain
45	no	adult	529827	NaN	120.0	NaN	cold	reduced	pale_cyanotic	more_3_sec	extreme_pain
46	yes	adult	529888	37.9	60.0	15.0	cool	NaN	pale_cyanotic	more_3_sec	extreme_pain
47	no	adult	529821	37.5	48.0	16.0	normal	normal	normal_pink	less_3_sec	alert
48	yes	adult	528890	38.9	80.0	44.0	cool	reduced	pale_pink	more_3_sec	depressed
49	no	adult	529642	37.2	84.0	48.0	cool	reduced	bright_red	more_3_sec	severe_pain
50	no	adult	529766	38.6	46.0	NaN	normal	normal	bright_pink	less_3_sec	alert
51	yes	adult	527706	37.4	84.0	36.0	normal	NaN	pale_pink	more_3_sec	mild_pain
52	no	adult	529483	NaN	NaN	NaN	normal	normal	pale_pink	less_3_sec	alert

Row	surgery	age	hospital_number	rectal_temp	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time	pain
53	no	adult	530544	38.6	40.0	20.0	NaN	NaN	NaN	less_3_sec	NaN
55	yes	young	5282839	38.6	160.0	20.0	cool	NaN	bright_red	less_3_sec	mild_pain
56	yes	adult	528872	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
57	yes	adult	529640	NaN	64.0	36.0	warm	NaN	bright_pink	less_3_sec	extreme_pain
58	yes	adult	528298	NaN	NaN	20.0	cold	reduced	pale_pink	NaN	extreme_pain
59	no	adult	528904	NaN	96.0	NaN	cool	reduced	pale_pink	more_3_sec	extreme_pain
60	no	adult	530438	37.8	48.0	32.0	normal	normal	pale_pink	less_3_sec	depressed
61	no	adult	527957	38.5	60.0	NaN	warm	increased	normal_pink	less_3_sec	alert
62	yes	adult	528630	37.8	88.0	22.0	warm	normal	bright_pink	less_3_sec	mild_pain
63	no	adult	534293	38.2	130.0	16.0	cold	reduced	pale_cyanotic	more_3_sec	depressed
65	yes	adult	534885	NaN	60.0	36.0	cool	normal	pale_pink	less_3_sec	mild_pain
66	no	adult	534784	37.9	72.0	NaN	normal	normal	bright_red	more_3_sec	mild_pain
67	no	adult	528931	38.4	54.0	24.0	normal	normal	normal_pink	less_3_sec	alert
68	no	adult	529777	NaN	52.0	16.0	normal	NaN	pale_pink	less_3_sec	NaN
69	no	adult	530276	38.0	48.0	12.0	normal	normal	normal_pink	less_3_sec	alert
70	no	adult	528214	37.0	60.0	20.0	cool	NaN	NaN	less_3_sec	mild_pain
71	yes	adult	529424	37.8	48.0	28.0	normal	normal	normal_pink	less_3_sec	alert
72	yes	adult	5299253	37.7	56.0	NaN	NaN	NaN	NaN	NaN	NaN
73	yes	adult	528469	38.1	52.0	24.0	normal	normal	bright_red	less_3_sec	severe_pain
74	yes	young	5292929	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
75	yes	young	534092	39.7	100.0	NaN	cool	reduced	bright_red	more_3_sec	depressed
76	yes	adult	534615	37.6	38.0	20.0	cool	reduced	normal_pink	less_3_sec	mild_pain
77	no	adult	534753	38.7	52.0	20.0	warm	NaN	normal_pink	less_3_sec	alert
78	yes	adult	530693	NaN	NaN	NaN	cool	reduced	pale_pink	3	extreme_pain
79	yes	adult	527463	37.5	96.0	18.0	normal	reduced	dark_cyanotic	more_3_sec	mild_pain
80	yes	adult	527518	36.4	98.0	35.0	cool	reduced	pale_cyanotic	less_3_sec	severe_pain

Row	surgery	age	hospital_number	rectal_temp	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time	pain
81	yes	adult	534756	37.3	40.0	NaN	NaN	reduced	normal_pink	less_3_sec	depressed
82	yes	young	5290759	38.1	100.0	80.0	cool	normal	bright_pink	less_3_sec	mild_pain
83	yes	adult	5279822	38.0	NaN	24.0	cool	reduced	dark_cyanotic	more_3_sec	extreme_pain
84	yes	adult	529849	37.8	60.0	80.0	normal	reduced	bright_pink	more_3_sec	depressed
85	no	adult	529304	38.0	54.0	30.0	warm	reduced	pale_pink	3	mild_pain
86	yes	adult	529388	NaN	88.0	40.0	cool	reduced	pale_cyanotic	more_3_sec	extreme_pain
87	no	adult	528006	NaN	40.0	16.0	NaN	NaN	NaN	less_3_sec	NaN
88	no	adult	529703	39.0	64.0	40.0	normal	normal	bright_red	less_3_sec	mild_pain
89	no	adult	535381	38.3	42.0	10.0	normal	normal	normal_pink	less_3_sec	alert
90	no	adult	534197	38.0	52.0	16.0	NaN	NaN	NaN	NaN	depressed
92	no	adult	530251	38.8	50.0	20.0	cool	normal	normal_pink	less_3_sec	alert
93	no	adult	530310	NaN	NaN	NaN	cool	reduced	normal_pink	less_3_sec	extreme_pain
94	no	adult	528729	37.5	48.0	30.0	cold	normal	pale_pink	less_3_sec	NaN
95	yes	adult	535158	37.3	48.0	20.0	NaN	normal	bright_pink	less_3_sec	mild_pain
96	no	adult	530319	NaN	84.0	36.0	NaN	NaN	pale_pink	less_3_sec	NaN
97	yes	adult	534145	38.1	88.0	32.0	cool	reduced	pale_cyanotic	less_3_sec	depressed
98	no	adult	534135	37.7	44.0	40.0	warm	normal	pale_pink	less_3_sec	alert
99	no	adult	530002	39.6	108.0	51.0	cool	reduced	dark_cyanotic	more_3_sec	depressed
100	yes	adult	535364	38.2	40.0	16.0	cool	reduced	normal_pink	less_3_sec	alert
101	yes	adult	534899	NaN	60.0	20.0	cold	reduced	pale_cyanotic	more_3_sec	extreme_pain
102	no	adult	534938	38.3	40.0	16.0	cool	NaN	normal_pink	less_3_sec	depressed
103	yes	young	5292489	38.0	140.0	68.0	normal	normal	normal_pink	less_3_sec	mild_pain
105	yes	adult	530381	NaN	70.0	36.0	normal	NaN	pale_pink	more_3_sec	depressed
106	yes	adult	528668	38.3	52.0	96.0	NaN	reduced	pale_pink	less_3_sec	NaN
107	no	adult	529764	37.3	50.0	32.0	normal	normal	pale_pink	less_3_sec	alert
108	yes	adult	533696	38.7	60.0	32.0	cold	reduced	bright_pink	more_3_sec	severe_pain

Row	surgery	age	hospital_number	rectal_temp	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time	pain
109	yes	young	5297379	38.4	84.0	40.0	cool	reduced	bright_pink	less_3_sec	mild_pain

Table columns 2 of 3

peristalsis	abdominal_distention	nasogastric_tube	nasogastric_reflux	nasogastric_reflux_ph	rectal_exam_feces	abdomen	packed_cell_volume	total_protein
absent	severe	NaN	NaN	NaN	decreased	distend_large	45.0	8.4
absent	slight	NaN	NaN	NaN	absent	other	50.0	85.0
hypomotile	none	NaN	NaN	NaN	normal	normal	33.0	6.7
absent	severe	none	less_1_liter	5.0	decreased	NaN	48.0	7.2
NaN	NaN	NaN	NaN	NaN	NaN	NaN	74.0	7.4
hypomotile	slight	slight	none	NaN	decreased	firm	NaN	NaN
hypomotile	moderate	none	none	NaN	decreased	distend_large	37.0	7.0
absent	slight	slight	none	NaN	decreased	distend_small	44.0	8.3
absent	severe	slight	none	NaN	decreased	distend_large	38.0	6.2
hypomotile	none	slight	none	NaN	decreased	NaN	40.0	6.2
hypermotile	slight	none	none	NaN	absent	distend_small	50.0	7.8
hypomotile	moderate	significant	none	NaN	absent	distend_large	NaN	7.0
hypomotile	slight	significant	NaN	7.2	normal	normal	37.0	6.1
absent	none	slight	less_1_liter	NaN	absent	distend_small	46.0	81.0
hypomotile	moderate	slight	more_1_liter	4.5	absent	NaN	45.0	6.8
absent	moderate	significant	NaN	NaN	absent	distend_large	53.0	7.8
NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
hypomotile	none	none	none	NaN	increased	distend_large	40.0	7.0
hypomotile	moderate	NaN	NaN	NaN	NaN	NaN	55.0	8.7
absent	severe	significant	none	NaN	absent	distend_small	46.0	6.1
hypomotile	none	slight	more_1_liter	5.5	absent	firm	49.0	6.8
hypomotile	none	NaN	NaN	NaN	normal	NaN	48.0	7.2

peristalsis	abdominal_distention	nasogastric_tube	nasogastric_reflux	nasogastric_reflux_ph	rectal_exam_feces	abdomen	packed_cell_volume	total_protein
absent	NaN	NaN	NaN	NaN	NaN	NaN	50.0	70.0
absent	moderate	significant	less_1_liter	2.0	NaN	NaN	51.0	65.0
NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
hypomotile	moderate	slight	none	NaN	decreased	distend_large	43.0	7.0
hypomotile	slight	none	none	NaN	normal	firm	37.0	5.5
NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
hypermotile	none	none	none	NaN	NaN	NaN	45.0	76.0
absent	severe	significant	less_1_liter	4.0	absent	distend_large	66.0	7.5
absent	slight	NaN	more_1_liter	6.0	decreased	firm	52.0	8.2
hypomotile	moderate	slight	none	NaN	absent	distend_large	43.0	6.6
hypermotile	NaN	NaN	NaN	NaN	absent	distend_small	49.0	8.6
absent	severe	significant	more_1_liter	NaN	absent	distend_small	52.0	6.6
absent	moderate	NaN	more_1_liter	NaN	NaN	other	73.0	8.4
NaN	none	none	less_1_liter	NaN	NaN	distend_large	51.0	6.0
hypomotile	none	NaN	none	NaN	normal	normal	56.0	80.0
hypomotile	slight	none	none	NaN	normal	firm	32.0	6.6
NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
hypomotile	moderate	none	more_1_liter	NaN	absent	distend_large	63.0	6.5
NaN	NaN	NaN	NaN	NaN	NaN	NaN	47.0	8.5
hypomotile	none	none	none	NaN	increased	other	47.0	7.0
absent	severe	none	none	NaN	NaN	distend_large	52.0	67.0
absent	NaN	slight	none	NaN	NaN	distend_large	57.0	69.0
absent	severe	none	none	NaN	absent	distend_large	60.0	6.5
absent	severe	slight	less_1_liter	NaN	absent	distend_large	65.0	7.5
hypermotile	none	none	none	NaN	normal	NaN	37.0	6.5
hypomotile	moderate	slight	less_1_liter	7.0	decreased	normal	54.0	6.5

peristalsis	abdominal_distention	nasogastric_tube	nasogastric_reflux	nasogastric_reflux_ph	rectal_exam_feces	abdomen	packed_cell_volume	total_protein
hypermotile	slight	none	less_1_liter	NaN	increased	normal	73.0	5.5
hypomotile	slight	none	none	NaN	NaN	other	49.0	9.1
hypomotile	slight	NaN	NaN	NaN	absent	distend_large	NaN	NaN
hypomotile	none	NaN	NaN	NaN	increased	other	43.0	7.7
NaN	NaN	NaN	NaN	NaN	NaN	NaN	41.0	6.4
hypomotile	severe	significant	NaN	NaN	absent	NaN	38.0	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN	24.0	6.7
hypomotile	moderate	slight	less_1_liter	NaN	NaN	NaN	42.0	7.7
absent	moderate	slight	NaN	NaN	absent	distend_small	53.0	5.9
absent	severe	none	less_1_liter	NaN	absent	distend_large	60.0	NaN
hypermotile	NaN	none	none	NaN	absent	distend_large	37.0	6.7
normal	slight	slight	none	NaN	normal	normal	44.0	7.7
NaN	NaN	slight	NaN	NaN	absent	NaN	64.0	8.0
absent	severe	none	none	NaN	NaN	NaN	65.0	82.0
hypomotile	slight	none	none	NaN	decreased	distend_small	26.0	72.0
hypomotile	none	none	more_1_liter	2.0	decreased	distend_small	58.0	74.0
hypomotile	none	slight	none	NaN	decreased	other	49.0	7.2
NaN	NaN	slight	more_1_liter	5.5	NaN	NaN	55.0	7.2
hypomotile	NaN	none	none	NaN	decreased	other	42.0	6.3
NaN	moderate	slight	less_1_liter	4.5	absent	distend_small	43.0	7.6
normal	none	slight	NaN	NaN	normal	normal	46.0	5.9
NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
hypomotile	none	slight	more_1_liter	7.0	normal	NaN	54.0	7.5
NaN	NaN	NaN	NaN	NaN	NaN	NaN	37.0	4.9
hypomotile	NaN	NaN	NaN	NaN	NaN	NaN	48.0	57.0
hypomotile	slight	NaN	NaN	NaN	decreased	NaN	37.0	68.0

peristalsis	abdominal_distention	nasogastric_tube	nasogastric_reflux	nasogastric_reflux_ph	rectal_exam_feces	abdomen	packed_cell_volume	total_protein
hypermotile	none	none	none	NaN	normal	normal	33.0	77.0
hypomotile	moderate	significant	less_1_liter	NaN	absent	distend_large	46.0	5.9
absent	slight	slight	more_1_liter	5.0	NaN	distend_small	69.0	8.9
hypomotile	slight	NaN	NaN	NaN	absent	distend_small	47.0	6.4
hypomotile	slight	significant	none	NaN	decreased	distend_large	36.0	NaN
absent	none	NaN	NaN	NaN	normal	NaN	36.0	5.7
NaN	severe	none	none	NaN	NaN	NaN	68.0	7.8
hypomotile	moderate	NaN	less_1_liter	5.5	absent	NaN	40.0	4.5
hypermotile	slight	slight	less_1_liter	NaN	NaN	distend_small	45.0	6.2
absent	moderate	significant	NaN	NaN	absent	distend_large	50.0	7.7
NaN	NaN	NaN	NaN	NaN	NaN	NaN	50.0	7.0
hypomotile	slight	slight	none	NaN	decreased	firm	42.0	7.5
hypermotile	none	NaN	NaN	NaN	NaN	NaN	38.0	61.0
NaN	NaN	NaN	more_1_liter	1.0	normal	normal	53.0	86.0
hypermotile	none	slight	none	NaN	decreased	normal	42.0	6.2
hypomotile	moderate	none	none	NaN	absent	distend_large	38.0	6.5
normal	none	none	none	NaN	normal	normal	48.0	8.6
hypomotile	moderate	slight	none	NaN	decreased	distend_large	41.0	69.0
hypomotile	none	slight	none	NaN	decreased	other	44.0	8.5
hypomotile	moderate	NaN	more_1_liter	1.0	absent	distend_large	55.0	60.0
hypomotile	slight	none	none	NaN	normal	distend_large	41.0	60.0
absent	moderate	none	less_1_liter	NaN	decreased	distend_large	59.0	8.0
hypomotile	NaN	NaN	NaN	NaN	normal	normal	34.0	66.0
absent	NaN	NaN	none	NaN	absent	distend_large	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN	37.0	57.0
hypomotile	slight	NaN	NaN	NaN	increased	normal	39.0	5.3

peristalsis	abdominal_distention	nasogastric_tube	nasogastric_reflux	nasogastric_reflux_ph	rectal_exam_feces	abdomen	packed_cell_volume	total_protein
hypomotile	slight	slight	NaN	NaN	absent	distend_large	36.0	7.3
NaN	NaN	none	none	NaN	normal	NaN	43.0	6.1
hypomotile	slight	NaN	NaN	NaN	normal	NaN	44.0	7.0
absent	severe	NaN	NaN	NaN	absent	distend_large	53.0	64.0
hypomotile	moderate	none	none	NaN	NaN	NaN	36.0	6.6

Table columns 3 of 3

abdomo_appearance	abdomo_protein	outcome	surgical_lesion	lesion_1	lesion_2	lesion_3	cp_data
NaN	NaN	died	no	11300.0	0.0	0.0	no
cloudy	2.0	euthanized	no	2208.0	0.0	0.0	no
NaN	NaN	lived	no	0.0	0.0	0.0	yes
serosanguious	5.3	died	yes	2208.0	0.0	0.0	yes
NaN	NaN	died	no	4300.0	0.0	0.0	no
NaN	NaN	lived	no	0.0	0.0	0.0	no
NaN	NaN	lived	yes	3124.0	0.0	0.0	no
NaN	NaN	died	yes	2208.0	0.0	0.0	no
NaN	NaN	euthanized	yes	3205.0	0.0	0.0	no
clear	2.2	lived	no	0.0	0.0	0.0	yes
NaN	NaN	lived	yes	2111.0	0.0	0.0	no
NaN	NaN	lived	no	4124.0	0.0	0.0	no
clear	NaN	died	no	0.0	0.0	0.0	yes
clear	2.0	lived	yes	2112.0	0.0	0.0	no
NaN	NaN	died	yes	3207.0	0.0	0.0	no
serosanguious	4.7	died	no	1400.0	0.0	0.0	yes
NaN	NaN	lived	no	0.0	0.0	0.0	no
clear	NaN	lived	yes	4205.0	0.0	0.0	yes
NaN	NaN	lived	no	0.0	0.0	0.0	no

abdomo_appearance	abdomo_protein	outcome	surgical_lesion	lesion_1	lesion_2	lesion_3	cp_data
cloudy	NaN	lived	yes	2111.0	0.0	0.0	no
NaN	NaN	lived	no	0.0	0.0	0.0	no
NaN	NaN	lived	yes	3111.0	0.0	0.0	no
NaN	NaN	lived	yes	3111.0	0.0	0.0	no
NaN	NaN	lived	yes	3111.0	0.0	0.0	no
NaN	NaN	lived	no	0.0	0.0	0.0	no
cloudy	3.9	lived	yes	3111.0	0.0	0.0	yes
cloudy	1.3	lived	no	4122.0	0.0	0.0	yes
NaN	NaN	died	no	4300.0	0.0	0.0	no
NaN	NaN	lived	no	0.0	0.0	0.0	no
NaN	NaN	died	yes	4205.0	0.0	0.0	no
serosanguious	7.4	euthanized	yes	2207.0	0.0	0.0	yes
NaN	NaN	lived	yes	2209.0	0.0	0.0	no
cloudy	6.6	lived	yes	2208.0	0.0	0.0	yes
NaN	NaN	lived	yes	1124.0	0.0	0.0	no
NaN	NaN	euthanized	yes	7111.0	0.0	0.0	no
cloudy	1.0	euthanized	no	5205.0	0.0	0.0	yes
clear	2.0	lived	yes	3111.0	0.0	0.0	no
clear	5.0	lived	no	0.0	0.0	0.0	yes
NaN	NaN	died	yes	2113.0	0.0	0.0	no
serosanguious	NaN	died	yes	4205.0	0.0	0.0	no
NaN	0.1	lived	yes	9400.0	0.0	0.0	yes
NaN	NaN	lived	no	3111.0	0.0	0.0	no
cloudy	2.0	euthanized	yes	3205.0	0.0	0.0	no
serosanguious	2.0	euthanized	yes	3205.0	0.0	0.0	no
serosanguious	NaN	died	yes	3205.0	0.0	0.0	no
NaN	NaN	lived	yes	2305.0	0.0	0.0	yes

abdomo_appearance	abdomo_protein	outcome	surgical_lesion	lesion_1	lesion_2	lesion_3	cp_data
NaN	NaN	lived	no	0.0	0.0	0.0	no
serosanguious	NaN	died	yes	7111.0	0.0	0.0	no
cloudy	4.1	died	no	4300.0	0.0	0.0	yes
clear	1.6	lived	no	3111.0	0.0	0.0	yes
serosanguious	NaN	died	yes	7209.0	0.0	0.0	yes
NaN	NaN	lived	no	3111.0	0.0	0.0	no
NaN	NaN	lived	no	3111.0	0.0	0.0	yes
cloudy	NaN	died	yes	7111.0	0.0	0.0	yes
NaN	NaN	lived	yes	3112.0	0.0	0.0	no
NaN	NaN	died	yes	3209.0	0.0	0.0	no
serosanguious	NaN	died	yes	3205.0	0.0	0.0	yes
NaN	NaN	died	yes	4205.0	0.0	0.0	no
NaN	NaN	lived	no	2124.0	0.0	0.0	no
NaN	NaN	lived	no	0.0	0.0	0.0	no
clear	6.0	died	yes	3205.0	0.0	0.0	yes
cloudy	2.0	euthanized	no	2209.0	1400.0	0.0	no
cloudy	1.0	lived	yes	2208.0	0.0	0.0	no
clear	2.0	lived	yes	2322.0	0.0	0.0	no
clear	NaN	lived	no	3111.0	0.0	0.0	yes
NaN	NaN	lived	no	2124.0	0.0	0.0	no
cloudy	4.1	lived	no	3111.0	0.0	0.0	yes
NaN	NaN	euthanized	yes	2209.0	0.0	0.0	yes
cloudy	7.0	lived	no	0.0	0.0	0.0	yes
NaN	NaN	died	yes	2113.0	0.0	0.0	no
cloudy	2.6	died	yes	2112.0	0.0	0.0	yes
NaN	NaN	died	yes	11124.0	0.0	0.0	no
cloudy	2.0	euthanized	yes	1400.0	0.0	0.0	no

abdomo_appearance	abdomo_protein	outcome	surgical_lesion	lesion_1	lesion_2	lesion_3	cp_data
NaN	NaN	lived	yes	3205.0	0.0	0.0	no
NaN	NaN	lived	no	0.0	0.0	0.0	no
NaN	NaN	died	yes	3025.0	0.0	0.0	no
serosanguious	NaN	lived	yes	2208.0	0.0	0.0	yes
serosanguious	3.6	died	yes	2205.0	0.0	0.0	yes
serosanguious	2.0	lived	yes	3111.0	0.0	0.0	no
NaN	NaN	lived	yes	3111.0	0.0	0.0	no
NaN	NaN	died	yes	3205.0	0.0	0.0	no
cloudy	NaN	lived	yes	5206.0	0.0	0.0	yes
NaN	NaN	lived	no	400.0	0.0	0.0	no
serosanguious	1.4	died	yes	4205.0	0.0	0.0	yes
cloudy	3.9	euthanized	yes	2208.0	0.0	0.0	yes
cloudy	2.3	lived	no	5000.0	0.0	0.0	yes
NaN	NaN	lived	no	0.0	0.0	0.0	no
NaN	NaN	lived	yes	2322.0	0.0	0.0	no
NaN	NaN	lived	no	0.0	0.0	0.0	no
NaN	NaN	died	yes	3205.0	0.0	0.0	no
NaN	NaN	lived	no	0.0	0.0	0.0	no
NaN	NaN	lived	yes	3205.0	0.0	0.0	no
NaN	NaN	lived	yes	3111.0	0.0	0.0	no
NaN	NaN	euthanized	no	3205.0	0.0	0.0	no
NaN	NaN	lived	no	0.0	0.0	0.0	no
cloudy	2.6	lived	no	4300.0	0.0	0.0	yes
NaN	NaN	lived	no	0.0	0.0	0.0	no
NaN	NaN	euthanized	yes	3205.0	0.0	0.0	no
NaN	NaN	lived	no	0.0	0.0	0.0	no
NaN	NaN	lived	yes	7111.0	0.0	0.0	no

abdomo_appearance	abdomo_protein	outcome	surgical_lesion	lesion_1	lesion_2	lesion_3	cp_data
NaN	NaN	lived	yes	31110.0	0.0	0.0	no
NaN	NaN	lived	yes	3209.0	0.0	0.0	yes
NaN	NaN	lived	no	0.0	0.0	0.0	no
serosanguious	2.0	euthanized	yes	3205.0	0.0	0.0	no
cloudy	2.0	NaN	NaN	NaN	NaN	NaN	NaN

```
Columns retained: ['surgery', 'age', 'hospital_number', 'rectal_temp', 'pulse', 'respiratory_rate', 'temp_of_extremities', 'peripheral_pulse', 'mucous_membrane',
    'capillary_refill_time', 'pain', 'peristalsis', 'abdominal_distention', 'nasogastric_tube', 'nasogastric_reflux', 'rectal_exam_feces', 'abdomen',
    'packed_cell_volume', 'total_protein', 'abdomo_appearance', 'abdomo_protein', 'outcome', 'surgical_lesion', 'lesion_1', 'lesion_2', 'lesion_3', 'cp_data']
Columns removed: ['nasogastric_reflux_ph']
```

Q6-Q10 Imputation and duplicates

Python code

```
num_cols = data.select_dtypes(include="number").columns
cat_cols = data.select_dtypes(exclude="number").columns
data_mean_imp = data.copy()
data_mean_imp[num_cols] = data[num_cols].fillna(data[num_cols].mean())
data_med_imp = data.copy()
data_med_imp[num_cols] = data[num_cols].fillna(data[num_cols].median())
data_mode_imp = data.copy()
for col in cat_cols:
    data_mode_imp[col] = data[col].fillna(first_mode(data[col]))
show("Mean imputation", data_mean_imp.head(6))
show("Median imputation", data_med_imp.head(6))
show("Mode imputation", data_mode_imp.head(6))
show("Duplicate rows", data.loc[data.duplicated()])
data_dedup = data.drop_duplicates()
print("Rows after removing duplicates:", len(data_dedup))
```

Output

```
Mean imputation
```

Table columns 1 of 3

	surgery	age	hospital_number	rectal_temp	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time	pain
0	no	adult	530101	38.500000	66.000000	28.000000	cool	reduced	NaN	more_3_sec	extreme_pain
1	yes	adult	534817	39.200000	88.000000	20.000000	NaN	NaN	pale_cyanotic	less_3_sec	mild_pain

	surgery	age	hospital_number	rectal_temp	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time	pain
2	no	adult	530334	38.300000	40.000000	24.000000	normal	normal	pale_pink	less_3_sec	mild_pain
3	yes	young	5290409	39.100000	164.000000	84.000000	cold	normal	dark_cyanotic	more_3_sec	depressed
4	no	adult	530255	37.300000	104.000000	35.000000	NaN	NaN	dark_cyanotic	more_3_sec	NaN
5	no	adult	528355	38.149412	75.544554	31.954023	warm	normal	pale_pink	less_3_sec	depressed

Table columns 2 of 3

	peristalsis	abdominal_distention	nasogastric_tube	nasogastric_reflux	nasogastric_reflux_ph	rectal_exam_feces	abdomen	packed_cell_volume	total_protein
0	absent	severe	NaN	NaN	4.87619	decreased	distend_large	45.000000	8.400000
1	absent	slight	NaN	NaN	4.87619	absent	other	50.000000	85.000000
2	hypomotile	none	NaN	NaN	4.87619	normal	normal	33.000000	6.700000
3	absent	severe	none	less_1_liter	5.00000	decreased	NaN	48.000000	7.200000
4	NaN	NaN	NaN	NaN	4.87619	NaN	NaN	74.000000	7.400000
5	hypomotile	slight	slight	none	4.87619	decreased	firm	47.336634	21.077778

Table columns 3 of 3

	abdomo_appearance	abdomo_protein	outcome	surgical_lesion	lesion_1	lesion_2	lesion_3	cp_data
0	NaN	3.184211	died	no	11300.0	0.0	0.0	no
1	cloudy	2.000000	euthanized	no	2208.0	0.0	0.0	no
2	NaN	3.184211	lived	no	0.0	0.0	0.0	yes
3	serosanguious	5.300000	died	yes	2208.0	0.0	0.0	yes
4	NaN	3.184211	died	no	4300.0	0.0	0.0	no
5	NaN	3.184211	lived	no	0.0	0.0	0.0	no

```
Median imputation
```

Table columns 1 of 3

	surgery	age	hospital_number	rectal_temp	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time	pain
0	no	adult	530101	38.5	66.0	28.0	cool	reduced	NaN	more_3_sec	extreme_pain

	surgery	age	hospital_number	rectal_temp	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time	pain
1	yes	adult	534817	39.2	88.0	20.0	NaN	NaN	pale_cyanotic	less_3_sec	mild_pain
2	no	adult	530334	38.3	40.0	24.0	normal	normal	pale_pink	less_3_sec	mild_pain
3	yes	young	5290409	39.1	164.0	84.0	cold	normal	dark_cyanotic	more_3_sec	depressed
4	no	adult	530255	37.3	104.0	35.0	NaN	NaN	dark_cyanotic	more_3_sec	NaN
5	no	adult	528355	38.1	64.0	28.0	warm	normal	pale_pink	less_3_sec	depressed

Table columns 2 of 3

	peristalsis	abdominal_distention	nasogastric_tube	nasogastric_reflux	nasogastric_reflux_ph	rectal_exam_feces	abdomen	packed_cell_volume	total_protein
0	absent	severe	NaN	NaN	5.5	decreased	distend_large	45.0	8.4
1	absent	slight	NaN	NaN	5.5	absent	other	50.0	85.0
2	hypomotile	none	NaN	NaN	5.5	normal	normal	33.0	6.7
3	absent	severe	none	less_1_liter	5.0	decreased	NaN	48.0	7.2
4	NaN	NaN	NaN	NaN	5.5	NaN	NaN	74.0	7.4
5	hypomotile	slight	slight	none	5.5	decreased	firm	46.0	7.5

Table columns 3 of 3

	abdomo_appearance	abdomo_protein	outcome	surgical_lesion	lesion_1	lesion_2	lesion_3	cp_data
0	NaN	2.45	died	no	11300.0	0.0	0.0	no
1	cloudy	2.00	euthanized	no	2208.0	0.0	0.0	no
2	NaN	2.45	lived	no	0.0	0.0	0.0	yes
3	serosanguious	5.30	died	yes	2208.0	0.0	0.0	yes
4	NaN	2.45	died	no	4300.0	0.0	0.0	no
5	NaN	2.45	lived	no	0.0	0.0	0.0	no

```
Mode imputation
```

Table columns 1 of 3

	surgery	age	hospital_number	rectal_temp	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time	pain
0	no	adult	530101	38.5	66.0	28.0	cool	reduced	normal_pink	more_3_sec	extreme_pain
1	yes	adult	534817	39.2	88.0	20.0	cool	reduced	pale_cyanotic	less_3_sec	mild_pain
2	no	adult	530334	38.3	40.0	24.0	normal	normal	pale_pink	less_3_sec	mild_pain
3	yes	young	5290409	39.1	164.0	84.0	cold	normal	dark_cyanotic	more_3_sec	depressed
4	no	adult	530255	37.3	104.0	35.0	cool	reduced	dark_cyanotic	more_3_sec	mild_pain
5	no	adult	528355	NaN	NaN	NaN	warm	normal	pale_pink	less_3_sec	depressed

Table columns 2 of 3

	peristalsis	abdominal_distention	nasogastric_tube	nasogastric_reflux	nasogastric_reflux_ph	rectal_exam_feces	abdomen	packed_cell_volume	total_protein
0	absent	severe	slight	none	NaN	decreased	distend_large	45.0	8.4
1	absent	slight	slight	none	NaN	absent	other	50.0	85.0
2	hypomotile	none	slight	none	NaN	normal	normal	33.0	6.7
3	absent	severe	none	less_1_liter	5.0	decreased	distend_large	48.0	7.2
4	hypomotile	slight	slight	none	NaN	absent	distend_large	74.0	7.4
5	hypomotile	slight	slight	none	NaN	decreased	firm	NaN	NaN

Table columns 3 of 3

	abdomo_appearance	abdomo_protein	outcome	surgical_lesion	lesion_1	lesion_2	lesion_3	cp_data
0	cloudy	NaN	died	no	11300.0	0.0	0.0	no
1	cloudy	2.0	euthanized	no	2208.0	0.0	0.0	no
2	cloudy	NaN	lived	no	0.0	0.0	0.0	yes
3	serosanguious	5.3	died	yes	2208.0	0.0	0.0	yes
4	cloudy	NaN	died	no	4300.0	0.0	0.0	no
5	cloudy	NaN	lived	no	0.0	0.0	0.0	no

```
Duplicate rows
```

Table columns 1 of 3

	surgery	age	hospital_number	rectal_temp	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time	pain	peristalsis

Table columns 2 of 3

	abdominal_distention	nasogastric_tube	nasogastric_reflux	nasogastric_reflux_ph	rectal_exam_feces	abdomen	packed_cell_volume	total_protein	abdomo_appearance

Table columns 3 of 3

	abdomo_protein	outcome	surgical_lesion	lesion_1	lesion_2	lesion_3	cp_data

```
Rows after removing duplicates: 110
```

Q11-Q13 Data types and text conversions

Python code

```
show("Current data types", data.dtypes)
numeric_text = [col for col in cat_cols if data[col].notna().any()
                and pd.to_numeric(data[col].dropna(), errors="coerce").notna().all()]
print("Text columns containing only numeric values:", numeric_text)
print("Numeric-coded fields needing semantic review:", [c for c in ["hospital_number", "lesion_1", "lesion_2", "lesion_3"] if c in data])
data_types_fixed = data.copy()
for col in numeric_text:
    data_types_fixed[col] = pd.to_numeric(data[col], errors="coerce")
data_factors = data_types_fixed.copy()
for col in data_factors.select_dtypes(exclude="number").columns:
    data_factors[col] = data_factors[col].astype("category")
show("Categorical conversion", data_factors.dtypes)
data_lower, data_upper = data.copy(), data.copy()
for col in cat_cols:
    data_lower[col] = data[col].str.lower()
    data_upper[col] = data[col].str.upper()
show("Lowercase", data_lower.head(6))
show("Uppercase", data_upper.head(6))
```

Output

```
Current data types

surgery                    str
age                        str
hospital_number          int64
rectal_temp            float64
pulse                  float64
respiratory_rate       float64
temp_of_extremities        str
peripheral_pulse           str
```

```
mucous_membrane              str
capillary_refill_time        str
pain                         str
peristalsis                  str
abdominal_distention         str
nasogastric_tube             str
nasogastric_reflux           str
nasogastric_reflux_ph    float64
rectal_exam_feces            str
abdomen                      str
packed_cell_volume       float64
total_protein            float64
abdomo_appearance            str
abdomo_protein           float64
outcome                      str
surgical_lesion              str
lesion_1                 float64
lesion_2                 float64
lesion_3                 float64
cp_data                      str
dtype: object

Text columns containing only numeric values: []
Numeric-coded fields needing semantic review: ['hospital_number', 'lesion_1', 'lesion_2', 'lesion_3']
Categorical conversion

surgery                  category
age                      category
hospital_number             int64
rectal_temp              float64
pulse                    float64
respiratory_rate         float64
temp_of_extremities      category
peripheral_pulse         category
mucous_membrane          category
capillary_refill_time    category
pain                     category
peristalsis              category
abdominal_distention     category
nasogastric_tube         category
nasogastric_reflux       category
nasogastric_reflux_ph    float64
rectal_exam_feces        category
abdomen                  category
packed_cell_volume       float64
total_protein            float64
abdomo_appearance        category
abdomo_protein           float64
outcome                  category
surgical_lesion          category
lesion_1                 float64
```

```
lesion_2                 float64
lesion_3                 float64
cp_data                 category
dtype: object
```

```
Lowercase
```

Table columns 1 of 3

	surgery	age	hospital_number	rectal_temp	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time	pain
0	no	adult	530101	38.5	66.0	28.0	cool	reduced	NaN	more_3_sec	extreme_pain
1	yes	adult	534817	39.2	88.0	20.0	NaN	NaN	pale_cyanotic	less_3_sec	mild_pain
2	no	adult	530334	38.3	40.0	24.0	normal	normal	pale_pink	less_3_sec	mild_pain
3	yes	young	5290409	39.1	164.0	84.0	cold	normal	dark_cyanotic	more_3_sec	depressed
4	no	adult	530255	37.3	104.0	35.0	NaN	NaN	dark_cyanotic	more_3_sec	NaN
5	no	adult	528355	NaN	NaN	NaN	warm	normal	pale_pink	less_3_sec	depressed

Table columns 2 of 3

	peristalsis	abdominal_distention	nasogastric_tube	nasogastric_reflux	nasogastric_reflux_ph	rectal_exam_feces	abdomen	packed_cell_volume	total_protein
0	absent	severe	NaN	NaN	NaN	decreased	distend_large	45.0	8.4
1	absent	slight	NaN	NaN	NaN	absent	other	50.0	85.0
2	hypomotile	none	NaN	NaN	NaN	normal	normal	33.0	6.7
3	absent	severe	none	less_1_liter	5.0	decreased	NaN	48.0	7.2
4	NaN	NaN	NaN	NaN	NaN	NaN	NaN	74.0	7.4
5	hypomotile	slight	slight	none	NaN	decreased	firm	NaN	NaN

Table columns 3 of 3

	abdomo_appearance	abdomo_protein	outcome	surgical_lesion	lesion_1	lesion_2	lesion_3	cp_data
0	NaN	NaN	died	no	11300.0	0.0	0.0	no
1	cloudy	2.0	euthanized	no	2208.0	0.0	0.0	no
2	NaN	NaN	lived	no	0.0	0.0	0.0	yes
3	serosanguious	5.3	died	yes	2208.0	0.0	0.0	yes

	abdomo_appearance	abdomo_protein	outcome	surgical_lesion	lesion_1	lesion_2	lesion_3	cp_data
4	NaN	NaN	died	no	4300.0	0.0	0.0	no
5	NaN	NaN	lived	no	0.0	0.0	0.0	no

`Uppercase`

Table columns 1 of 3

	surgery	age	hospital_number	rectal_temp	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time	pain
0	NO	ADULT	530101	38.5	66.0	28.0	COOL	REDUCED	NaN	MORE_3_SEC	EXTREME_PAIN
1	YES	ADULT	534817	39.2	88.0	20.0	NaN	NaN	PALE_CYANOTIC	LESS_3_SEC	MILD_PAIN
2	NO	ADULT	530334	38.3	40.0	24.0	NORMAL	NORMAL	PALE_PINK	LESS_3_SEC	MILD_PAIN
3	YES	YOUNG	5290409	39.1	164.0	84.0	COLD	NORMAL	DARK_CYANOTIC	MORE_3_SEC	DEPRESSED
4	NO	ADULT	530255	37.3	104.0	35.0	NaN	NaN	DARK_CYANOTIC	MORE_3_SEC	NaN
5	NO	ADULT	528355	NaN	NaN	NaN	WARM	NORMAL	PALE_PINK	LESS_3_SEC	DEPRESSED

Table columns 2 of 3

	peristalsis	abdominal_distention	nasogastric_tube	nasogastric_reflux	nasogastric_reflux_ph	rectal_exam_feces	abdomen	packed_cell_volume	total_protein
0	ABSENT	SEVERE	NaN	NaN	NaN	DECREASED	DISTEND_LARGE	45.0	8.4
1	ABSENT	SLIGHT	NaN	NaN	NaN	ABSENT	OTHER	50.0	85.0
2	HYPOMOTILE	NONE	NaN	NaN	NaN	NORMAL	NORMAL	33.0	6.7
3	ABSENT	SEVERE	NONE	LESS_1_LITER	5.0	DECREASED	NaN	48.0	7.2
4	NaN	NaN	NaN	NaN	NaN	NaN	NaN	74.0	7.4
5	HYPOMOTILE	SLIGHT	SLIGHT	NONE	NaN	DECREASED	FIRM	NaN	NaN

Table columns 3 of 3

	abdomo_appearance	abdomo_protein	outcome	surgical_lesion	lesion_1	lesion_2	lesion_3	cp_data
0	NaN	NaN	DIED	NO	11300.0	0.0	0.0	NO
1	CLOUDY	2.0	EUTHANIZED	NO	2208.0	0.0	0.0	NO
2	NaN	NaN	LIVED	NO	0.0	0.0	0.0	YES
3	SEROSANGUIOUS	5.3	DIED	YES	2208.0	0.0	0.0	YES

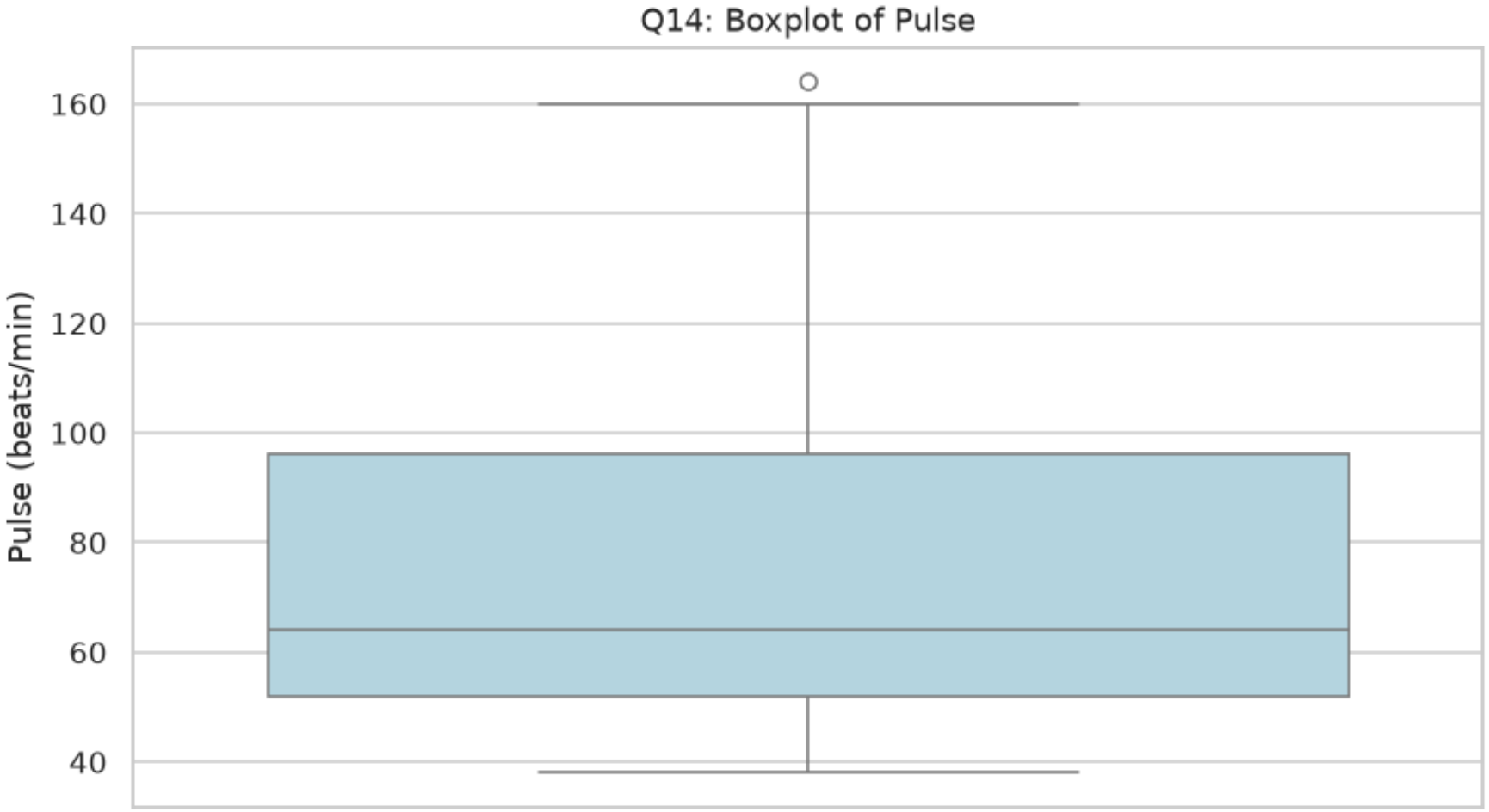
	abdomo_appearance	abdomo_protein	outcome	surgical_lesion	lesion_1	lesion_2	lesion_3	cp_data
4	NaN	NaN	DIED	NO	4300.0	0.0	0.0	NO
5	NaN	NaN	LIVED	NO	0.0	0.0	0.0	NO

Q14-Q16 Detect and remove extreme pulse outliers

Python code

```
sns.boxplot(y=data["pulse"], color="lightblue")
finish("Q14: Boxplot of Pulse", ylabel="Pulse (beats/min)")
q1, q3 = data["pulse"].quantile([.25, .75])
iqr_val = q3 - q1
outlier_mask = data["pulse"].lt(q1 - 1.5 * iqr_val) | data["pulse"].gt(q3 + 1.5 * iqr_val)
show("Q15: Pulse outliers using 1.5 × IQR", data.loc[outlier_mask, "pulse"])
keep = data["pulse"].isna() | data["pulse"].between(q1 - 3 * iqr_val, q3 + 3 * iqr_val)
data_no_outliers = data.loc[keep].copy()
print("Q16: Extreme outliers removed using 3 × IQR:", int((~keep).sum()))
show("Data without extreme pulse outliers", data_no_outliers.head(6))
```

Output




```
Q15: Pulse outliers using 1.5 × IQR

3    164.0
Name: pulse, dtype: float64

Q16: Extreme outliers removed using 3 × IQR: 0
Data without extreme pulse outliers
```

Table columns 1 of 3

	surgery	age	hospital_number	rectal_temp	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time	pain
0	no	adult	530101	38.5	66.0	28.0	cool	reduced	NaN	more_3_sec	extreme_pain
1	yes	adult	534817	39.2	88.0	20.0	NaN	NaN	pale_cyanotic	less_3_sec	mild_pain
2	no	adult	530334	38.3	40.0	24.0	normal	normal	pale_pink	less_3_sec	mild_pain
3	yes	young	5290409	39.1	164.0	84.0	cold	normal	dark_cyanotic	more_3_sec	depressed
4	no	adult	530255	37.3	104.0	35.0	NaN	NaN	dark_cyanotic	more_3_sec	NaN
5	no	adult	528355	NaN	NaN	NaN	warm	normal	pale_pink	less_3_sec	depressed

Table columns 2 of 3

	peristalsis	abdominal_distention	nasogastric_tube	nasogastric_reflux	nasogastric_reflux_ph	rectal_exam_feces	abdomen	packed_cell_volume	total_protein
0	absent	severe	NaN	NaN	NaN	decreased	distend_large	45.0	8.4
1	absent	slight	NaN	NaN	NaN	absent	other	50.0	85.0
2	hypomotile	none	NaN	NaN	NaN	normal	normal	33.0	6.7
3	absent	severe	none	less_1_liter	5.0	decreased	NaN	48.0	7.2
4	NaN	NaN	NaN	NaN	NaN	NaN	NaN	74.0	7.4
5	hypomotile	slight	slight	none	NaN	decreased	firm	NaN	NaN

Table columns 3 of 3

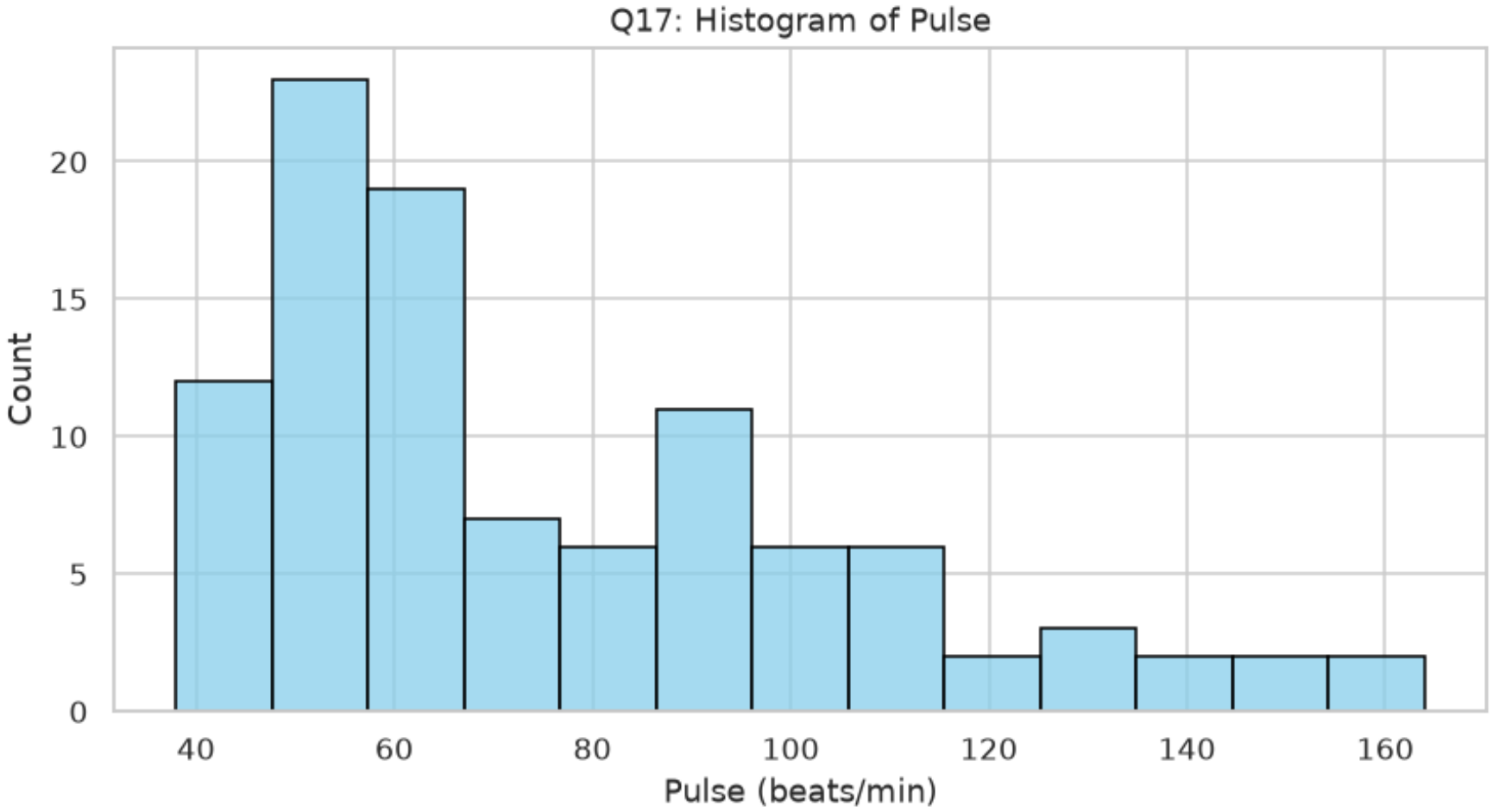
	abdomo_appearance	abdomo_protein	outcome	surgical_lesion	lesion_1	lesion_2	lesion_3	cp_data
0	NaN	NaN	died	no	11300.0	0.0	0.0	no
1	cloudy	2.0	euthanized	no	2208.0	0.0	0.0	no
2	NaN	NaN	lived	no	0.0	0.0	0.0	yes
3	serosanguious	5.3	died	yes	2208.0	0.0	0.0	yes
4	NaN	NaN	died	no	4300.0	0.0	0.0	no
5	NaN	NaN	lived	no	0.0	0.0	0.0	no

Q17-Q20 Histograms, density, violin, and dot plots

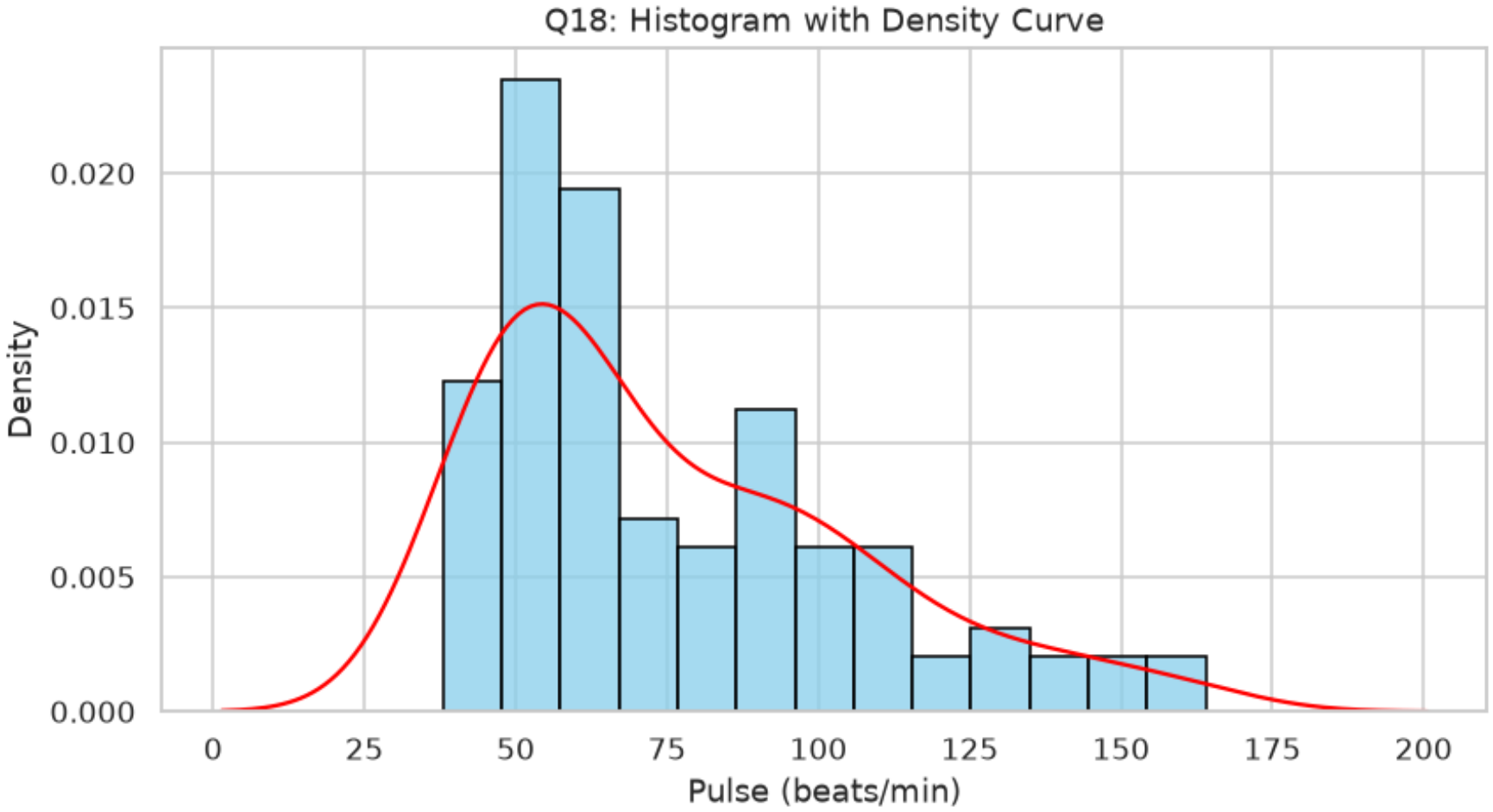
Python code

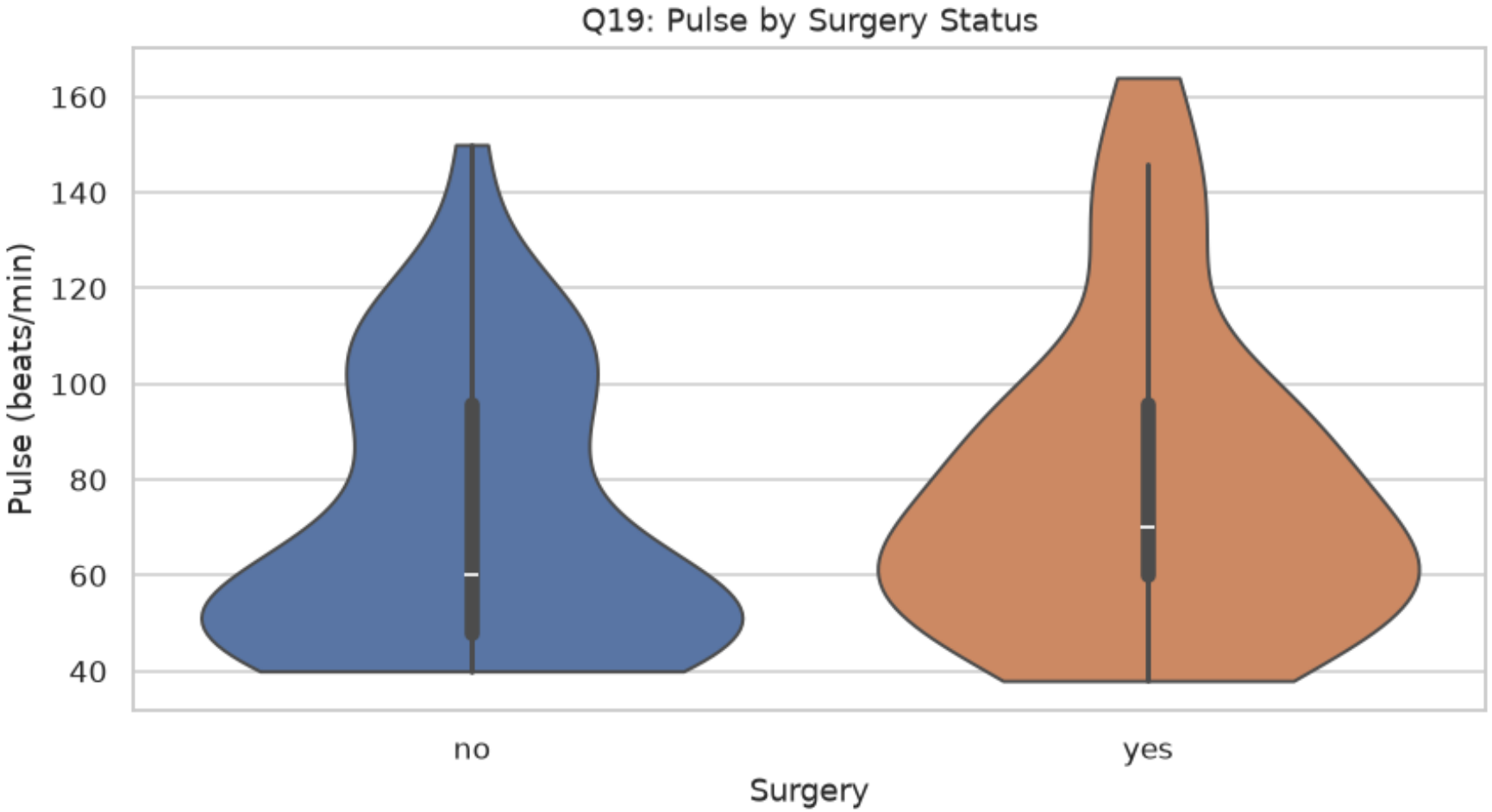
```
sns.histplot(data=data, x="pulse", binwidth=10, color="skyblue", edgecolor="black")
finish("Q17: Histogram of Pulse", "Pulse (beats/min)", "Count")
describe_distribution(data["pulse"], "Pulse distribution")
sns.histplot(data=data, x="pulse", binwidth=10, stat="density", color="skyblue", edgecolor="black")
sns.kdeplot(data=data, x="pulse", color="red")
finish("Q18: Histogram with Density Curve", "Pulse (beats/min)", "Density")
sns.violinplot(data=data.dropna(subset=["surgery", "pulse"]), x="surgery", y="pulse", hue="surgery", legend=False, cut=0)
finish("Q19: Pulse by Surgery Status", "Surgery", "Pulse (beats/min)")
small_data = data["pulse"].dropna().head(15)
small_bins = (small_data / 2).round() * 2
stack_positions = small_bins.groupby(small_bins).cumcount() + 1
plt.scatter(small_bins, stack_positions, color="darkblue", s=60)
plt.yticks(range(1, int(stack_positions.max()) + 1))
finish("Q20: Dot Plot of First 15 Observed Pulse Values", "Pulse (2 bpm bins)", "Stacked count")
```

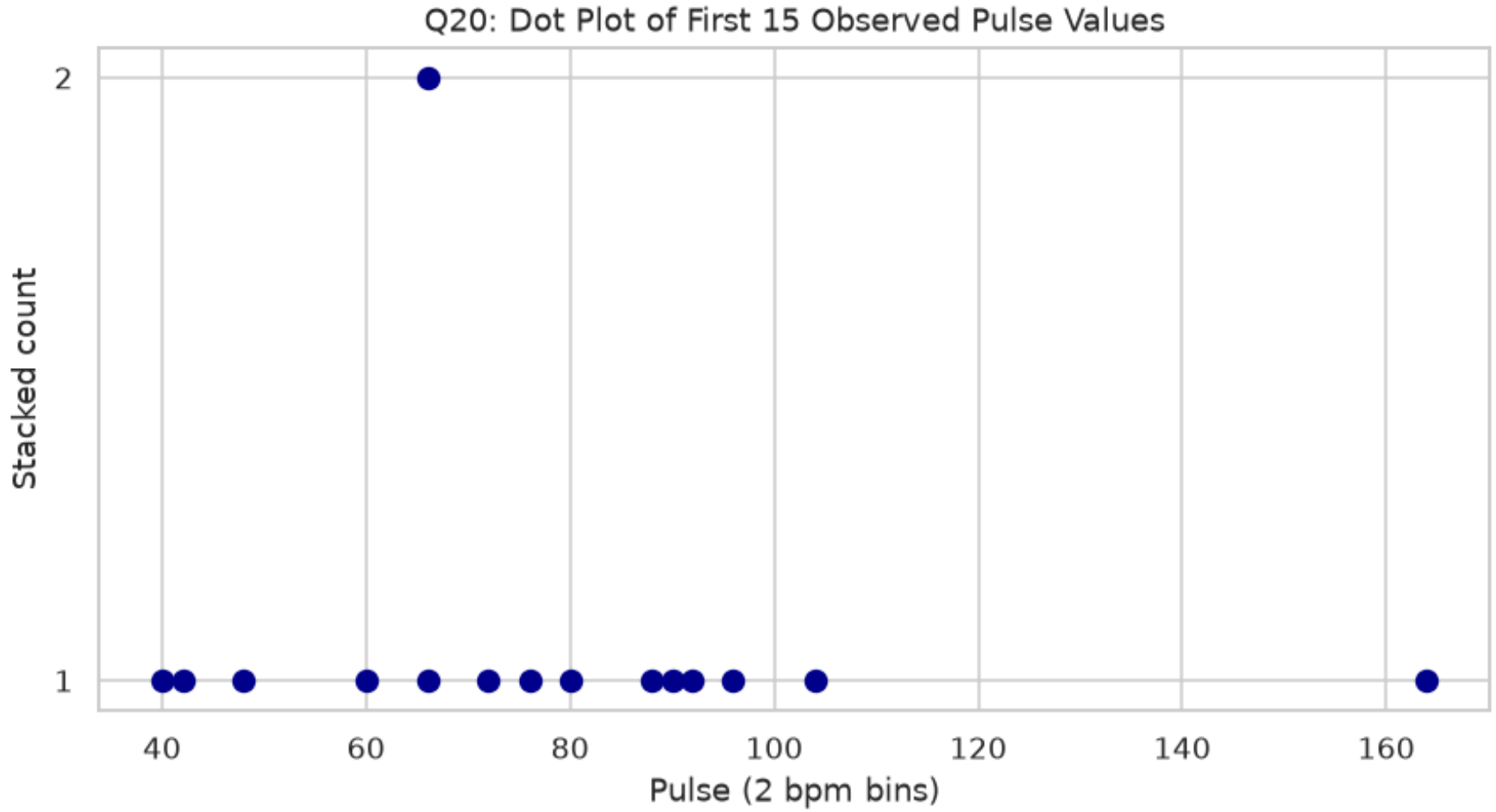
Output




```
Pulse distribution: n=101, median=64.0, middle 50%=52.0-96.0; right-skewed (skewness=0.97).
```


Q18: Histogram with Density Curve
Density
0.020
0.015
0.010
0.005
0.000
0
25
50
75
100
125
150
175
200
Pulse (beats/min)


Q19: Pulse by Surgery Status
Pulse (beats/min)
160
140
120
100
80
60
40
no
yes
Surgery



Q21-Q26 Categorical frequencies and charts

Python code

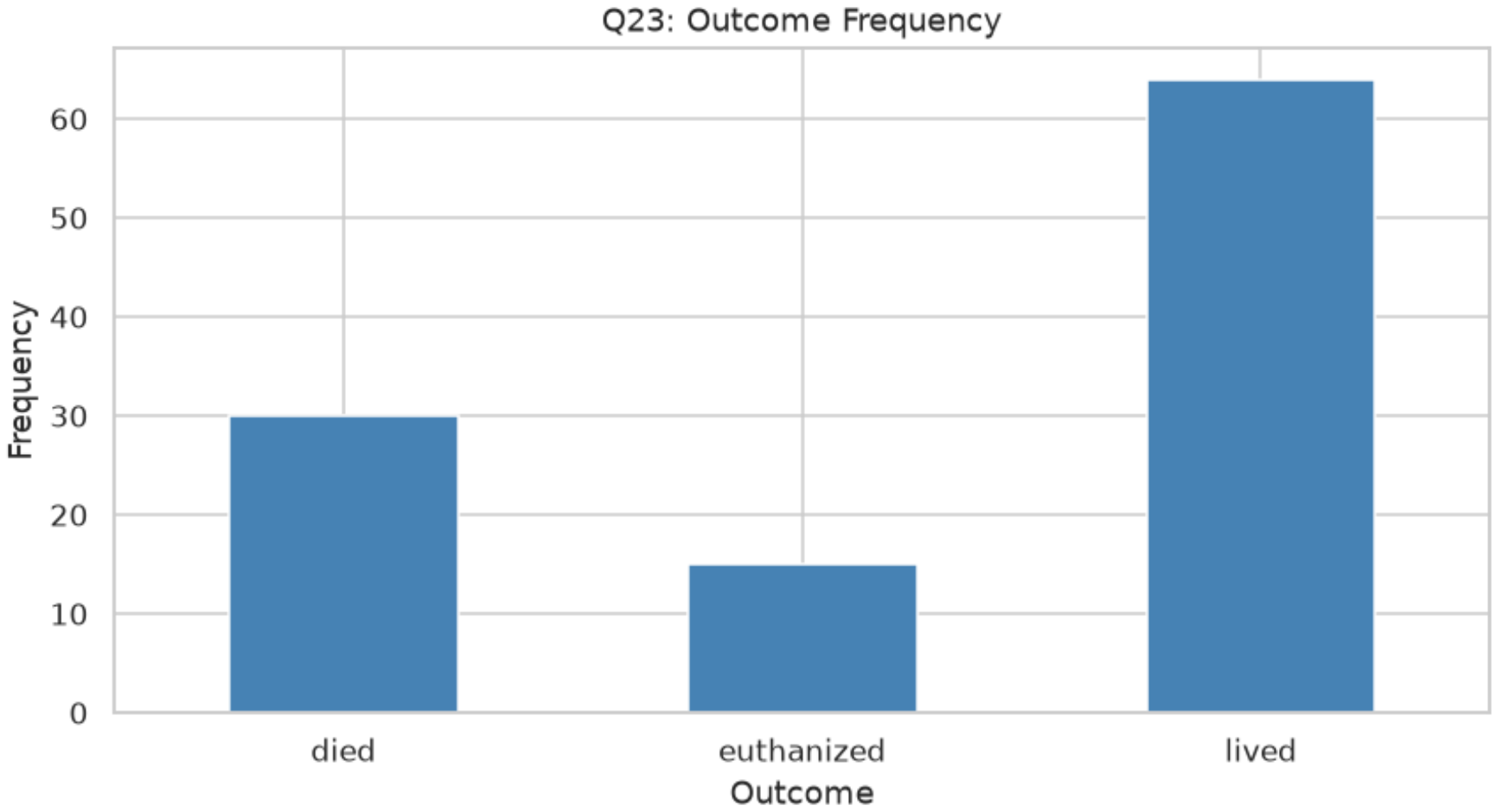
```
freq_table = data["outcome"].value_counts().sort_index()
prop_table = freq_table / freq_table.sum()
perc_table = prop_table * 100
show("Q21-Q22: Outcome distribution", pd.DataFrame({"Frequency": freq_table, "Proportion": prop_table, "Percentage": perc_table}))
freq_table.plot.bar(color="steelblue", rot=0)
finish("Q23: Outcome Frequency", "Outcome", "Frequency")
data["mucous_membrane"].value_counts().sort_values().plot.barh(color="teal")
```

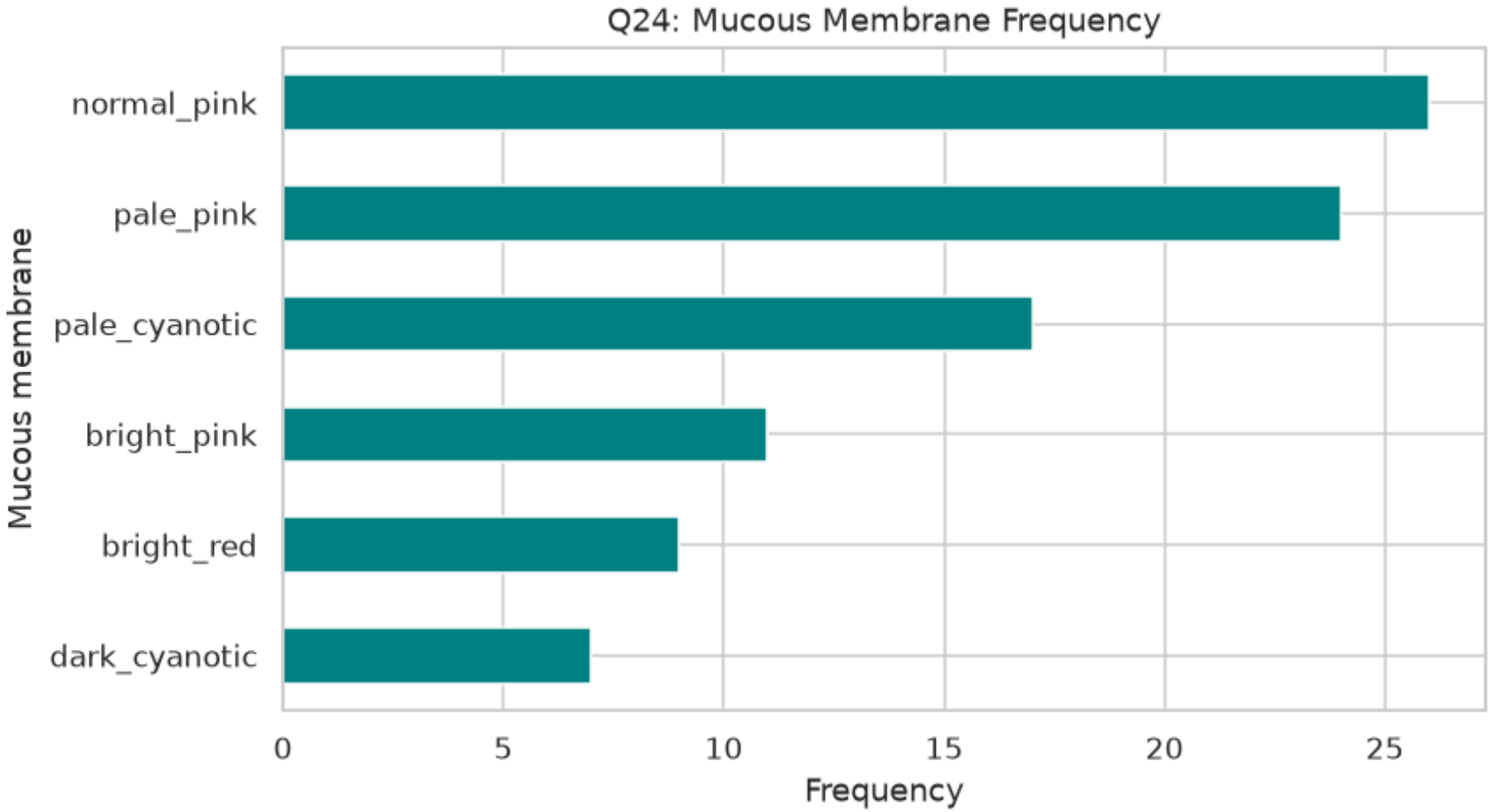
```
finish("Q24: Mucous Membrane Frequency", "Frequency", "Mucous membrane")
perc_table.plot.bar(color="mediumseagreen", rot=0)
finish("Q25: Percentage of Each Outcome", "Outcome", "Percentage (%)")
surgery_counts = data["surgery"].value_counts()
plt.pie(surgery_counts, labels=surgery_counts.index, autopct="%1.1f%%", startangle=90)
plt.axis("equal")
finish("Q26: Distribution of Surgery Status")
```

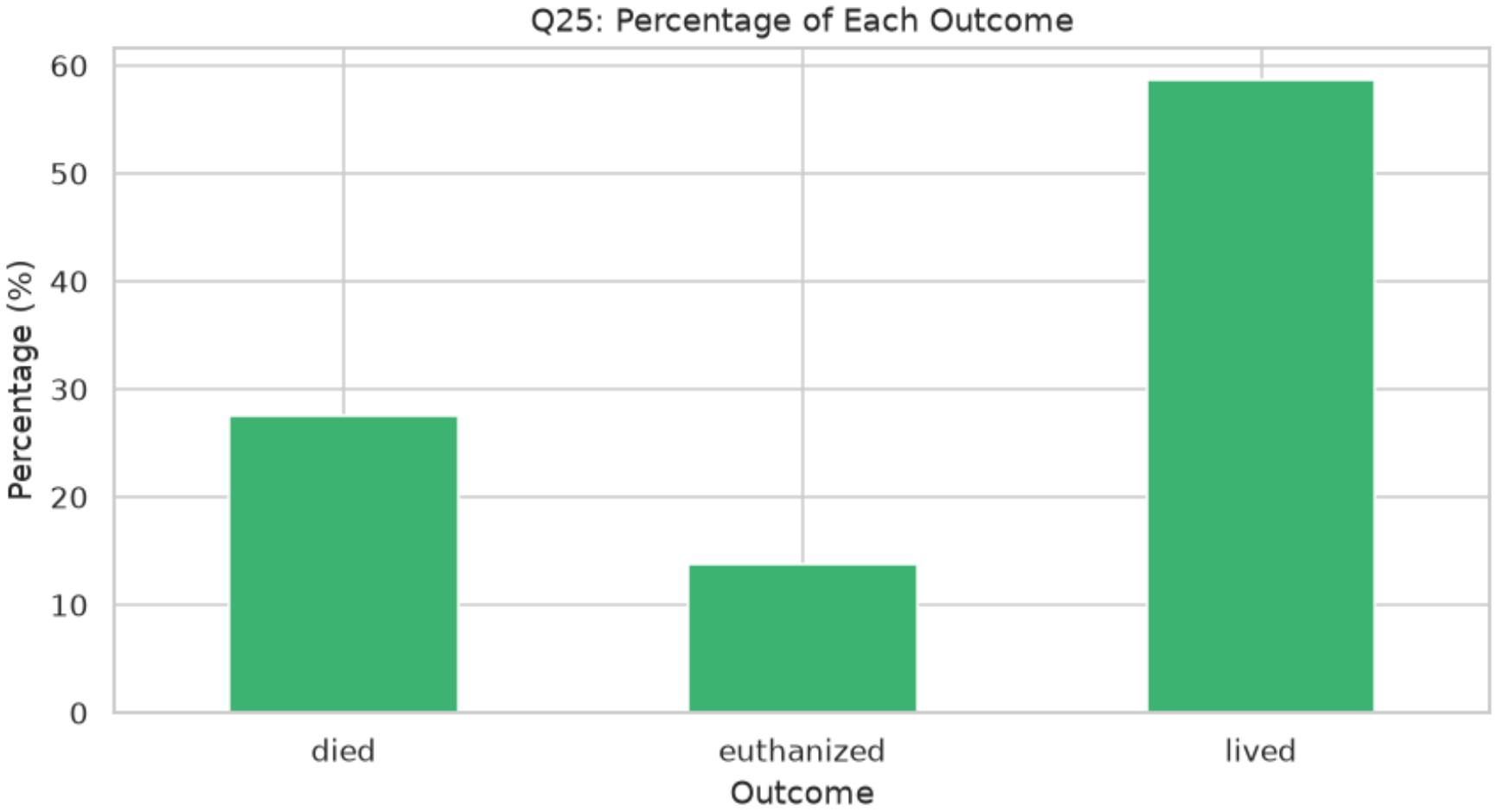
Output

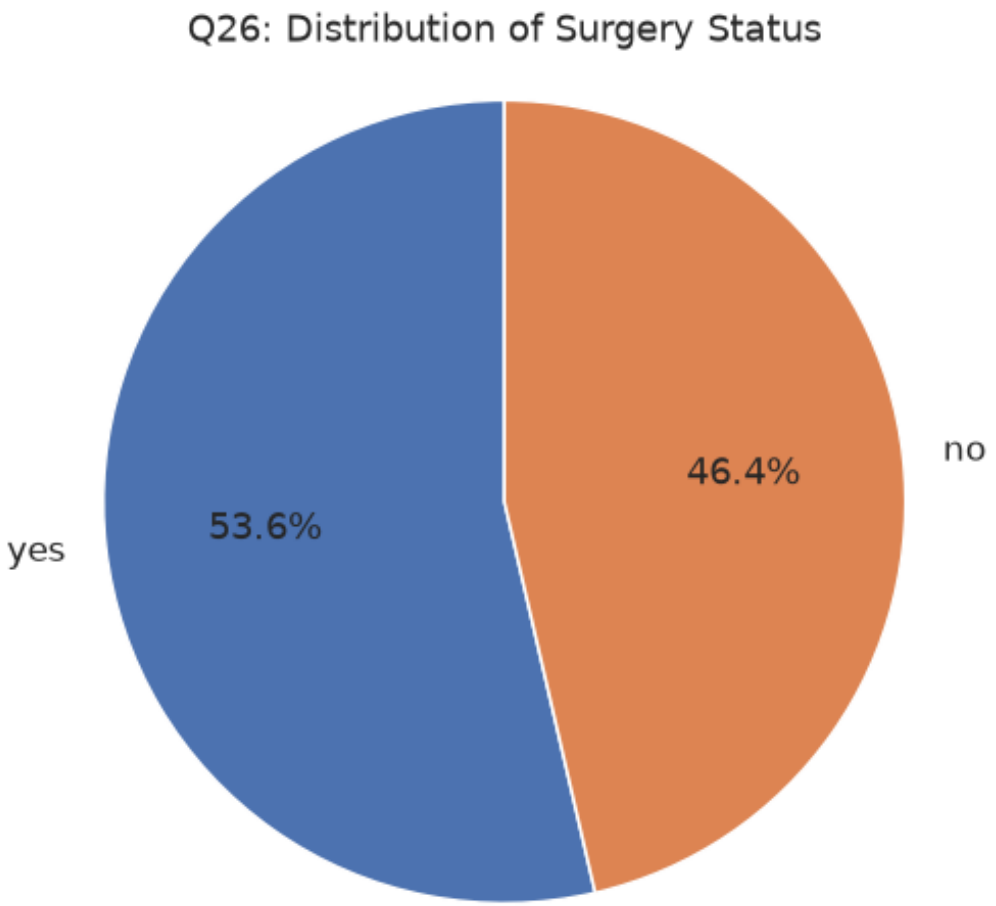
```
Q21-Q22: Outcome distribution
```

	Frequency	Proportion	Percentage
outcome			
died	30	0.275229	27.522936
euthanized	15	0.137615	13.761468
lived	64	0.587156	58.715596


Q23: Outcome Frequency
Frequency
60
50
40
30
20
10
0
died
euthanized
lived
Outcome


Q24: Mucous Membrane Frequency
Mucous membrane
normal_pink
pale_pink
pale_cyanotic
bright_pink
bright_red
dark_cyanotic
0
5
10
15
20
25
Frequency


Q25: Percentage of Each Outcome
Percentage (%)
60
50
40
30
20
10
0
died
euthanized
lived
Outcome



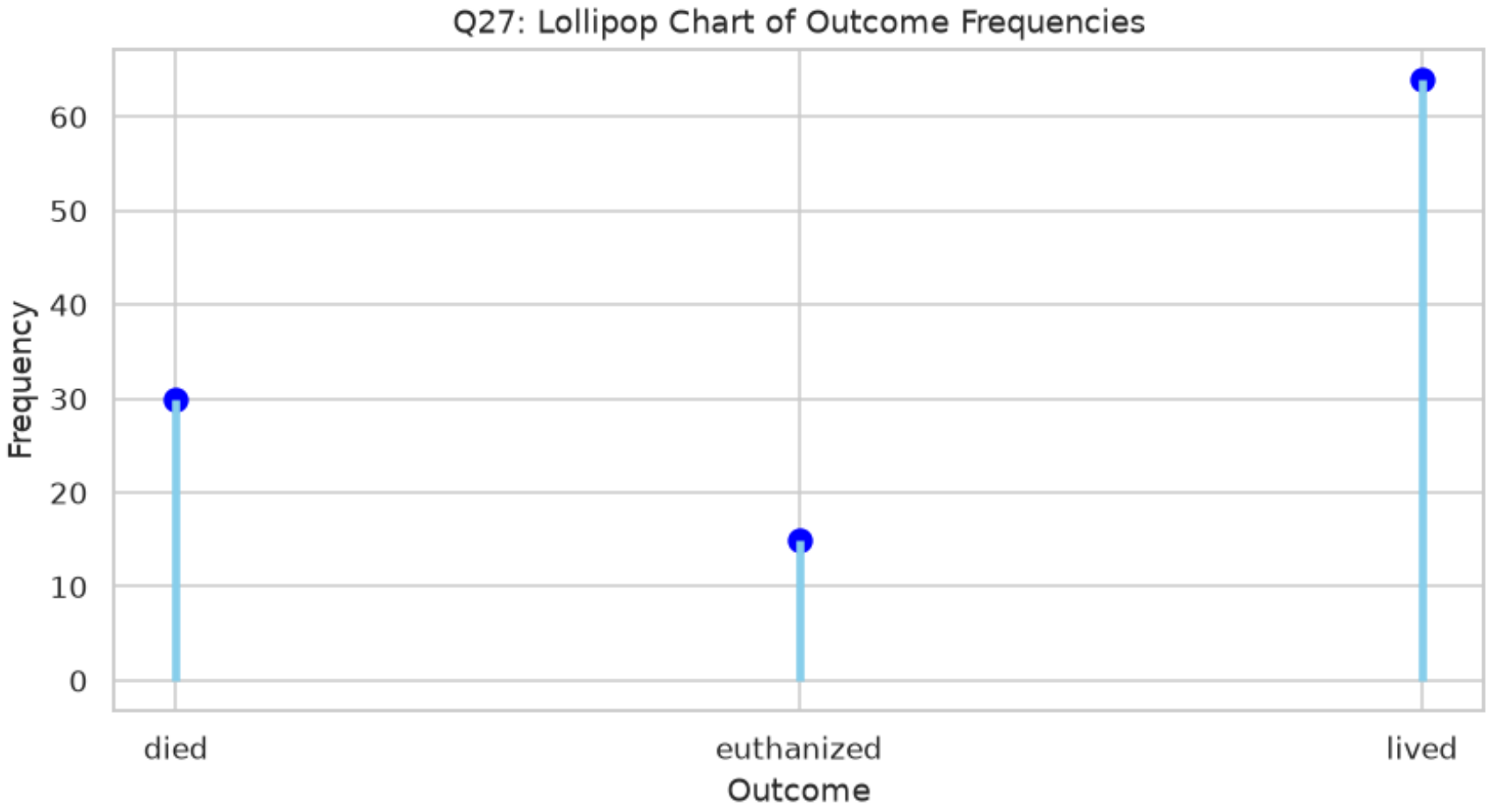
Q27-Q30 Lollipop, dot, density, and faceted charts

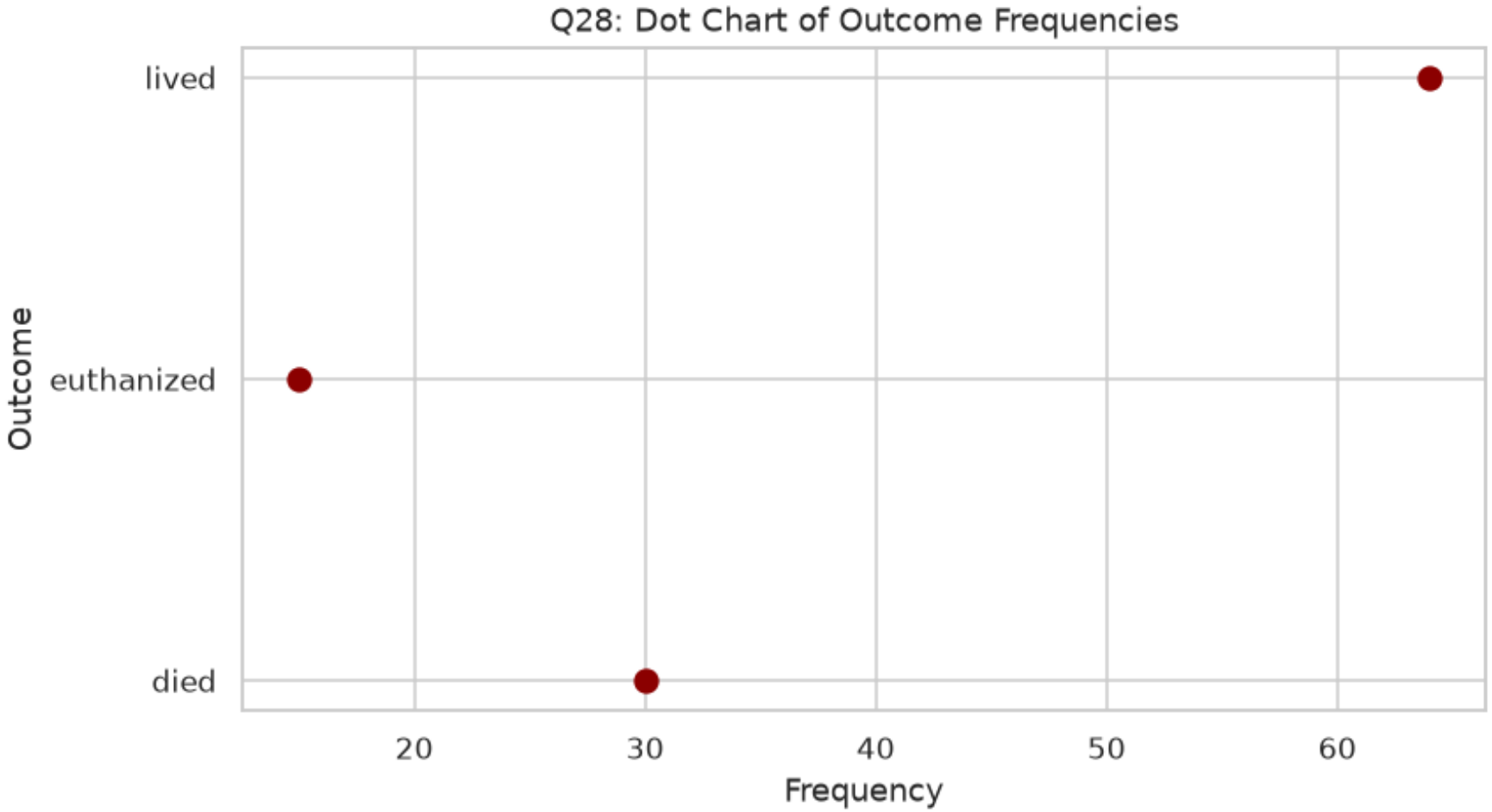
Python code

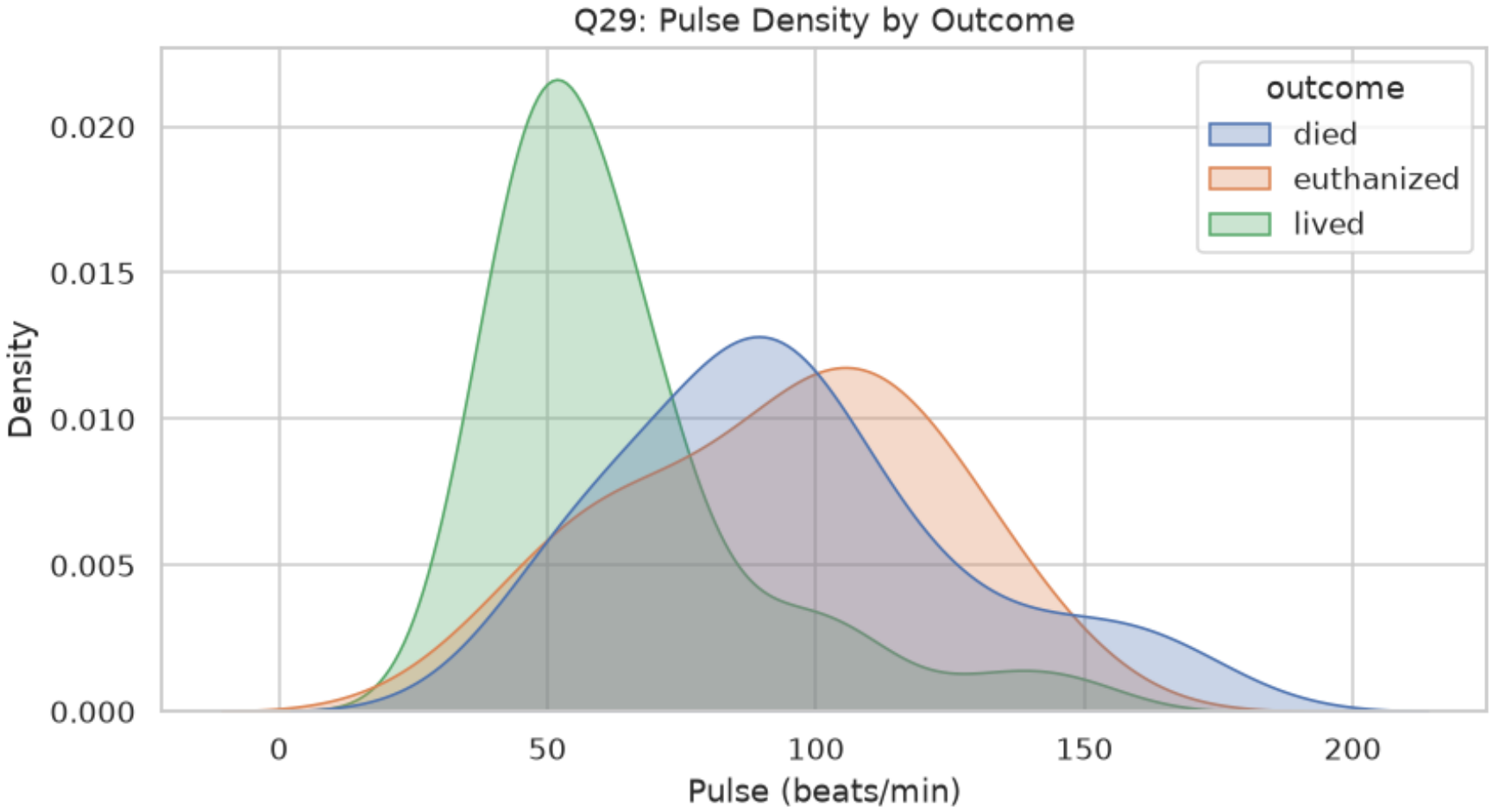
```
plt.vlines(freq_table.index, 0, freq_table.values, color="skyblue", linewidth=3)
plt.scatter(freq_table.index, freq_table.values, color="blue", s=70)
finish("Q27: Lollipop Chart of Outcome Frequencies", "Outcome", "Frequency")
plt.scatter(freq_table.values, freq_table.index, color="darkred", s=70)
finish("Q28: Dot Chart of Outcome Frequencies", "Frequency", "Outcome")
sns.kdeplot(data=data.dropna(subset=["pulse", "outcome"]), x="pulse", hue="outcome", fill=True, common_norm=False, alpha=.3)
finish("Q29: Pulse Density by Outcome", "Pulse (beats/min)", "Density")
```

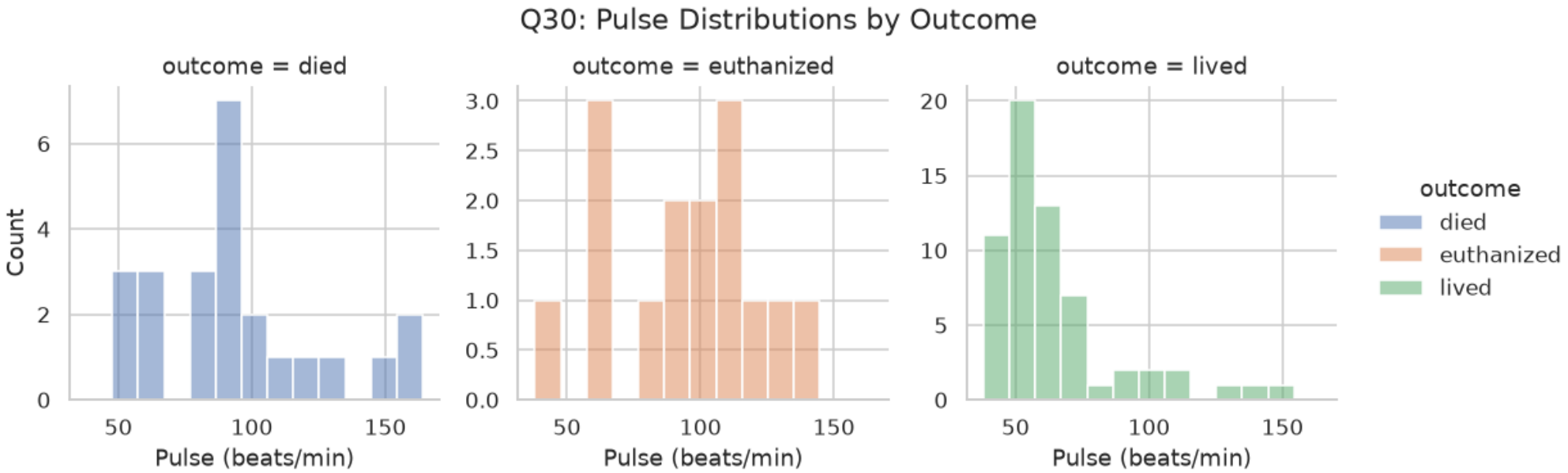
```
g = sns.displot(data=data.dropna(subset=["pulse", "outcome"]), x="pulse", col="outcome", hue="outcome", binwidth=10, height=3.3, facet_kws={"sharey":
    False})
g.set_axis_labels("Pulse (beats/min)", "Count")
g.figure.suptitle("Q30: Pulse Distributions by Outcome", y=1.05)
plt.show()
for outcome, group in data.dropna(subset=["outcome"]).groupby("outcome"):
    describe_distribution(group["pulse"], str(outcome))
```

Output




Q28: Dot Chart of Outcome Frequencies
lived
euthanized
died
Outcome
20
30
40
50
60
Frequency


Q29: Pulse Density by Outcome
outcome
died
euthanized
lived
Density
0.020
0.015
0.010
0.005
0.000
0
50
100
150
200
Pulse (beats/min)




```
died: n=24, median=90.0, middle 50%=76.5-106.5; right-skewed (skewness=0.74).
euthanized: n=15, median=100.0, middle 50%=70.0-113.0; approximately symmetric (skewness=-0.29).
lived: n=61, median=54.0, middle 50%=48.0-72.0; right-skewed (skewness=1.79).
```

Assignment 4 - Bivariate Analysis

This assignment examines relationships between two variables using contingency tables, bar charts, scatter and line plots, density plots, and grouped box and violin plots. It summarizes relationships among five clinical measurements with a correlation heatmap and a correlogram.

Questions

1. Load the dataset shared in your mail into the R programming environment and perform the following bivariate analysis and visualization tasks.

2. Select two suitable categorical variables, create a contingency table, and display their relationship using: a simple stacked bar plot using Base R, and a stacked bar plot using ggplot2.

3. Select two suitable categorical variables and create a grouped/side- by-side bar plot to compare their frequency distributions.

4. Select two suitable numerical variables and generate a scatter plot to analyze the relationship between them.

5. Select two appropriate variables and construct a line plot to visualize their relationship or trend.

6. Select suitable numerical variables and generate a heatmap to represent the relationship among the variables.

7. Select suitable variables and construct a bar chart to compare the values or frequencies between two variables.

8. Select one numerical variable and one categorical variable and generate a density plot to compare the distribution of the numerical variable across different categories.

9. Select one categorical variable and one numerical variable and create a box plot to compare the distribution of the numerical variable across different groups.

10. Select one categorical variable and one numerical variable and construct a violin plot to visualize the distribution and density of the numerical variable across categories.

11. Select suitable categorical and numerical variables and visualize their distribution by combining a violin plot and box plot in a single graphical representation.

12. Select suitable numerical variables, compute their correlation matrix, and generate a correlogram to study the strength and direction of relationships among the variables.

13. Provide suitable titles, axis labels, legends, and graphical parameters for all the plots generated.

Q1-Q3 Load, cross-tabulate, and compare categories

Python code

```
data = load_horse()
show("Dataset preview", data.head(6))
cat_tab = pd.crosstab(data["outcome"], data["surgery"])
show("Outcome × Surgery contingency table", cat_tab)
```

```
fig, ax = plt.subplots()
bottom = np.zeros(len(cat_tab.columns))
for outcome, row in cat_tab.iterrows():
    ax.bar(cat_tab.columns, row.to_numpy(), bottom=bottom, label=outcome)
    bottom += row.to_numpy()
ax.legend(title="Outcome")
finish("Q2a: Stacked Bar Plot Using Matplotlib", "Surgery", "Count")
cat_tab.T.plot.bar(stacked=True, colormap="Set2", rot=0)
plt.legend(title="Outcome")
finish("Q2b: Stacked Bar Plot Using Pandas", "Surgery", "Count")
cat_tab.T.plot.bar(stacked=False, colormap="Set1", rot=0)
plt.legend(title="Outcome")
finish("Q3: Grouped Outcome Frequencies by Surgery", "Surgery", "Count")
```

Output

```
Dataset preview
```

Table columns 1 of 3

	surgery	age	hospital_number	rectal_temp	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time	pain
0	no	adult	530101	38.5	66.0	28.0	cool	reduced	NaN	more_3_sec	extreme_pain
1	yes	adult	534817	39.2	88.0	20.0	NaN	NaN	pale_cyanotic	less_3_sec	mild_pain
2	no	adult	530334	38.3	40.0	24.0	normal	normal	pale_pink	less_3_sec	mild_pain
3	yes	young	5290409	39.1	164.0	84.0	cold	normal	dark_cyanotic	more_3_sec	depressed
4	no	adult	530255	37.3	104.0	35.0	NaN	NaN	dark_cyanotic	more_3_sec	NaN
5	no	adult	528355	NaN	NaN	NaN	warm	normal	pale_pink	less_3_sec	depressed

Table columns 2 of 3

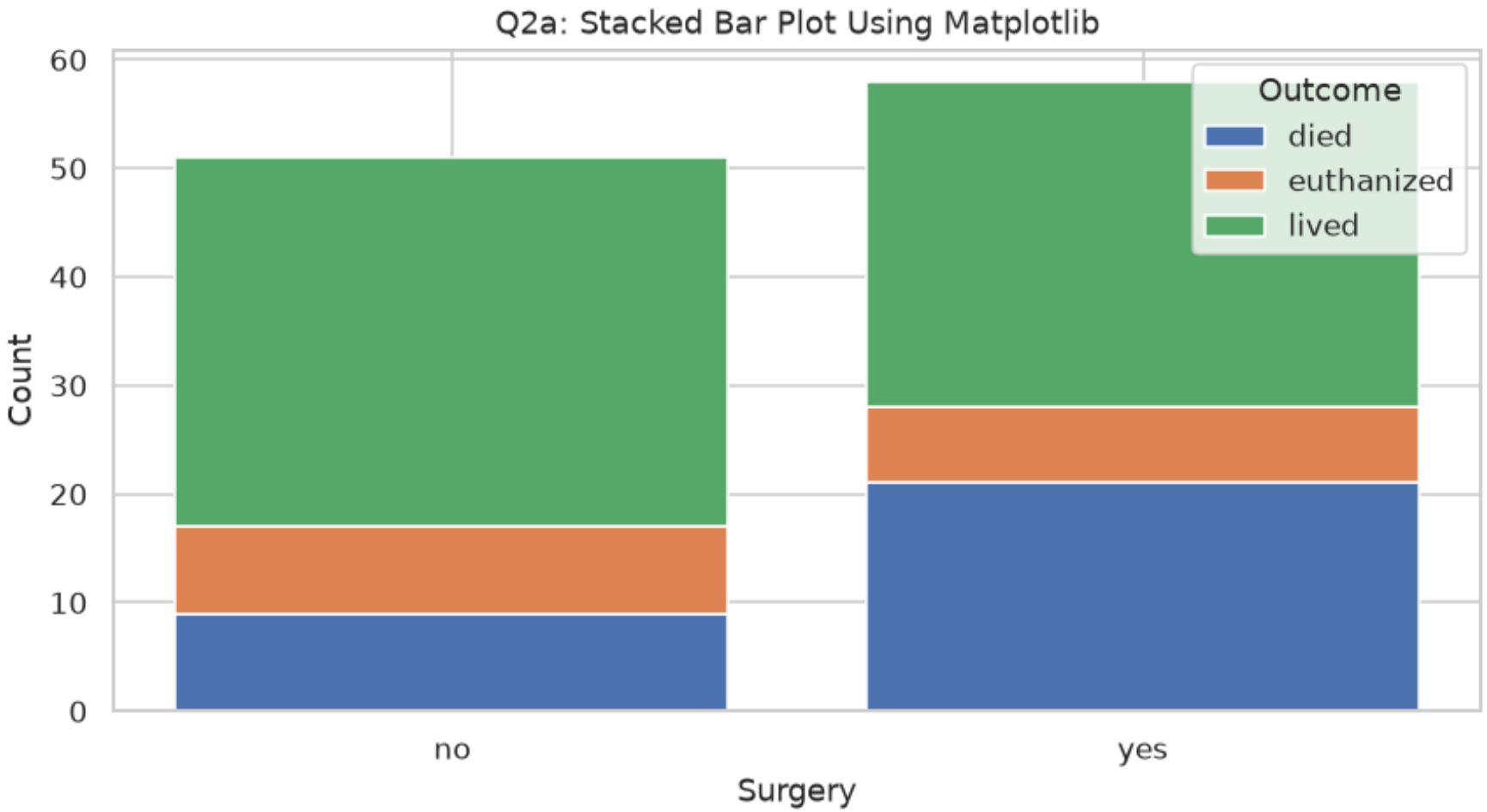
	peristalsis	abdominal_distention	nasogastric_tube	nasogastric_reflux	nasogastric_reflux_ph	rectal_exam_feces	abdomen	packed_cell_volume	total_protein
0	absent	severe	NaN	NaN	NaN	decreased	distend_large	45.0	8.4
1	absent	slight	NaN	NaN	NaN	absent	other	50.0	85.0
2	hypomotile	none	NaN	NaN	NaN	normal	normal	33.0	6.7
3	absent	severe	none	less_1_liter	5.0	decreased	NaN	48.0	7.2
4	NaN	NaN	NaN	NaN	NaN	NaN	NaN	74.0	7.4
5	hypomotile	slight	slight	none	NaN	decreased	firm	NaN	NaN

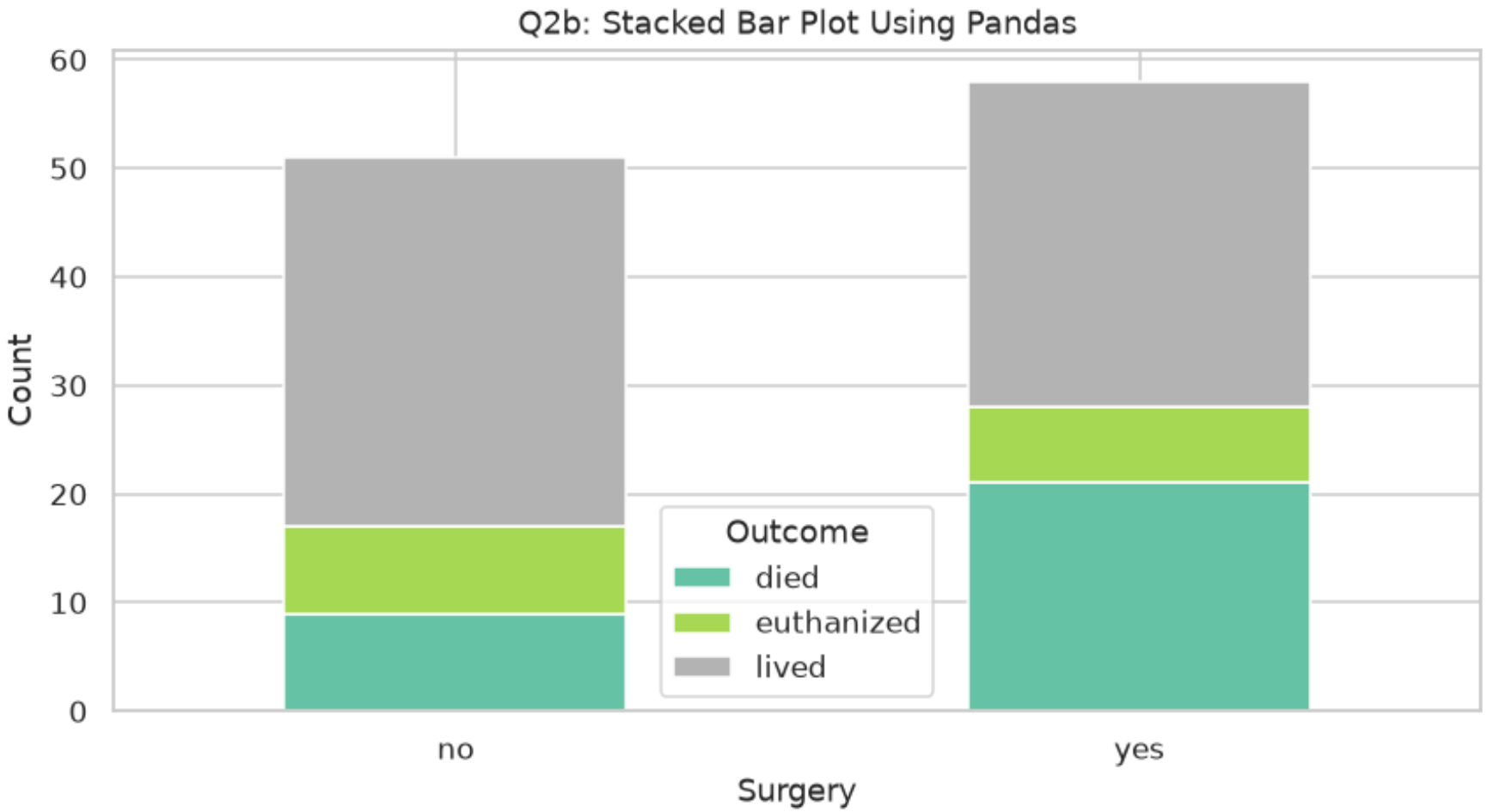
Table columns 3 of 3

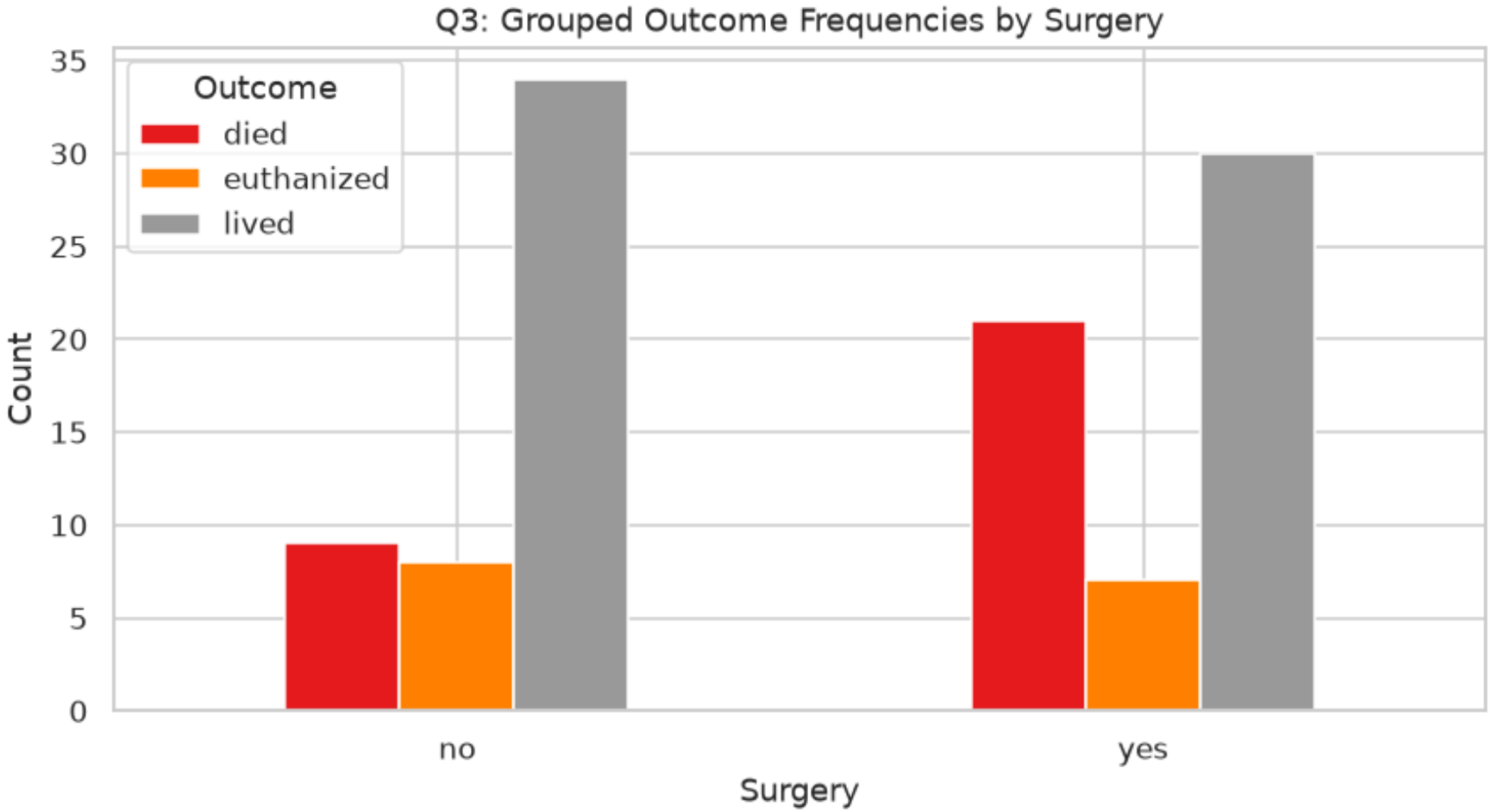
	abdomo_appearance	abdomo_protein	outcome	surgical_lesion	lesion_1	lesion_2	lesion_3	cp_data
0	NaN	NaN	died	no	11300.0	0.0	0.0	no
1	cloudy	2.0	euthanized	no	2208.0	0.0	0.0	no
2	NaN	NaN	lived	no	0.0	0.0	0.0	yes
3	serosanguious	5.3	died	yes	2208.0	0.0	0.0	yes
4	NaN	NaN	died	no	4300.0	0.0	0.0	no
5	NaN	NaN	lived	no	0.0	0.0	0.0	no

`Outcome × Surgery contingency table`

surgery	no	yes
outcome		
died	9	21
euthanized	8	7
lived	34	30


Q2a: Stacked Bar Plot Using Matplotlib
Outcome
died
euthanized
lived
Count
60
50
40
30
20
10
0
no
yes
Surgery


Q2b: Stacked Bar Plot Using Pandas
Count
60
50
40
30
20
10
0
Outcome
died
euthanized
lived
no
yes
Surgery



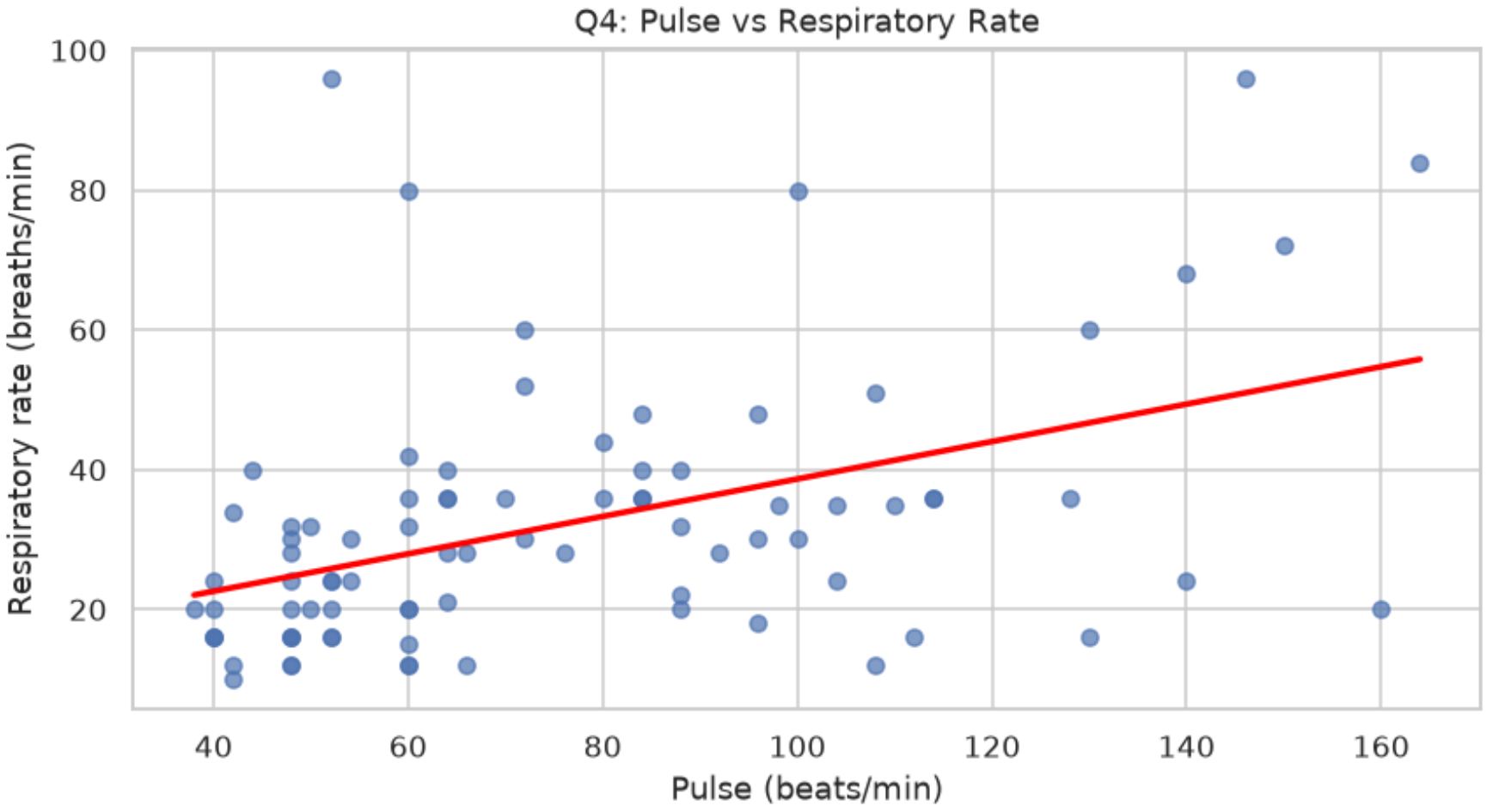
Q4-Q7 Numerical relationships and group means

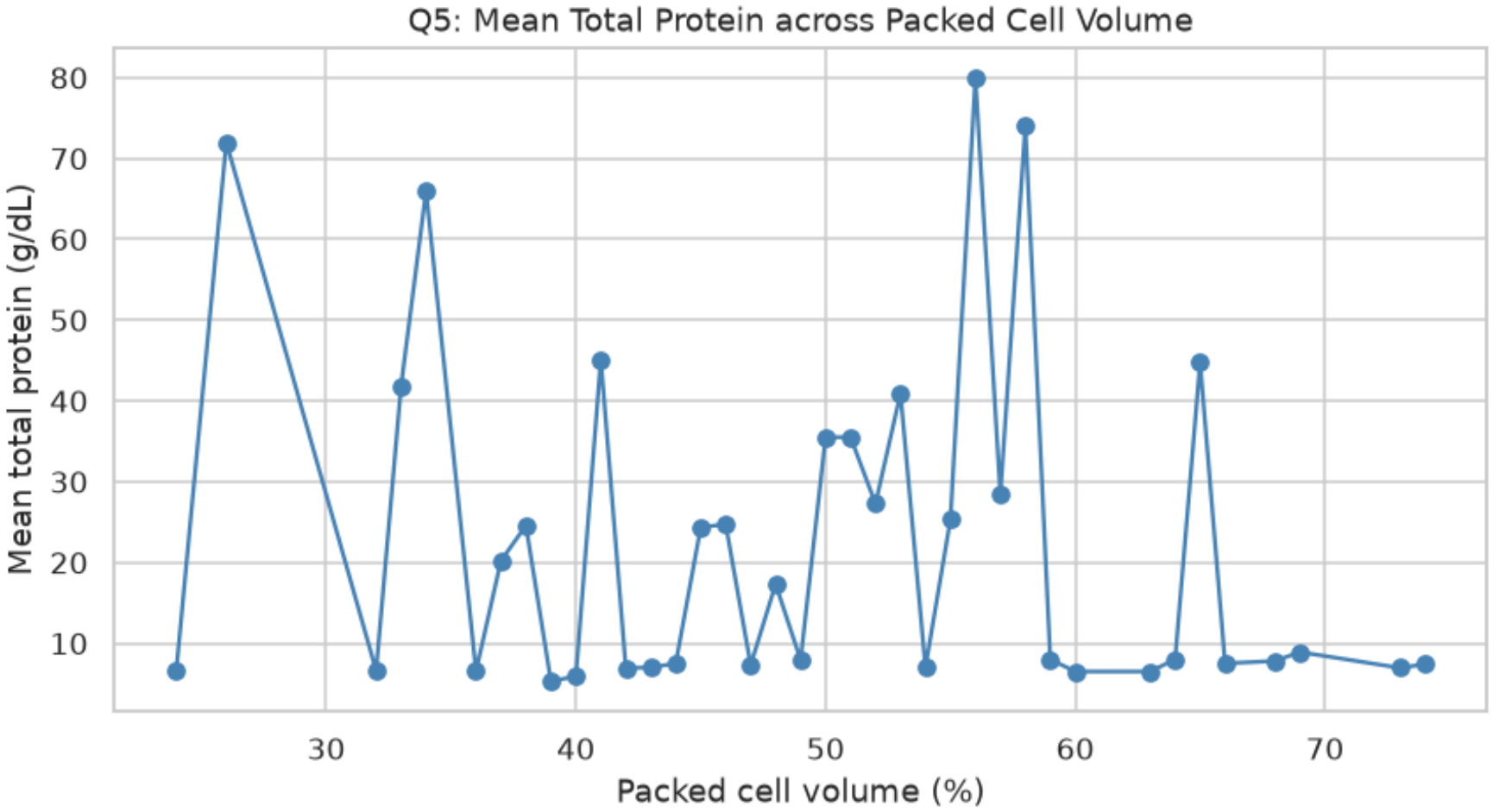
Python code

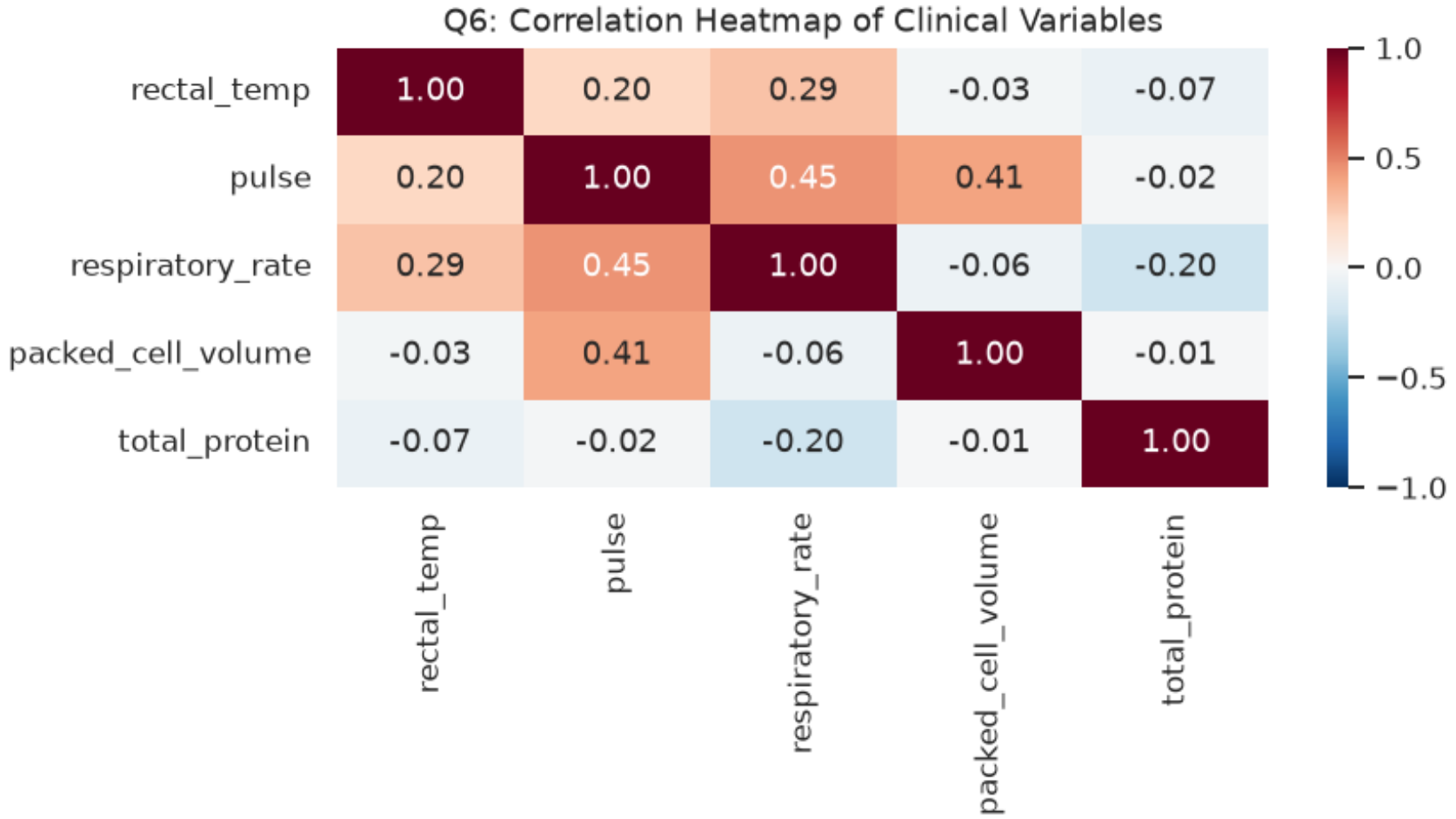
```
sns.regplot(data=data, x="pulse", y="respiratory_rate", ci=None, scatter_kws={"alpha": .7}, line_kws={"color": "red"})
finish("Q4: Pulse vs Respiratory Rate", "Pulse (beats/min)", "Respiratory rate (breaths/min)")
mean_trend = data.dropna(subset=["packed_cell_volume", "total_protein"]).groupby("packed_cell_volume")["total_protein"].mean().sort_index()
plt.plot(mean_trend.index, mean_trend, marker="o", color="steelblue")
finish("Q5: Mean Total Protein across Packed Cell Volume", "Packed cell volume (%)", "Mean total protein (g/dL)")
cor_matrix = data[clinical_columns].corr()
sns.heatmap(cor_matrix, annot=True, fmt=".2f", cmap="RdBu_r", vmin=-1, vmax=1, center=0)
```

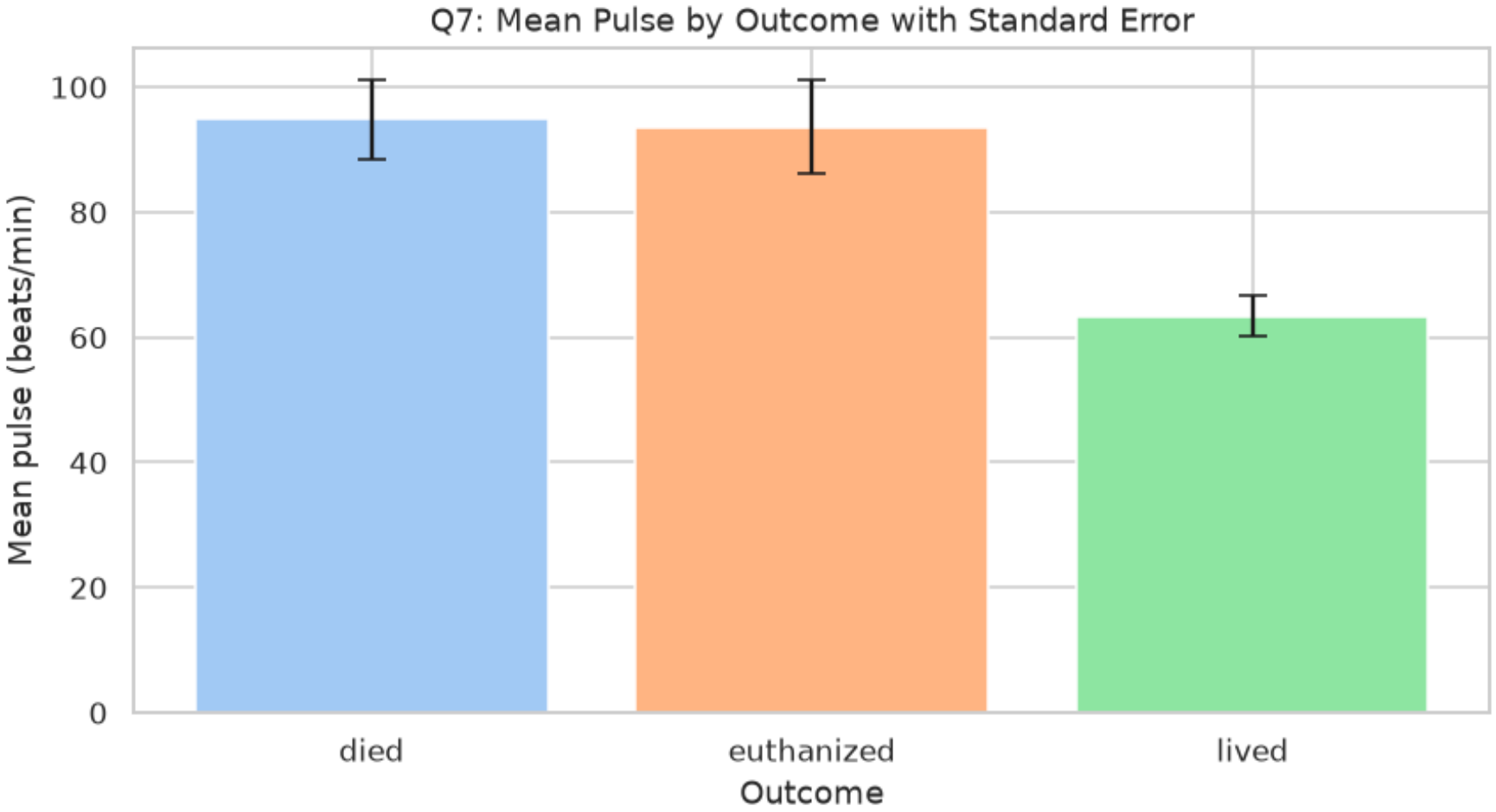
```
finish("Q6: Correlation Heatmap of Clinical Variables")
grouped = data.groupby("outcome")["pulse"].agg(["mean", "sem"])
plt.bar(grouped.index, grouped["mean"], yerr=grouped["sem"], capsize=5, color=sns.color_palette("pastel", len(grouped)))
finish("Q7: Mean Pulse by Outcome with Standard Error", "Outcome", "Mean pulse (beats/min)")
```

Output




Q5: Mean Total Protein across Packed Cell Volume
Mean total protein (g/dL)
80
70
60
50
40
30
20
10
30
40
50
60
70
Packed cell volume (%)


Q6: Correlation Heatmap of Clinical Variables
rectal_temp
pulse
respiratory_rate
packed_cell_volume
total_protein
1.00
0.20
0.29
-0.03
-0.07
0.20
1.00
0.45
0.41
-0.02
0.29
0.45
1.00
-0.06
-0.20
-0.03
0.41
-0.06
1.00
-0.01
-0.07
-0.02
-0.20
-0.01
1.00
1.0
0.5
0.0
−0.5
−1.0



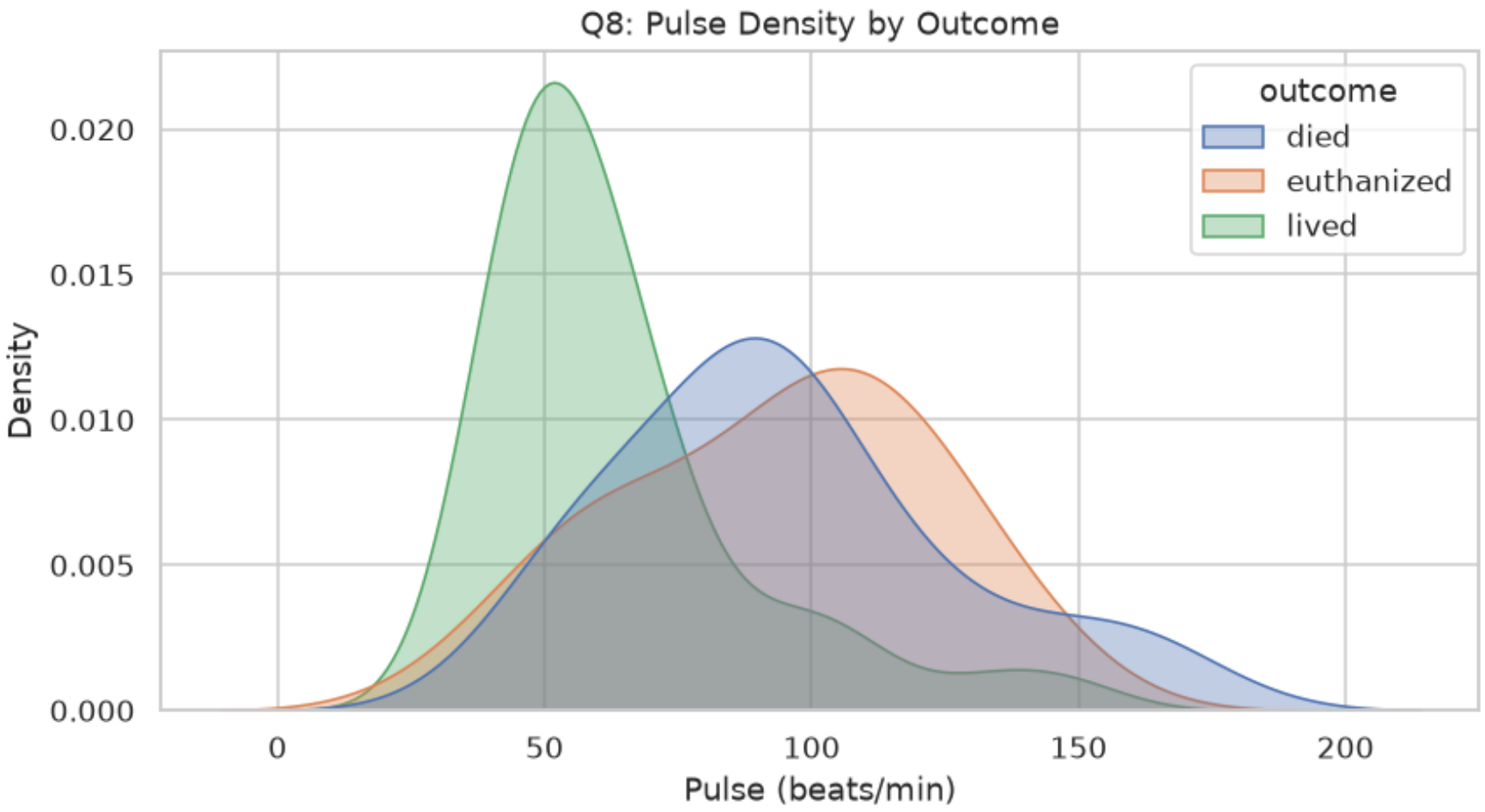
Q8-Q11 Grouped distributions

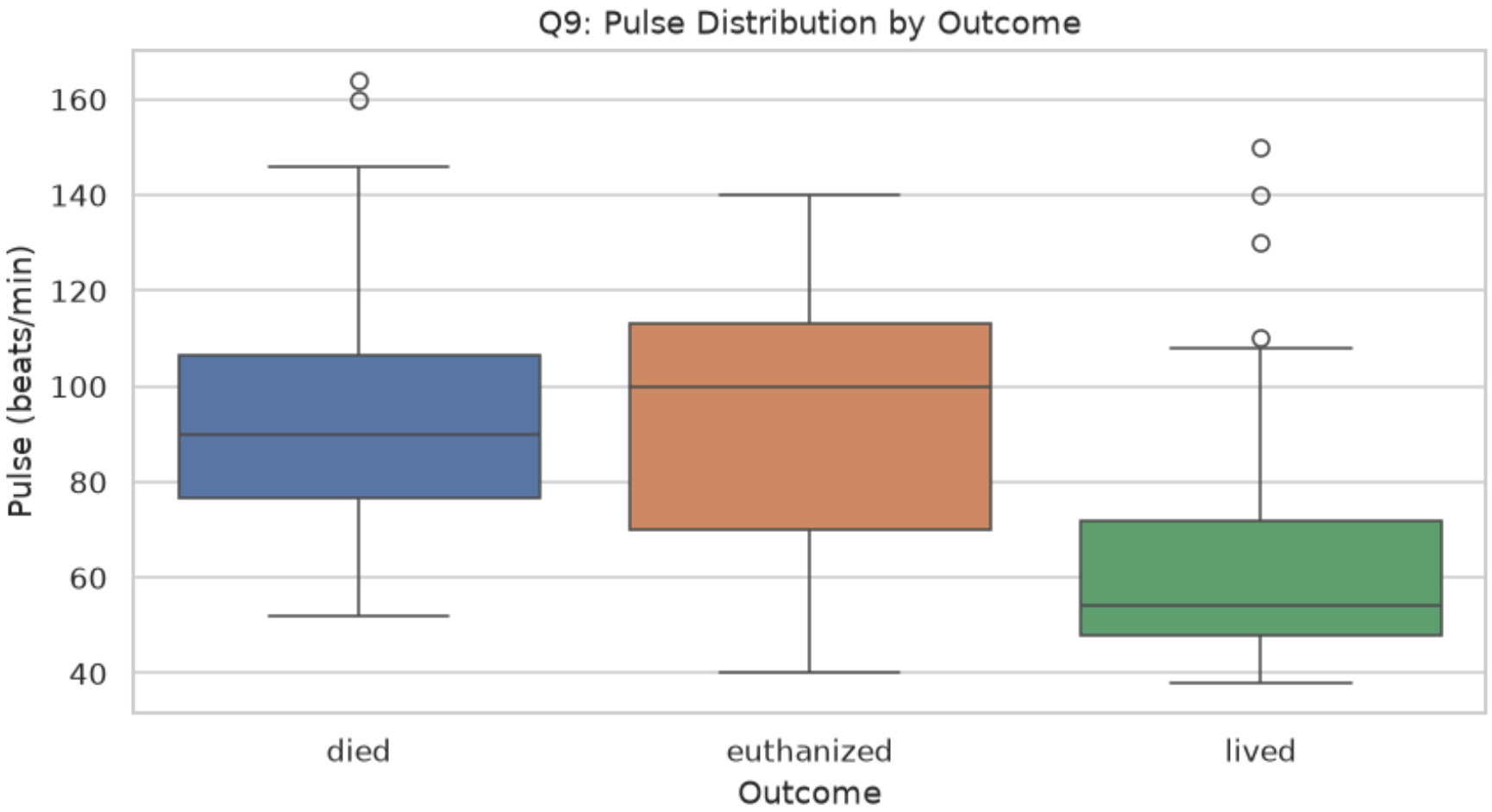
Python code

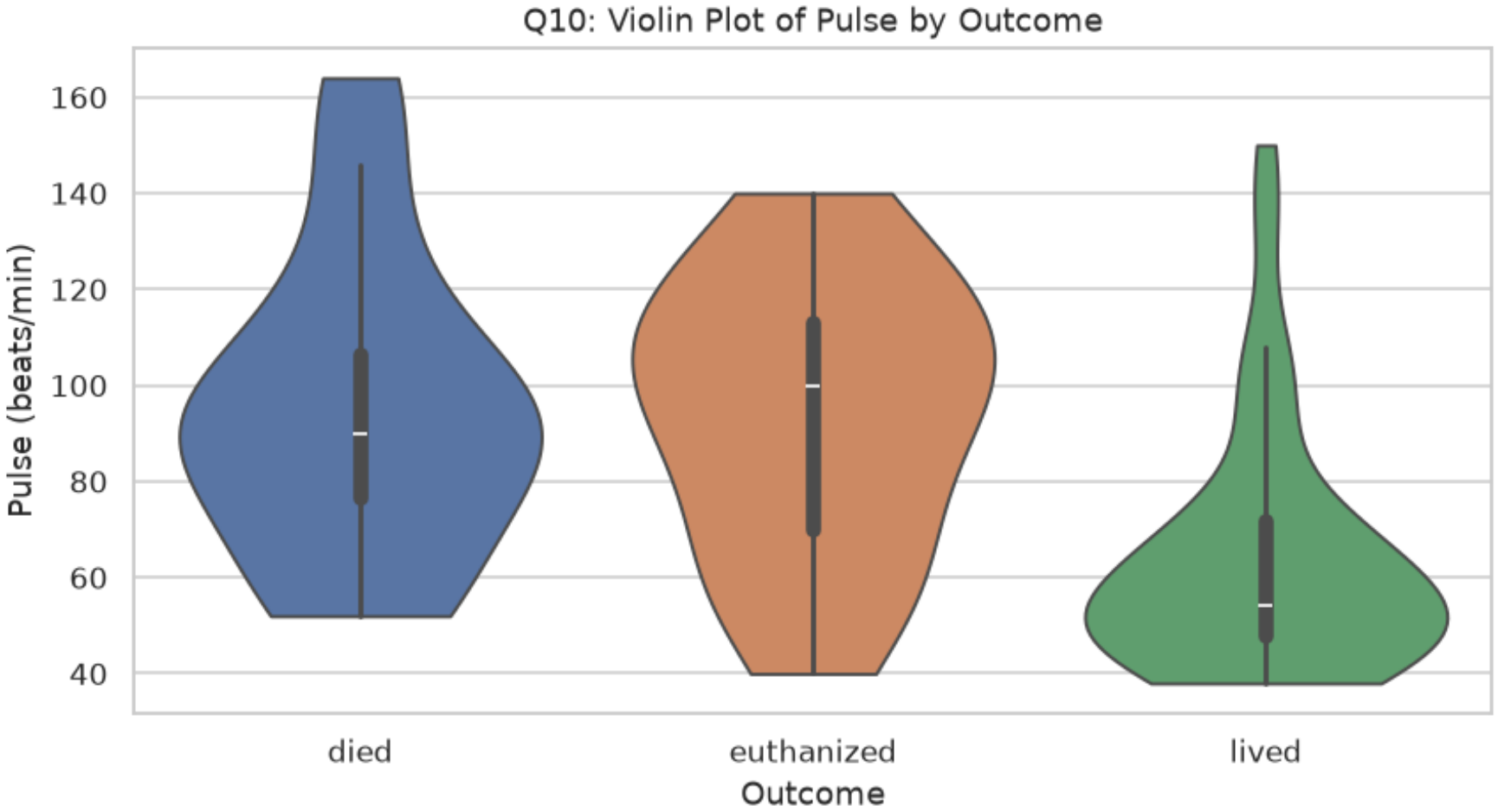
```
plot_data = data.dropna(subset=["outcome", "pulse"])
sns.kdeplot(data=plot_data, x="pulse", hue="outcome", fill=True, alpha=.35, common_norm=False)
finish("Q8: Pulse Density by Outcome", "Pulse (beats/min)", "Density")
sns.boxplot(data=plot_data, x="outcome", y="pulse", hue="outcome", legend=False)
finish("Q9: Pulse Distribution by Outcome", "Outcome", "Pulse (beats/min)")
sns.violinplot(data=plot_data, x="outcome", y="pulse", hue="outcome", legend=False, cut=0)
finish("Q10: Violin Plot of Pulse by Outcome", "Outcome", "Pulse (beats/min)")
```

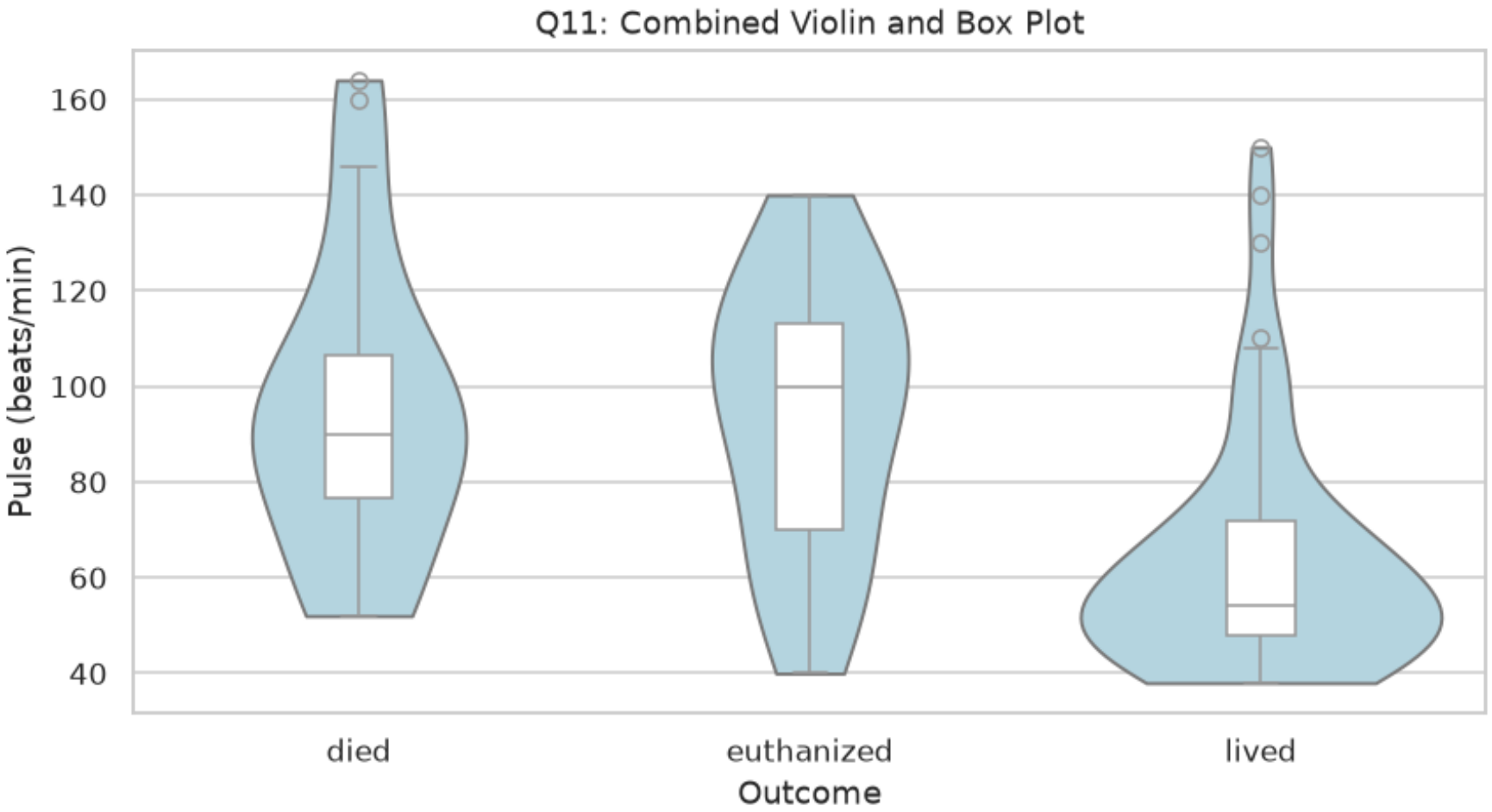
```
sns.violinplot(data=plot_data, x="outcome", y="pulse", color="lightblue", inner=None, cut=0)
sns.boxplot(data=plot_data, x="outcome", y="pulse", width=.15, color="white")
finish("Q11: Combined Violin and Box Plot", "Outcome", "Pulse (beats/min)")
```

Output




Q9: Pulse Distribution by Outcome
Pulse (beats/min)
160
140
120
100
80
60
40
died
euthanized
lived
Outcome


Q10: Violin Plot of Pulse by Outcome
Pulse (beats/min)
160
140
120
100
80
60
40
died
euthanized
lived
Outcome



Q12-Q13 Correlation matrix and correlogram

Python code

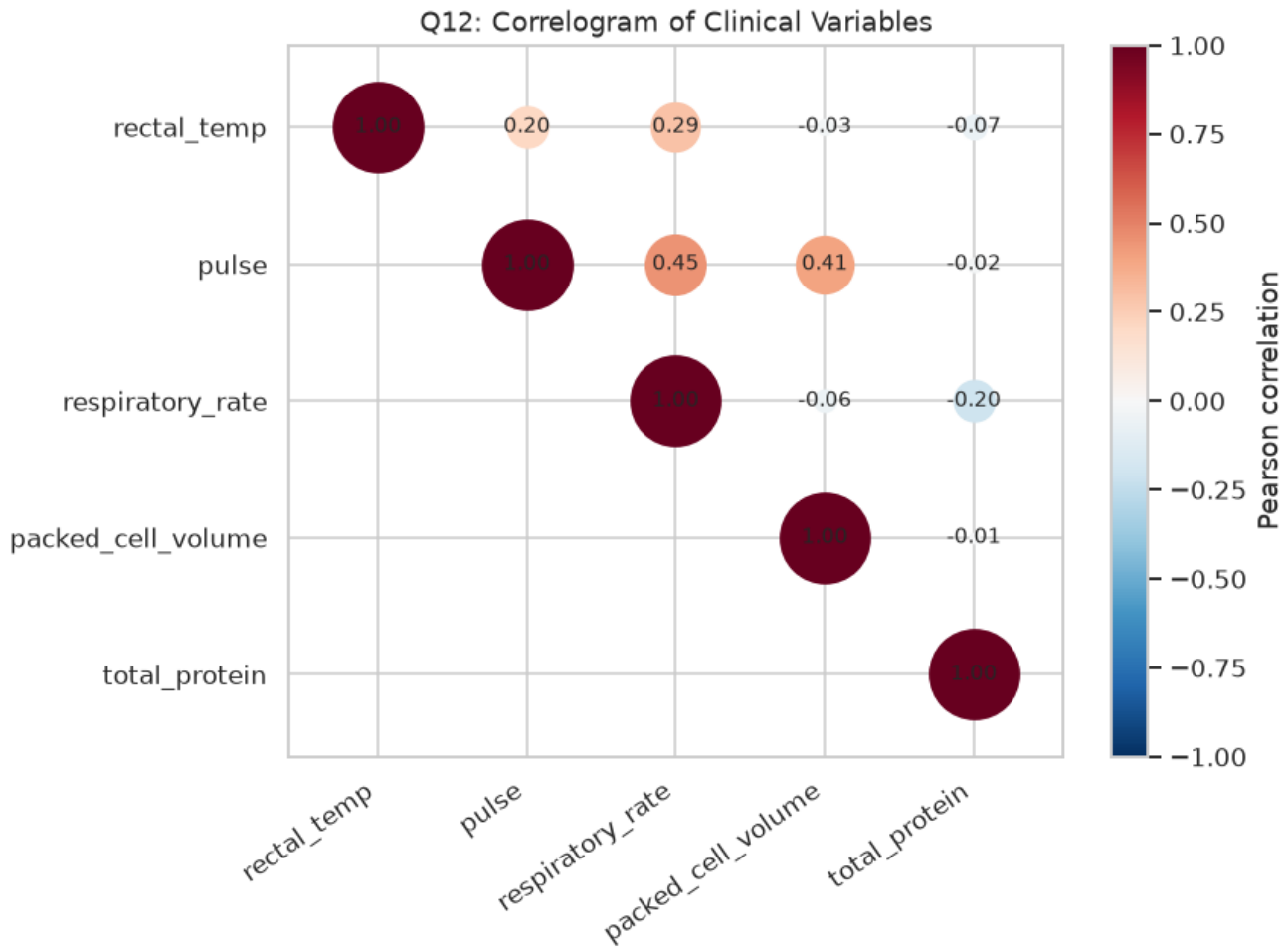
```
show("Correlation matrix", cor_matrix.round(3))
order = linkage(pdist(cor_matrix.to_numpy()), method="complete")
ordered_idx = dendrogram(order, no_plot=True)["leaves"]
ordered_cor = cor_matrix.iloc[ordered_idx, ordered_idx]
fig, ax = plt.subplots(figsize=(8, 6))
for i in range(len(ordered_cor)):
    for j in range(i, len(ordered_cor)):
```

```
        value = ordered_cor.iloc[i, j]
        ax.scatter(j, i, s=abs(value) * 1500, c=[value], cmap="RdBu_r", vmin=-1, vmax=1)
        ax.text(j, i, f"{value:.2f}", ha="center", va="center", fontsize=9)
ax.set_xticks(range(len(ordered_cor)), ordered_cor.columns, rotation=35, ha="right")
ax.set_yticks(range(len(ordered_cor)), ordered_cor.index)
ax.set_xlim(-.6, len(ordered_cor) - .4)
ax.set_ylim(len(ordered_cor) - .4, -.6)
fig.colorbar(plt.cm.ScalarMappable(norm=plt.Normalize(-1, 1), cmap="RdBu_r"), ax=ax, label="Pearson correlation")
finish("Q12: Correlogram of Clinical Variables")
print("Q13: Titles, axis labels, legends, readable sizes, and consistent styling are applied throughout.")
```

Output

```
Correlation matrix
```

	rectal_temp	pulse	respiratory_rate	packed_cell_volume	total_protein
rectal_temp	1.000	0.204	0.290	-0.028	-0.070
pulse	0.204	1.000	0.446	0.406	-0.022
respiratory_rate	0.290	0.446	1.000	-0.058	-0.204
packed_cell_volume	-0.028	0.406	-0.058	1.000	-0.012
total_protein	-0.070	-0.022	-0.204	-0.012	1.000



Q13: Titles, axis labels, legends, readable sizes, and consistent styling are applied throughout.

Assignment 5 - Multivariate Analysis

This assignment maps several clinical variables to position, color, shape, and point size. Faceted scatter plots compare groups, while a pairs plot and parallel coordinate plot show relationships across multiple measurements.

Questions

1. Load the dataset shared through e-mail into the R environment. Examine the dataset and identify the numerical and categorical variables.

2. Create a scatter plot involving three variables, where two numerical variables arerepresented on the X- and Y-axes, and the third variable is represented using color.

3. Create a scatter plot involving four variables, where two numerical variables arerepresented on the X- and Y-axes, the third variable is represented using color, and the fourth variable is represented using shape.

4. Create a scatter plot involving four variables, where two numerical variables arerepresented on the X- and Y-axes, the third variable is represented using color, and the fourthnumerical variable is represented using point size.

5. Construct a bubble plot using three numerical variables, where two variables arerepresented on the X- and Y-axes, and the third variable determines the size of the bubbles.

6. Generate subplots using facet_wrap() by selecting an appropriate categorical variablefrom the dataset.

7. Generate subplots using facet_grid() by selecting two appropriate categoricalvariables from the dataset. Arrange the plots in rows and columns and compare the patterns observed across the different groups.

8. Construct a pairs plot using at least four numerical variables from the dataset. Use the plot to identify possible relationships and correlations among the selected variables.

9. Create a parallel coordinate plot using at least four numerical variables from thedataset. Differentiate the observations using an appropriate categorical variable.

Q1 Identify numerical and categorical variables

Python code

```
data = load_horse()
print("Numerical variables:", data.select_dtypes(include="number").columns.tolist())
print("Categorical variables:", data.select_dtypes(exclude="number").columns.tolist())
```

Output

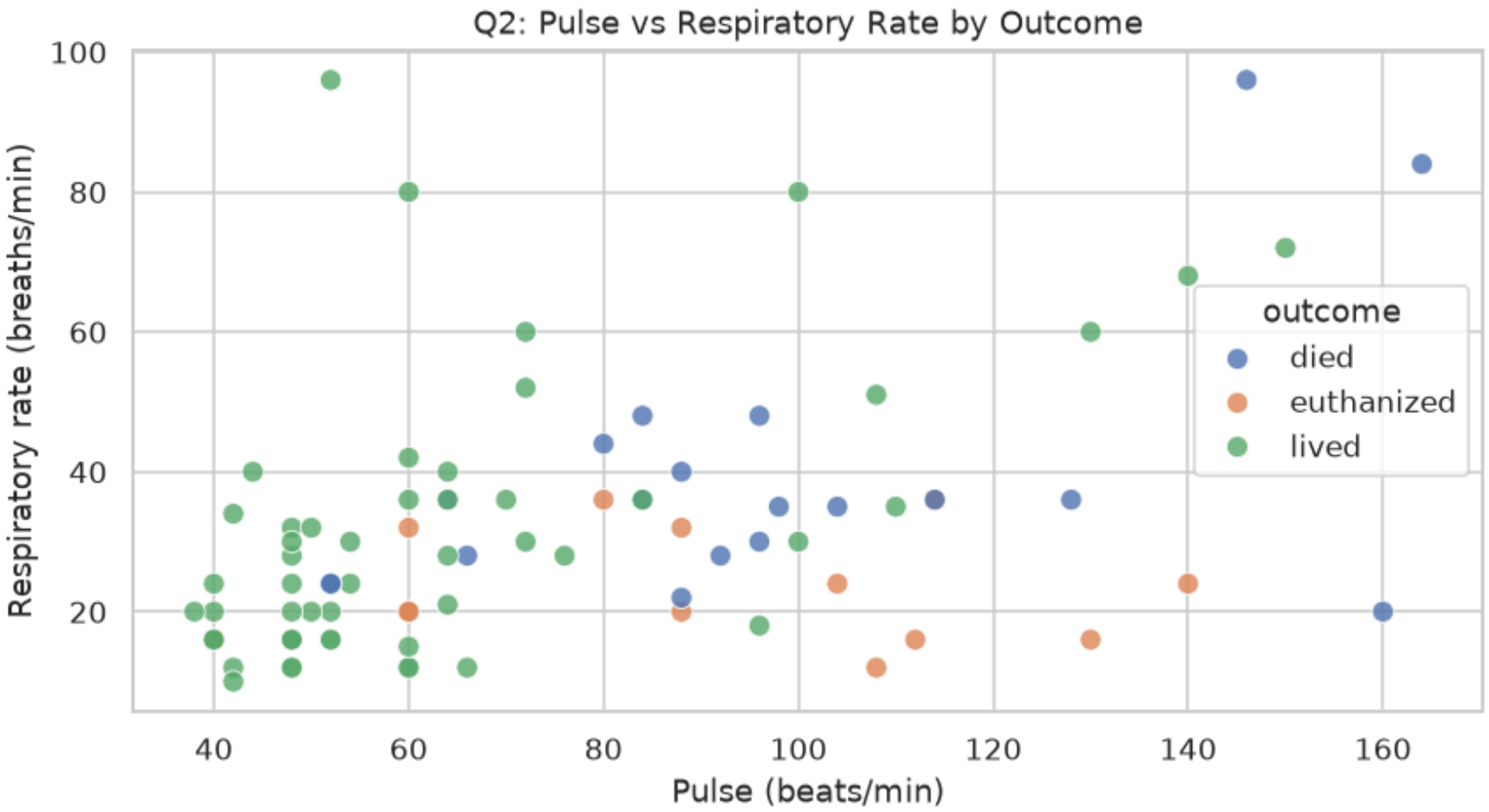
```
Numerical variables: ['hospital_number', 'rectal_temp', 'pulse', 'respiratory_rate', 'nasogastric_reflux_ph', 'packed_cell_volume', 'total_protein', 'abdomo_protein',
     'lesion_1', 'lesion_2', 'lesion_3']
Categorical variables: ['surgery', 'age', 'temp_of_extremities', 'peripheral_pulse', 'mucous_membrane', 'capillary_refill_time', 'pain', 'peristalsis',
    'abdominal_distention', 'nasogastric_tube', 'nasogastric_reflux', 'rectal_exam_feces', 'abdomen', 'abdomo_appearance', 'outcome', 'surgical_lesion', 'cp_data']
```

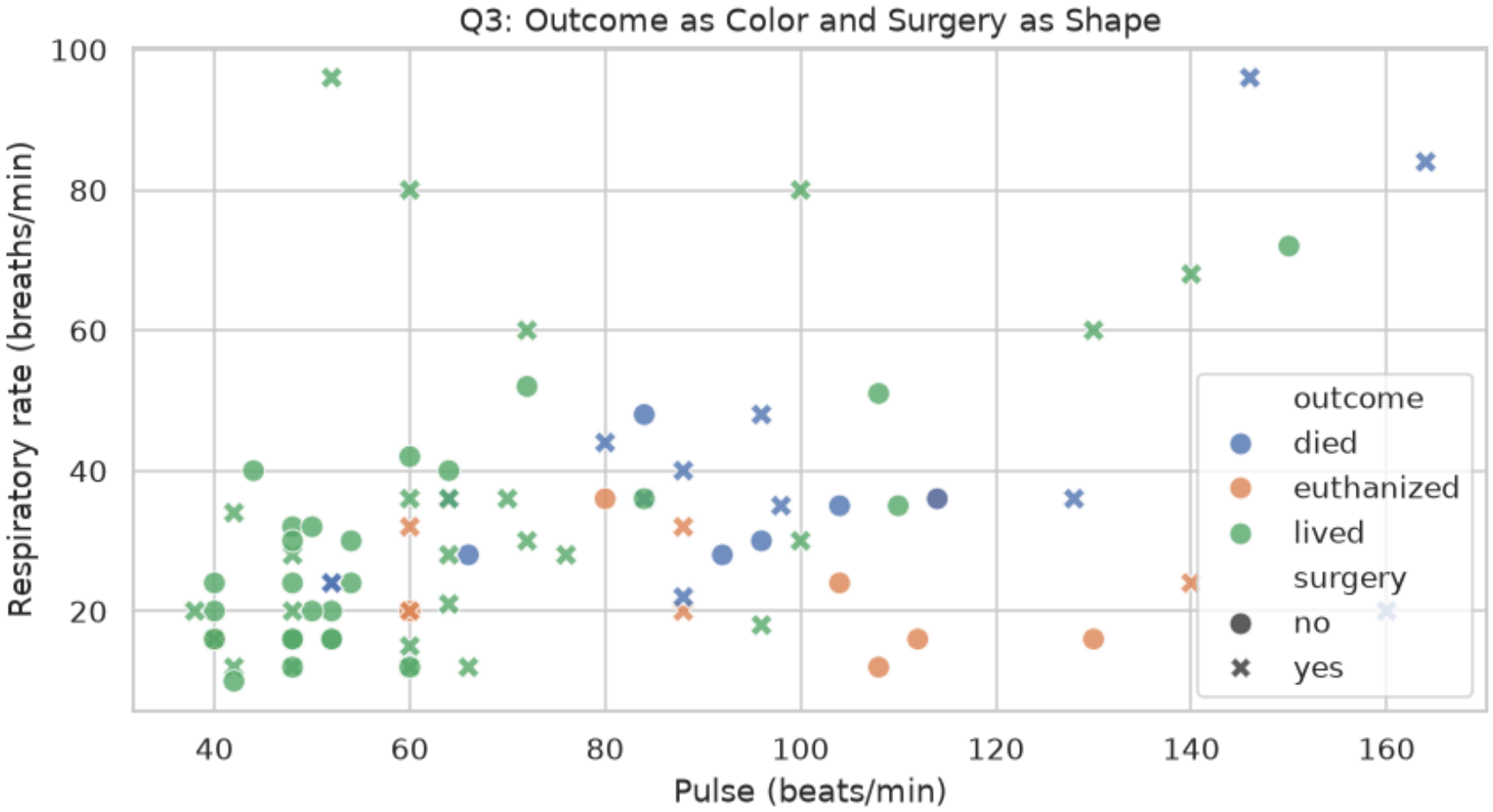
Q2-Q5 Scatter and bubble plots

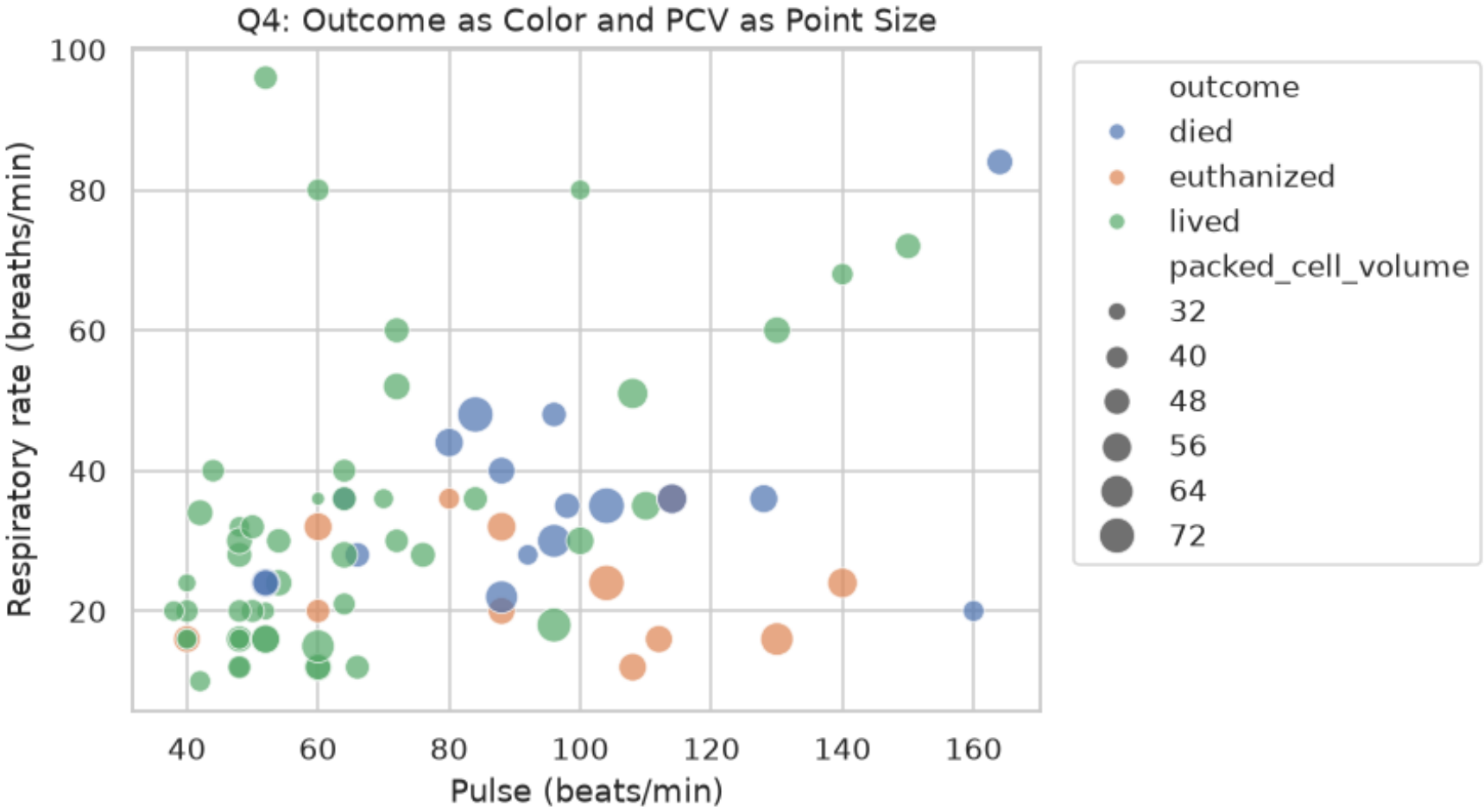
Python code

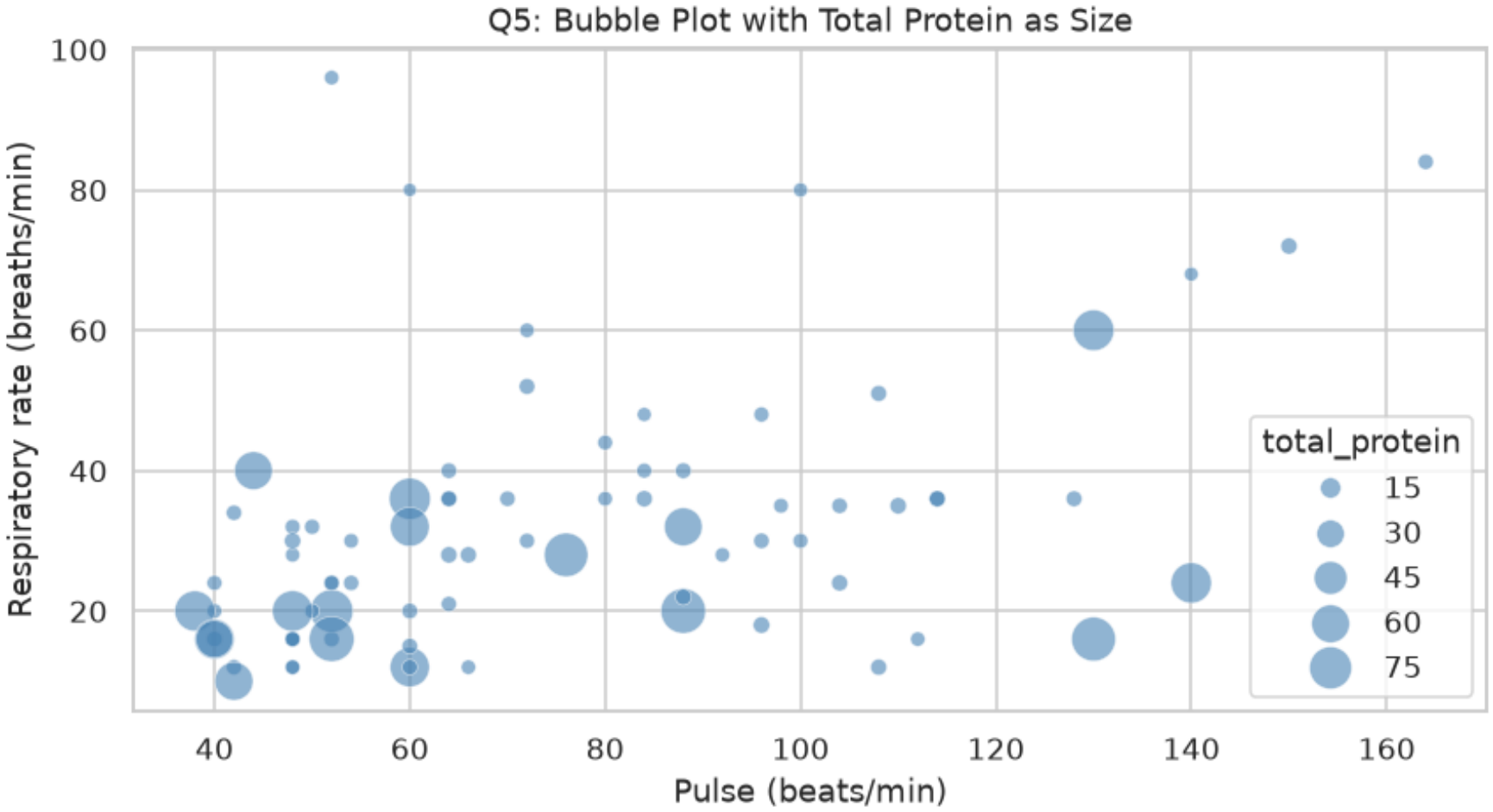
```
sns.scatterplot(data=data.dropna(subset=["pulse", "respiratory_rate", "outcome"]), x="pulse", y="respiratory_rate", hue="outcome", s=65, alpha=.8)
finish("Q2: Pulse vs Respiratory Rate by Outcome", "Pulse (beats/min)", "Respiratory rate (breaths/min)")
sns.scatterplot(data=data.dropna(subset=["pulse", "respiratory_rate", "outcome", "surgery"]), x="pulse", y="respiratory_rate", hue="outcome",
    style="surgery", s=70, alpha=.8)
finish("Q3: Outcome as Color and Surgery as Shape", "Pulse (beats/min)", "Respiratory rate (breaths/min)")
sns.scatterplot(data=data.dropna(subset=["pulse", "respiratory_rate", "outcome", "packed_cell_volume"]), x="pulse", y="respiratory_rate", hue="outcome",
    size="packed_cell_volume", sizes=(25, 180), alpha=.7)
plt.legend(bbox_to_anchor=(1.02, 1), loc="upper left")
finish("Q4: Outcome as Color and PCV as Point Size", "Pulse (beats/min)", "Respiratory rate (breaths/min)")
sns.scatterplot(data=data.dropna(subset=["pulse", "respiratory_rate", "total_protein"]), x="pulse", y="respiratory_rate", size="total_protein", sizes=(25,
    280), color="steelblue", alpha=.6)
finish("Q5: Bubble Plot with Total Protein as Size", "Pulse (beats/min)", "Respiratory rate (breaths/min)")
```

Output


Q2: Pulse vs Respiratory Rate by Outcome
Respiratory rate (breaths/min)
Pulse (beats/min)
100
80
60
40
20
40
60
80
100
120
140
160
outcome
died
euthanized
lived


Q3: Outcome as Color and Surgery as Shape
Respiratory rate (breaths/min)
Pulse (beats/min)
100
80
60
40
20
40
60
80
100
120
140
160
outcome
died
euthanized
lived
surgery
no
yes


Q4: Outcome as Color and PCV as Point Size
Respiratory rate (breaths/min)
Pulse (beats/min)
100
80
60
40
20
40
60
80
100
120
140
160
outcome
died
euthanized
lived
packed_cell_volume
32
40
48
56
64
72



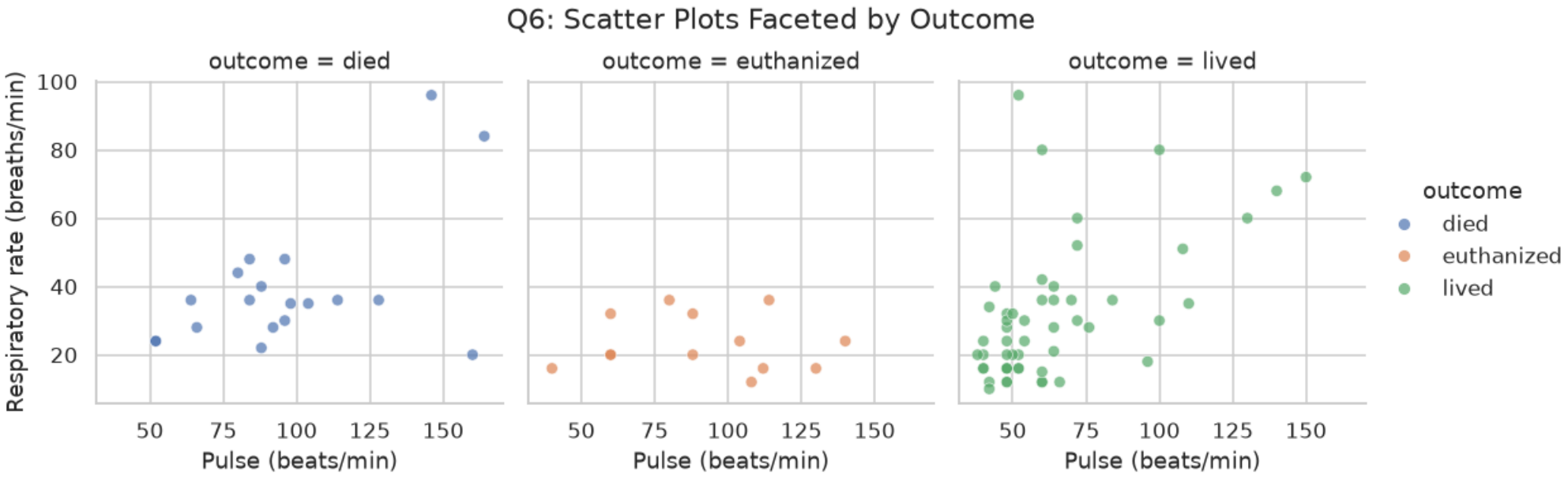
Q6-Q7 Faceted scatter plots

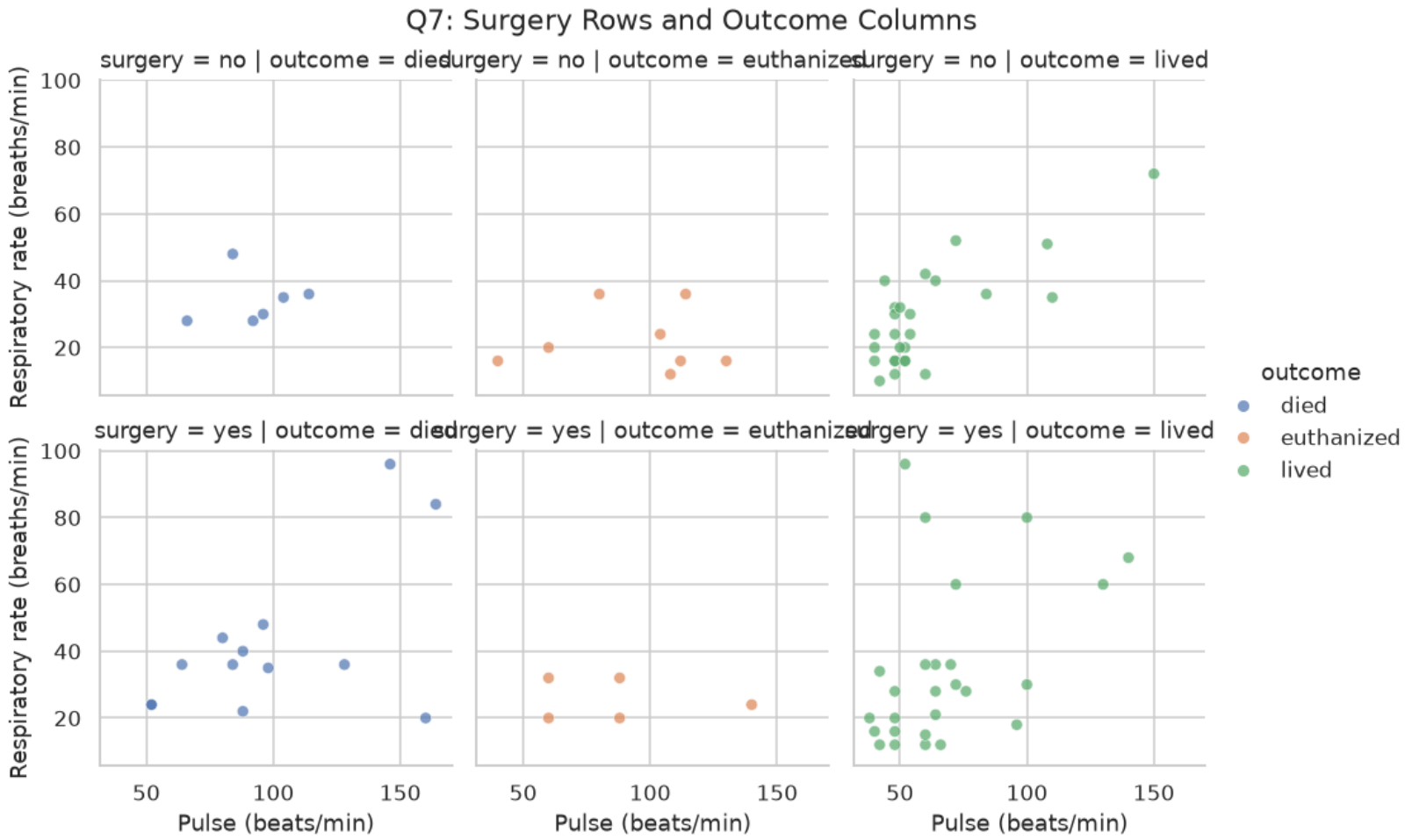
Python code

```
g = sns.relplot(data=data.dropna(subset=["pulse", "respiratory_rate", "outcome"]), x="pulse", y="respiratory_rate", col="outcome", hue="outcome",
    height=3.4, alpha=.7)
g.set_axis_labels("Pulse (beats/min)", "Respiratory rate (breaths/min)")
g.figure.suptitle("Q6: Scatter Plots Faceted by Outcome", y=1.04)
plt.show()
g = sns.relplot(data=data.dropna(subset=["pulse", "respiratory_rate", "surgery", "outcome"]), x="pulse", y="respiratory_rate", row="surgery", col="outcome",
     hue="outcome", height=3, alpha=.7)
```

```
g.set_axis_labels("Pulse (beats/min)", "Respiratory rate (breaths/min)")
g.figure.suptitle("Q7: Surgery Rows and Outcome Columns", y=1.02)
plt.show()
show("Group patterns: mean pulse and respiratory rate", data.groupby(["surgery", "outcome"])[["pulse", "respiratory_rate"]].agg(["count", "mean"]))
```

Output





Group patterns: mean pulse and respiratory rate

		pulse	respiratory_rate		
		count	mean	count	mean
surgery	outcome				
no	died	8	96.500000	6	34.166667
euthanized	8	93.500000	8	22.000000	
lived	32	60.437500	26	28.384615	

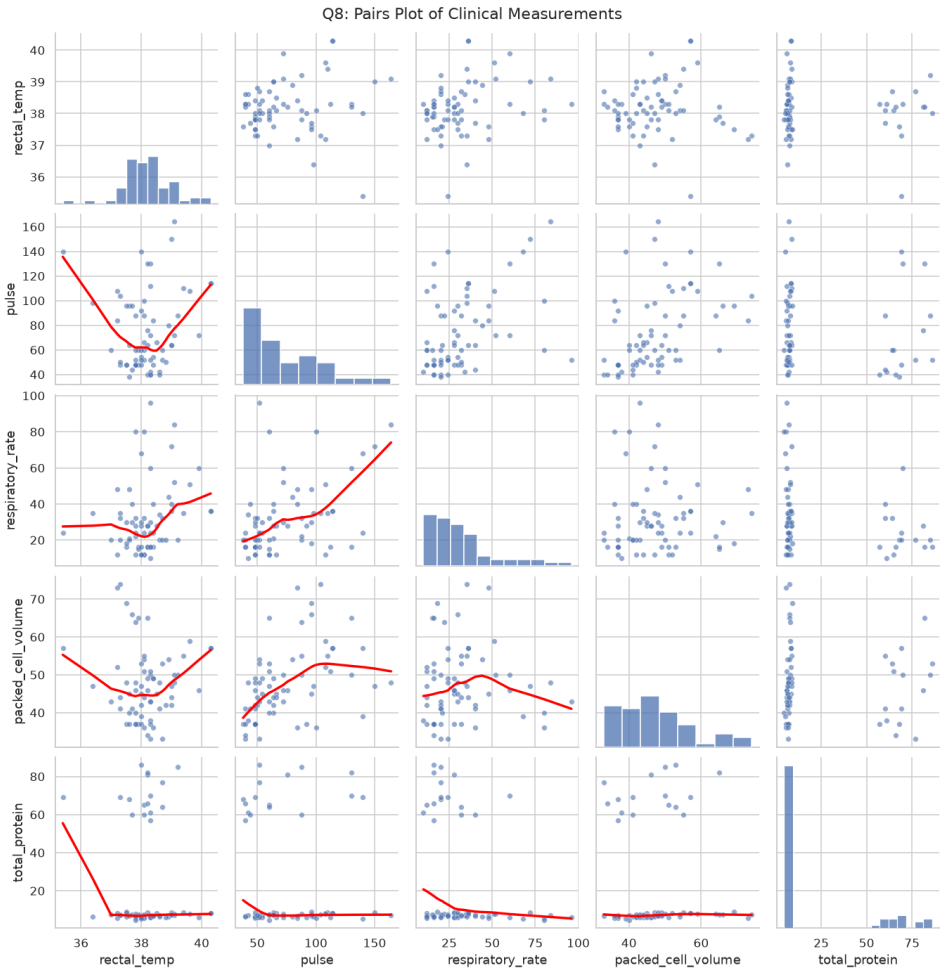
		pulse	respiratory_rate		
yes	died	16	94.000000	15	39.266667
euthanized	7	93.714286	5	25.600000	
lived	29	66.620690	26	34.769231	

Q8 Pairs plot

Python code

```
num_subset = data[clinical_columns].dropna()
g = sns.pairplot(num_subset, diag_kind="hist", plot_kws={"alpha": .6, "s": 25})
g.map_lower(sns.regplot, scatter=False, ci=None, lowess=True, line_kws={"color": "red"})
g.figure.suptitle("Q8: Pairs Plot of Clinical Measurements", y=1.01)
plt.show()
pair_correlations = num_subset.corr()
upper = pair_correlations.where(np.triu(np.ones(pair_correlations.shape), k=1).astype(bool)).stack()
strongest = upper.abs().idxmax()
print(f"Largest absolute Pearson correlation: {strongest[0]} vs {strongest[1]} = {upper.loc[strongest]:.3f}")
```

Output




```
Largest absolute Pearson correlation: pulse vs packed_cell_volume = 0.449
```

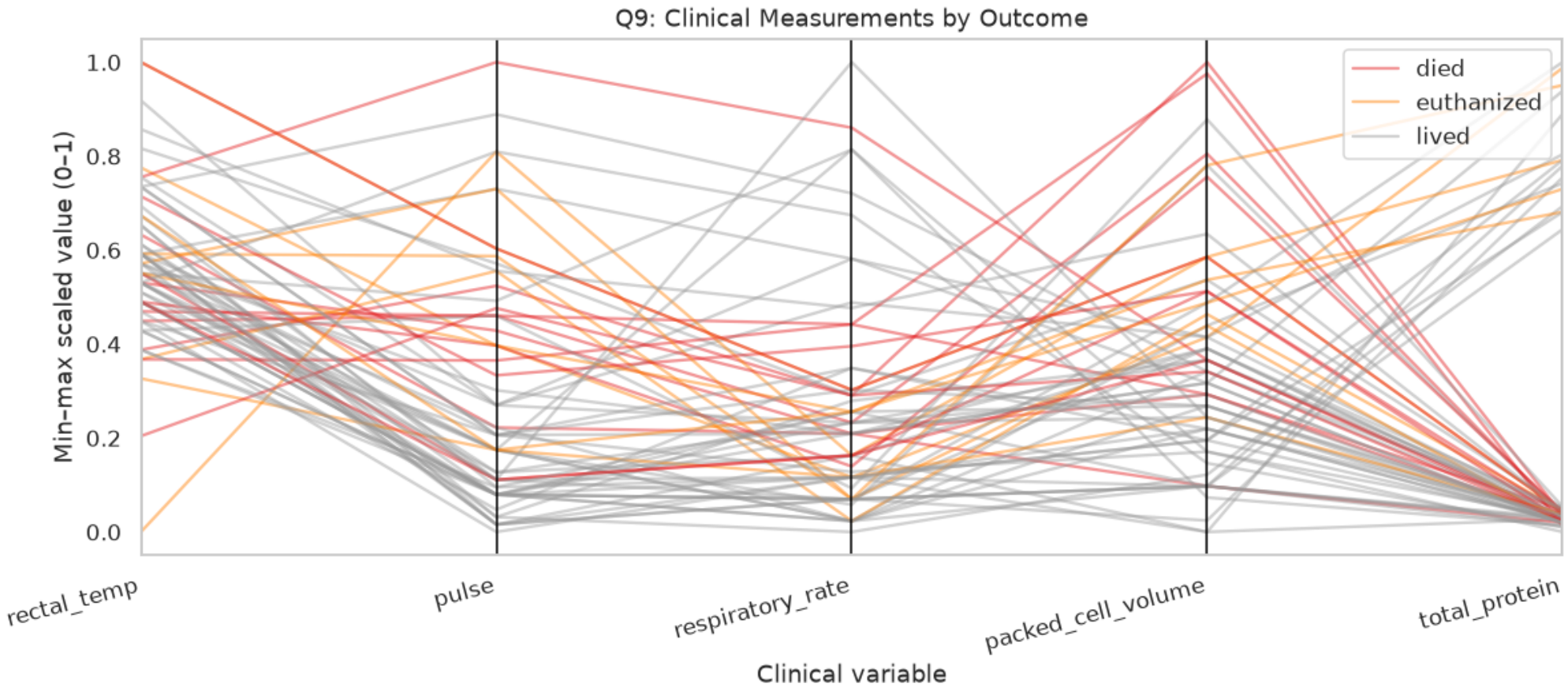
Q9 Parallel coordinate plot

Python code

```
parcoord_data = data[clinical_columns + ["outcome"]].dropna().copy()
span = parcoord_data[clinical_columns].max() - parcoord_data[clinical_columns].min()
parcoord_data[clinical_columns] = (parcoord_data[clinical_columns] - parcoord_data[clinical_columns].min()) / span.replace(0, 1)
plt.figure(figsize=(11, 5))
pd.plotting.parallel_coordinates(parcoord_data, "outcome", cols=clinical_columns, alpha=.45, colormap="Set1")
plt.xticks(rotation=15, ha="right")
```

```
finish("Q9: Clinical Measurements by Outcome", "Clinical variable", "Min-max scaled value (0-1)")
```

Output



Assignment 6 - Time Series Analysis

This assignment explores the monthly AirPassengers series, decomposes its multiplicative components, and inspects autocorrelation. It fits the same seasonal ARIMA specification as the R solution and forecasts the next 12 months with prediction intervals.

Questions

Perform the following steps using the built-in AirPassengers dataset in R:

1. Load the built-in AirPassengers dataset in R.

2. Explore the first five and last five observations of the dataset.

3. Identify the starting and ending time periods of the dataset. (start, end)

4. Determine the frequency and time-series characteristics of the dataset. (frequency, class)

5. Visualize the monthly passenger trend over the entire period. (plot)

6. Decompose the time series into trend, seasonal, and random components. (decompose)

7. Plot the decomposed components to examine the underlying patterns.

8. Analyze the autocorrelation and partial autocorrelation of the time series. (acf, pacf)

9. Fit an ARIMA model to the AirPassengers time series. (arima)

10. Display and interpret the fitted ARIMA model parameters and model summary.(summary)

11. Forecast the number of passengers for the next 12 months using the fitted ARIMA model.(predict)

12. Visualize the actual and forecasted passenger values with appropriate labels and title.(plot, lines)

Q1-Q4 Load and inspect AirPassengers

Python code

```
air_passenger_values = [
    112,118,132,129,121,135,148,148,136,119,104,118,
    115,126,141,135,125,149,170,170,158,133,114,140,
    145,150,178,163,172,178,199,199,184,162,146,166,
    171,180,193,181,183,218,230,242,209,191,172,194,
    196,196,236,235,229,243,264,272,237,211,180,201,
    204,188,235,227,234,264,302,293,259,229,203,229,
    242,233,267,269,270,315,364,347,312,274,237,278,
    284,277,317,313,318,374,413,405,355,306,271,306,
```

```
    315,301,356,348,355,422,465,467,404,347,305,336,
    340,318,362,348,363,435,491,505,404,359,310,337,
    360,342,406,396,420,472,548,559,463,407,362,405,
    417,391,419,461,472,535,622,606,508,461,390,432
]
AirPassengers = pd.Series(air_passenger_values, index=pd.date_range("1949-01-01", periods=144, freq="MS"), name="Passengers (thousands)")
show("First five observations", AirPassengers.head(5))
show("Last five observations", AirPassengers.tail(5))
print("Start:", AirPassengers.index[0].strftime("%Y-%m"))
print("End:", AirPassengers.index[-1].strftime("%Y-%m"))
print("Frequency: 12 observations per year; index frequency:", AirPassengers.index.freqstr)
print("Class:", type(AirPassengers).__name__, "with", type(AirPassengers.index).__name__)
```

Output

```
First five observations

1949-01-01    112
1949-02-01    118
1949-03-01    132
1949-04-01    129
1949-05-01    121
Freq: MS, Name: Passengers (thousands), dtype: int64

Last five observations

1960-08-01    606
1960-09-01    508
1960-10-01    461
1960-11-01    390
1960-12-01    432
Freq: MS, Name: Passengers (thousands), dtype: int64

Start: 1949-01
End: 1960-12
Frequency: 12 observations per year; index frequency: MS
Class: Series with DatetimeIndex
```

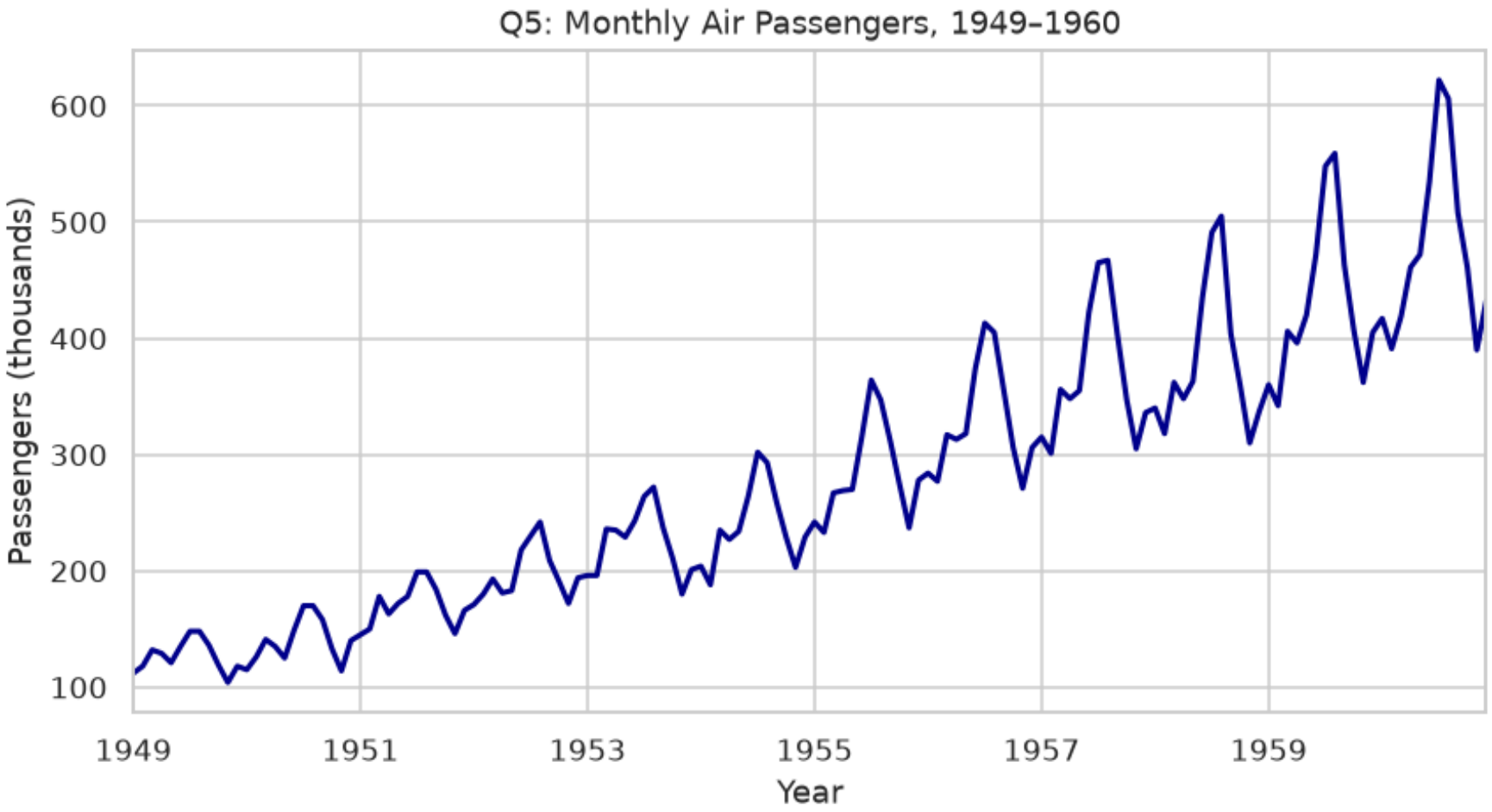
Q5-Q8 Trend, decomposition, ACF, and PACF

Python code

```
AirPassengers.plot(color="darkblue", linewidth=2)
finish("Q5: Monthly Air Passengers, 1949-1960", "Year", "Passengers (thousands)")
decomp = seasonal_decompose(AirPassengers, model="multiplicative", period=12)
show("Q6: Decomposed components", pd.DataFrame({"Observed": AirPassengers, "Trend": decomp.trend, "Seasonal": decomp.seasonal, "Random":
    decomp.resid}).head(14))
fig = decomp.plot()
fig.set_size_inches(10, 8)
fig.suptitle("Q7: Multiplicative Time Series Decomposition", y=1.01)
```

```
fig.tight_layout()
plt.show()
fig, axes = plt.subplots(1, 2, figsize=(11, 4))
plot_acf(AirPassengers, lags=36, ax=axes[0], title="Q8: Autocorrelation")
plot_pacf(AirPassengers, lags=36, method="ywmle", ax=axes[1], title="Q8: Partial Autocorrelation")
fig.tight_layout()
plt.show()
```

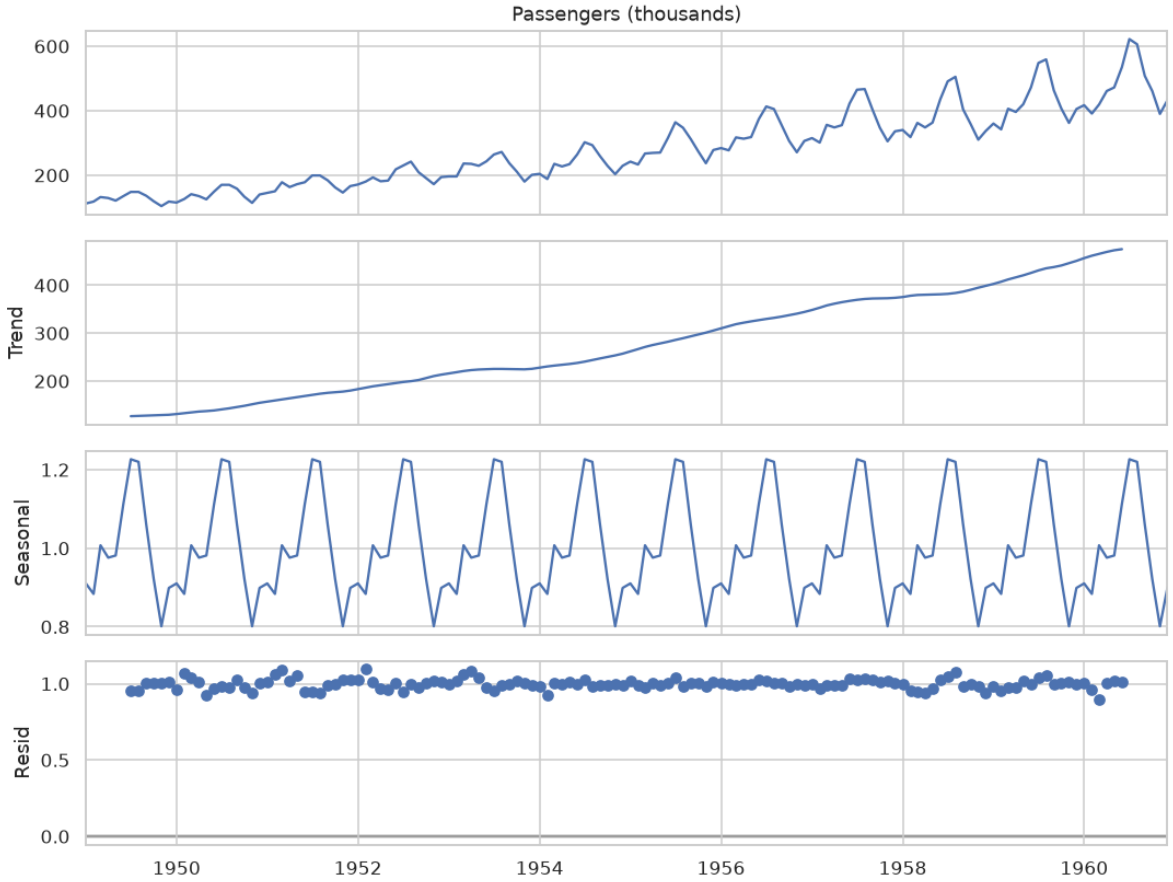
Output

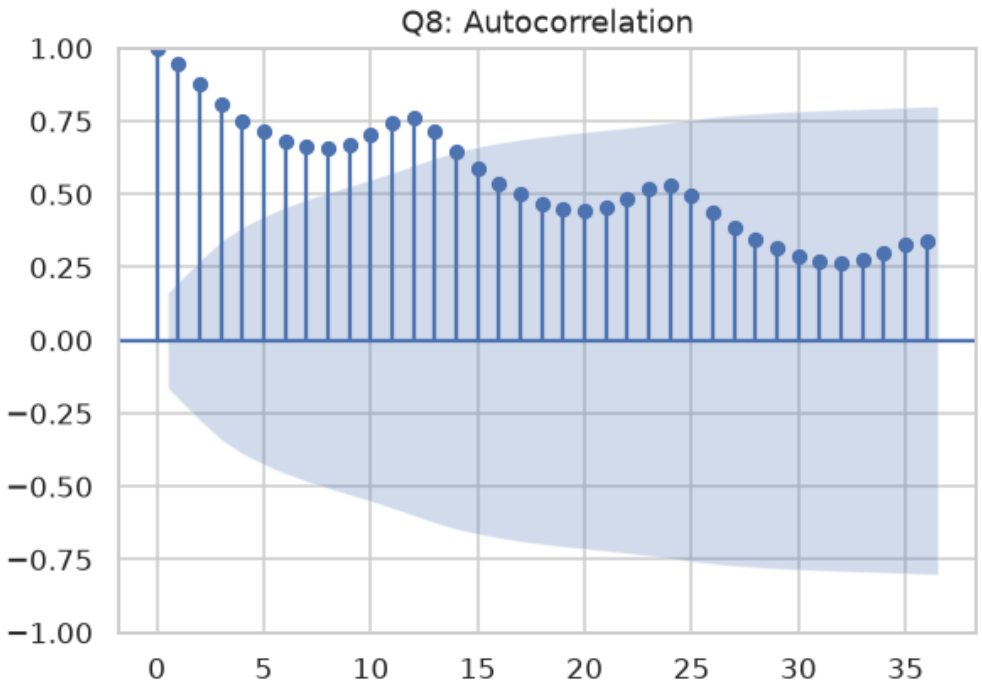


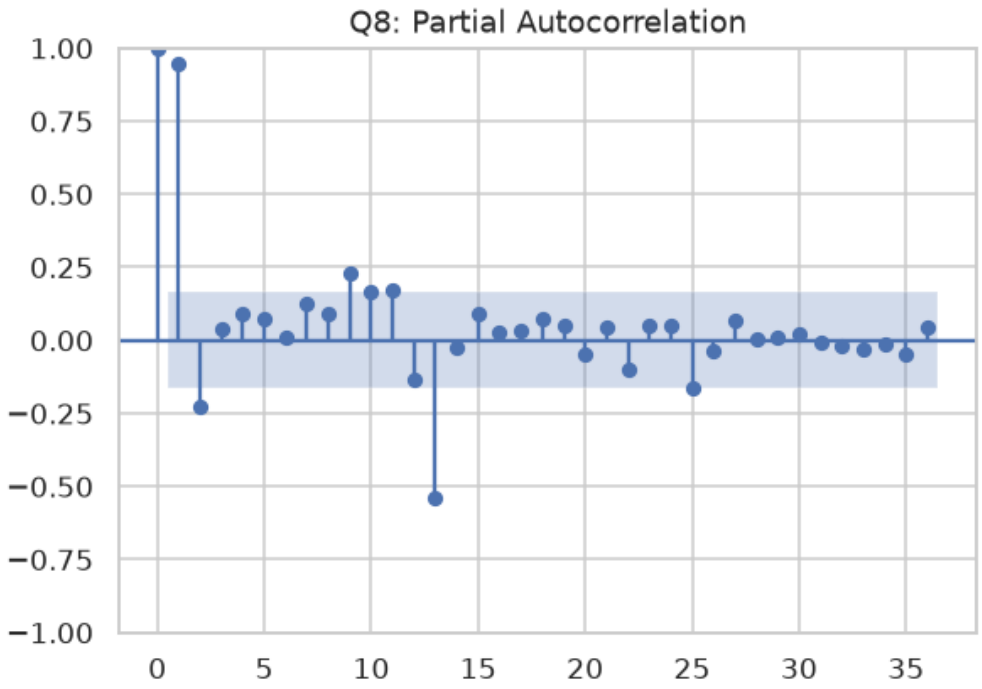

```
Q6: Decomposed components
```

	Observed	Trend	Seasonal	Random
1949-01-01	112	NaN	0.910230	NaN
1949-02-01	118	NaN	0.883625	NaN
1949-03-01	132	NaN	1.007366	NaN
1949-04-01	129	NaN	0.975906	NaN
1949-05-01	121	NaN	0.981378	NaN
1949-06-01	135	NaN	1.112776	NaN
1949-07-01	148	126.791667	1.226556	0.951664
1949-08-01	148	127.250000	1.219911	0.953401
1949-09-01	136	127.958333	1.060492	1.002220
1949-10-01	119	128.583333	0.921757	1.004028
1949-11-01	104	129.000000	0.801178	1.006270
1949-12-01	118	129.750000	0.898824	1.011812
1950-01-01	115	131.250000	0.910230	0.962603
1950-02-01	126	133.083333	0.883625	1.071467

Q7: Multiplicative Time Series Decomposition







Q9-Q10 Fit and interpret the seasonal ARIMA model

Python code

```
model = SARIMAX(AirPassengers, order=(0, 1, 1), seasonal_order=(0, 1, 1, 12), trend="n")
fit = model.fit(disp=False, maxiter=500)
print(fit.summary())
print("SARIMA(0,1,1)(0,1,1)[12] uses first and annual seasonal differences.")
print("ma.L1 describes short-term shocks; ma.S.L12 describes shocks recurring after 12 months.")
parameter_table = pd.DataFrame({"Estimate": fit.params, "Standard error": fit.bse, "p-value": fit.pvalues})
show("Estimated parameters", parameter_table)
print(f"AIC = {fit.aic:.2f}; compare AIC only with models fitted to the same series.")
print("Coefficient uncertainty is assessed using confidence intervals and p-values; small standard errors alone do not establish significance.")
```

Output

```
SARIMAX Results
==========================================================================================
Dep. Variable:             Passengers (thousands)   No. Observations:                  144
Model:             SARIMAX(0, 1, 1)x(0, 1, 1, 12)   Log Likelihood                -507.501
Date:                            Mon, 19 Oct 2026   AIC                            1021.003
```

```
Time:                                14:35:51   BIC                          1029.628
Sample:                            01-01-1949   HQIC                         1024.507
                                 - 12-01-1960
Covariance Type:                          opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ma.L1         -0.3087      0.068     -4.507      0.000      -0.443      -0.174
ma.S.L12      -0.1075      0.091     -1.184      0.236      -0.285       0.070
sigma2       135.4236     15.443      8.769      0.000     105.156     165.691
===================================================================================
Ljung-Box (L1) (Q):                   0.00   Jarque-Bera (JB):                 5.93
Prob(Q):                              0.97   Prob(JB):                         0.05
Heteroskedasticity (H):               2.60   Skew:                             0.07
Prob(H) (two-sided):                  0.00   Kurtosis:                         4.03
===================================================================================

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).
SARIMA(0,1,1)(0,1,1)[12] uses first and annual seasonal differences.
ma.L1 describes short-term shocks; ma.S.L12 describes shocks recurring after 12 months.
Estimated parameters
```

	Estimate	Standard error	p-value
ma.L1	-0.308682	0.068497	6.589410e-06
ma.S.L12	-0.107451	0.090743	2.363641e-01
sigma2	135.423644	15.442921	1.797767e-18

```
AIC = 1021.00; compare AIC only with models fitted to the same series.
Coefficient uncertainty is assessed using confidence intervals and p-values; small standard errors alone do not establish significance.
```

Q11-Q12 Forecast the next 12 months

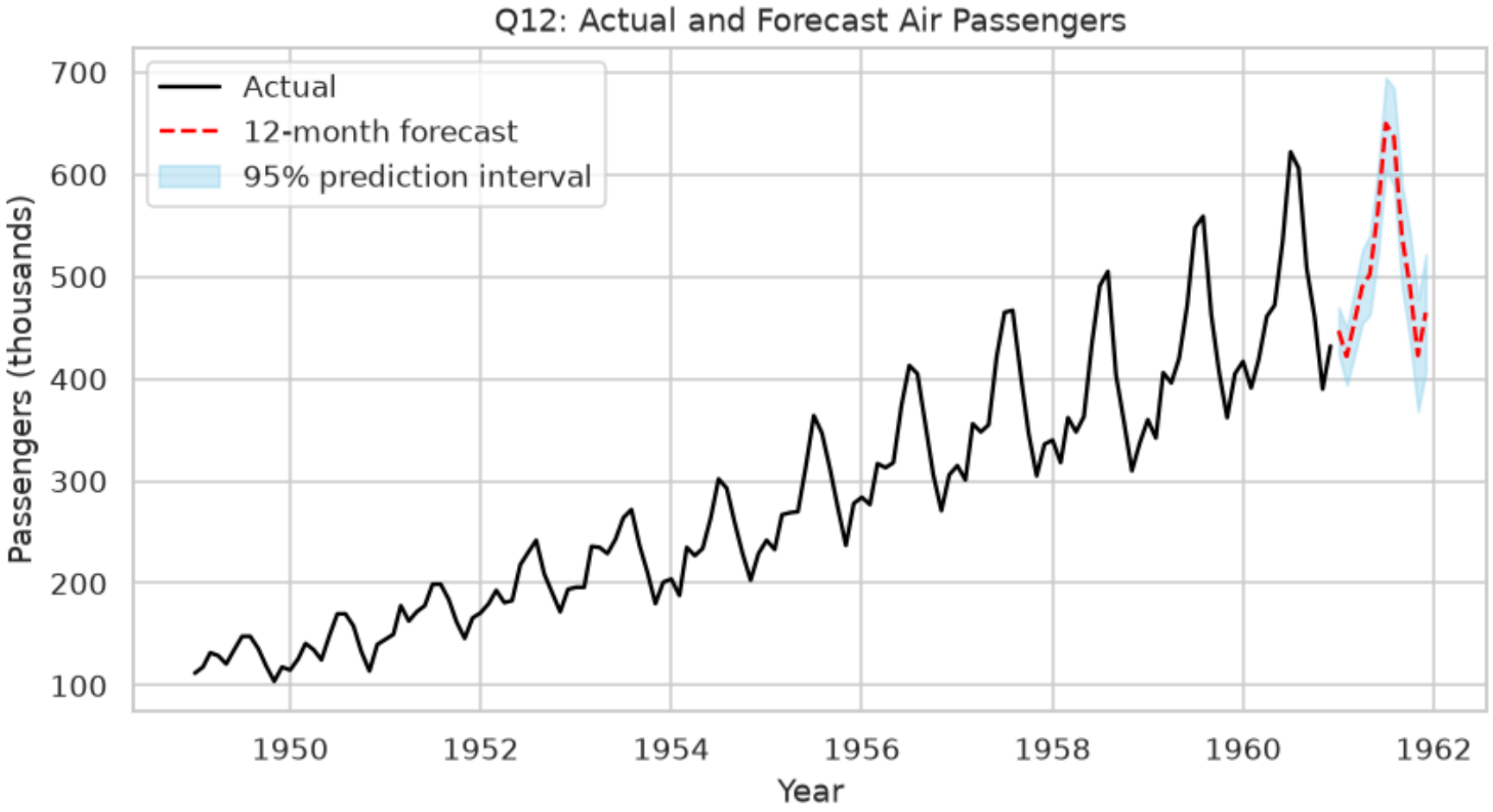
Python code

```
forecast_result = fit.get_forecast(steps=12)
forecast_vals = forecast_result.predicted_mean
intervals = forecast_result.conf_int(alpha=.05)
forecast_table = pd.DataFrame({"Forecast": forecast_vals,
                               "Lower 95%": intervals.iloc[:, 0], "Upper 95%": intervals.iloc[:, 1]})
show("Q11: Forecasts for 1961 (thousands of passengers)", forecast_table)
plt.plot(AirPassengers.index, AirPassengers, color="black", label="Actual")
plt.plot(forecast_vals.index, forecast_vals, color="red", linestyle="--", label="12-month forecast")
plt.fill_between(forecast_vals.index, intervals.iloc[:, 0], intervals.iloc[:, 1], color="skyblue", alpha=.4, label="95% prediction interval")
plt.legend()
finish("Q12: Actual and Forecast Air Passengers", "Year", "Passengers (thousands)")
```

Output

```
Q11: Forecasts for 1961 (thousands of passengers)
```

	Forecast	Lower 95%	Upper 95%
1961-01-01	447.053344	424.244917	469.861771
1961-02-01	421.877590	394.149446	449.605734
1961-03-01	453.526489	421.628596	485.424382
1961-04-01	489.900906	454.318590	525.483222
1961-05-01	502.183615	463.264123	541.103107
1961-06-01	564.224717	522.232426	606.217007
1961-07-01	649.795305	604.940226	694.650384
1961-08-01	636.714783	589.168977	684.260590
1961-09-01	538.921049	488.828840	589.013257
1961-10-01	491.067343	438.552060	543.582626
1961-11-01	422.824489	367.993106	477.655872
1961-12-01	464.752767	407.699228	521.806305


Q12: Actual and Forecast Air Passengers
Actual
12-month forecast
95% prediction interval
Passengers (thousands)
700
600
500
400
300
200
100
1950
1952
1954
1956
1958
1960
1962
Year

Assignment 7 - One, Two and Multiple Variable Analysis

This assignment combines univariate summaries, binned frequencies, categorical cross-tabulations, and grouped statistics. It then explores numerical relationships with correlations, regression, faceted box plots, grouped error bars, a heatmap, and a scatterplot matrix.

Questions

1. Load the dataset shared via email

2. Compute summary statistics: Mean, Median, Minimum, Maximum, Range, Variance, Standard Deviation, Interquartile Range (IQR), Skewness (using e1071::skewness), and Kurtosis (using e1071::kurtosis).

3. Store these values in a single structured data.frame or named vector and print it.

4. Create a frequency distribution table and a relative frequency distribution table for a categorical variable. Generate a Pie Chart and a Bar Plot to represent the distribution.

5. Create a frequency distribution table with bin intervals for a numerical variable. Plot its Histogram and a Box Plot to check for outliers.

6. Choose two categorical variables: Construct a Contingency Table (Cross-tabulation) and compute the row/column percentages using prop.table()

7. Create a Stacked Bar Plot or Grouped Bar Plot to visualize the relationship.

8. Select one categorical variable and one numerical variable: Compute the mean, median, and standard deviation of the numerical variable grouped by each category using aggregate()

9. Generate Side-by-Side Box Plots and a Violin Plot to visualize the numerical distribution across categories.

10. Select two numerical variables: Calculate the Pearson and Spearman Correlation Coefficients (cor()) and Construct a Scatter Plot with a fitted linear regression trend line.

11. Select two categorical features and one numerical feature: Calculate summary statistics(Mean and Standard Deviation) for the numerical variable across all combinations of the two categorical variables and Create a Faceted Box Plot and a Grouped Bar Chart with Error Bars.

12. Select one categorical variable and two numerical variables: Construct a Scatter Plot ofthe two numerical features where points are colored/shaped by the categorical feature. Compute a correlation matrix between the two numerical variables separately for each category level.

13. Select at least 3 numerical features from the dataset: Generate a full Correlation Matrixand plot it as a Heatmap

14. Produce a Pairs Plot / Scatterplot Matrix to examine pairwise relationships and distributions simultaneously

Q1-Q3 Numerical summary statistics

Python code

```
df = load_horse()
pulse_clean = df["pulse"].dropna()
n = len(pulse_clean)
centered = pulse_clean - pulse_clean.mean()
sample_sd = pulse_clean.std(ddof=1)
summary_df = pd.DataFrame({"Statistic": ["Mean", "Median", "Minimum", "Maximum", "Range", "Variance", "SD", "IQR", "Skewness", "Kurtosis"],
    "Value": [pulse_clean.mean(), pulse_clean.median(), pulse_clean.min(), pulse_clean.max(),
              np.ptp(pulse_clean), pulse_clean.var(ddof=1), sample_sd,
              pulse_clean.quantile(.75) - pulse_clean.quantile(.25),
              (centered**3).mean() / sample_sd**3,
              (centered**4).mean() / sample_sd**4 - 3]})
show("Pulse statistics (e1071 default type 3 skewness and excess kurtosis)", summary_df)
```

Output

```
Pulse statistics (e1071 default type 3 skewness and excess kurtosis)
```

	Statistic	Value
0	Mean	75.544554
1	Median	64.000000
2	Minimum	38.000000
3	Maximum	164.000000
4	Range	126.000000
5	Variance	940.790495
6	SD	30.672308
7	IQR	44.000000
8	Skewness	0.937582
9	Kurtosis	0.089239

Q4-Q5 Frequency distributions

Python code

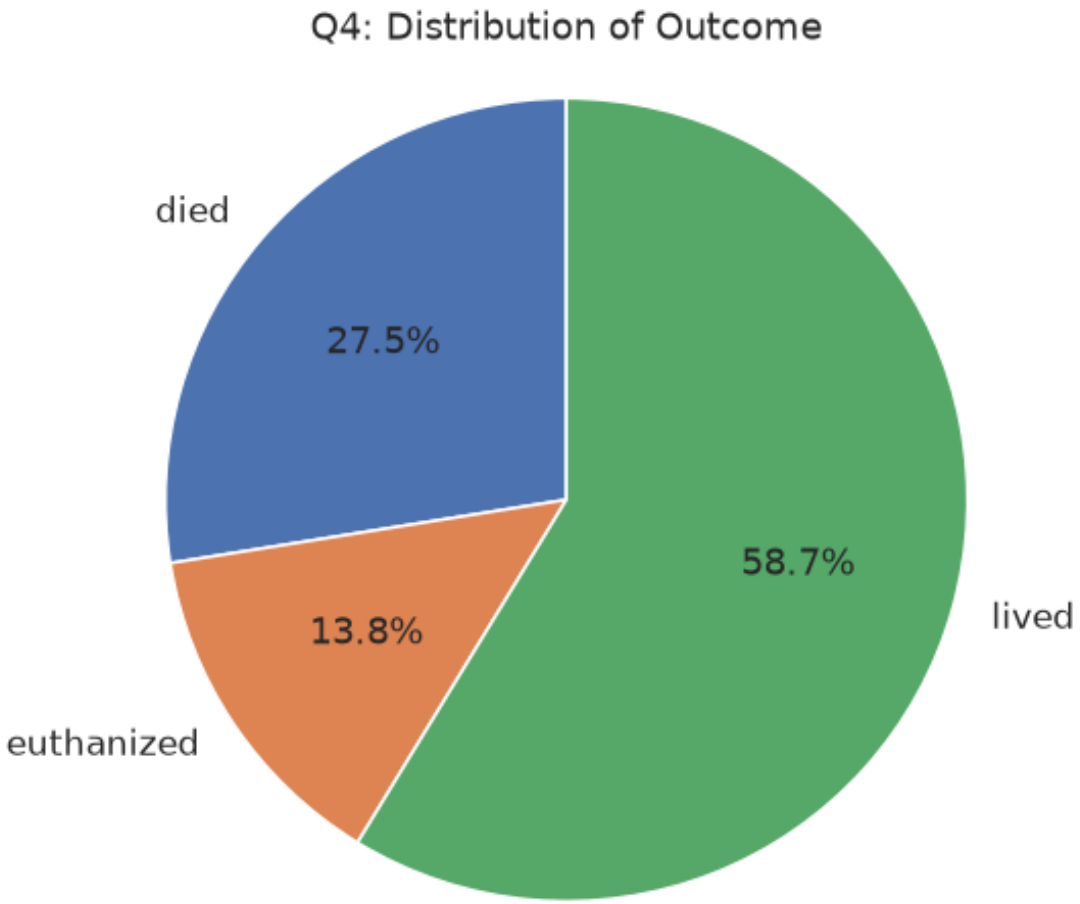
```
freq_table = df["outcome"].value_counts().sort_index()
rel_freq_table = freq_table / freq_table.sum()
show("Outcome frequencies", pd.DataFrame({"Frequency": freq_table, "Relative frequency": rel_freq_table}))
plt.pie(freq_table, labels=freq_table.index, autopct="%1.1f%%", startangle=90)
plt.axis("equal")
```

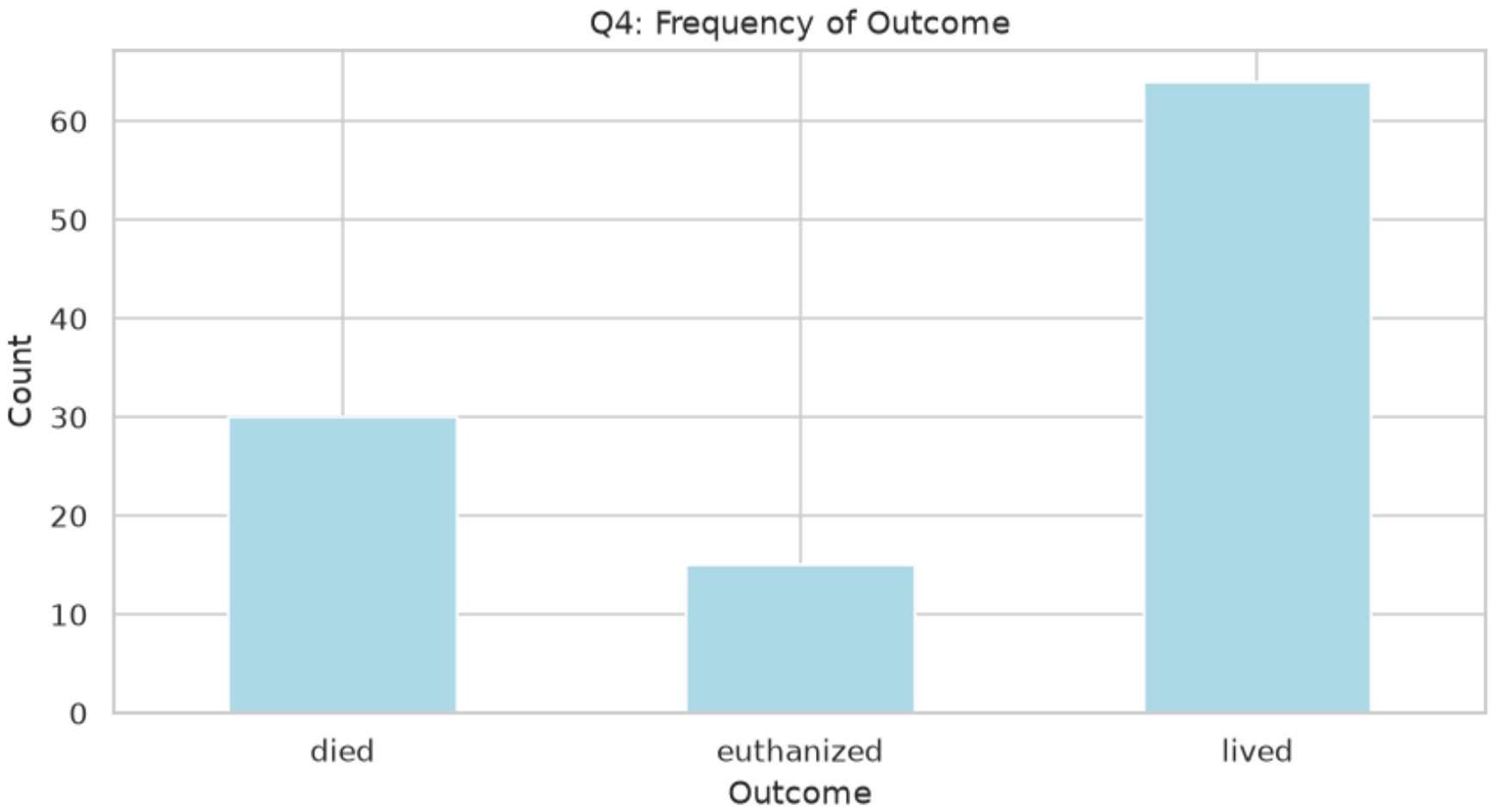
```
finish("Q4: Distribution of Outcome")
freq_table.plot.bar(rot=0, color="lightblue")
finish("Q4: Frequency of Outcome", "Outcome", "Count")
pulse_bins = pd.cut(df["pulse"], bins=np.arange(30, 191, 20), right=False)
pulse_freq = pulse_bins.value_counts(sort=False)
show("Q5: Pulse frequency by interval", pulse_freq)
print("Non-missing pulse values outside [30,190):", int((df["pulse"].notna() & pulse_bins.isna()).sum()))
sns.histplot(data=df, x="pulse", bins=np.arange(30, 191, 20), color="skyblue")
finish("Q5: Histogram of Pulse", "Pulse (beats/min)", "Count")
sns.boxplot(y=df["pulse"], color="lightgreen")
finish("Q5: Box Plot of Pulse", ylabel="Pulse (beats/min)")
```

Output

```
Outcome frequencies
```

	Frequency	Relative frequency
outcome		
died	30	0.275229
euthanized	15	0.137615
lived	64	0.587156

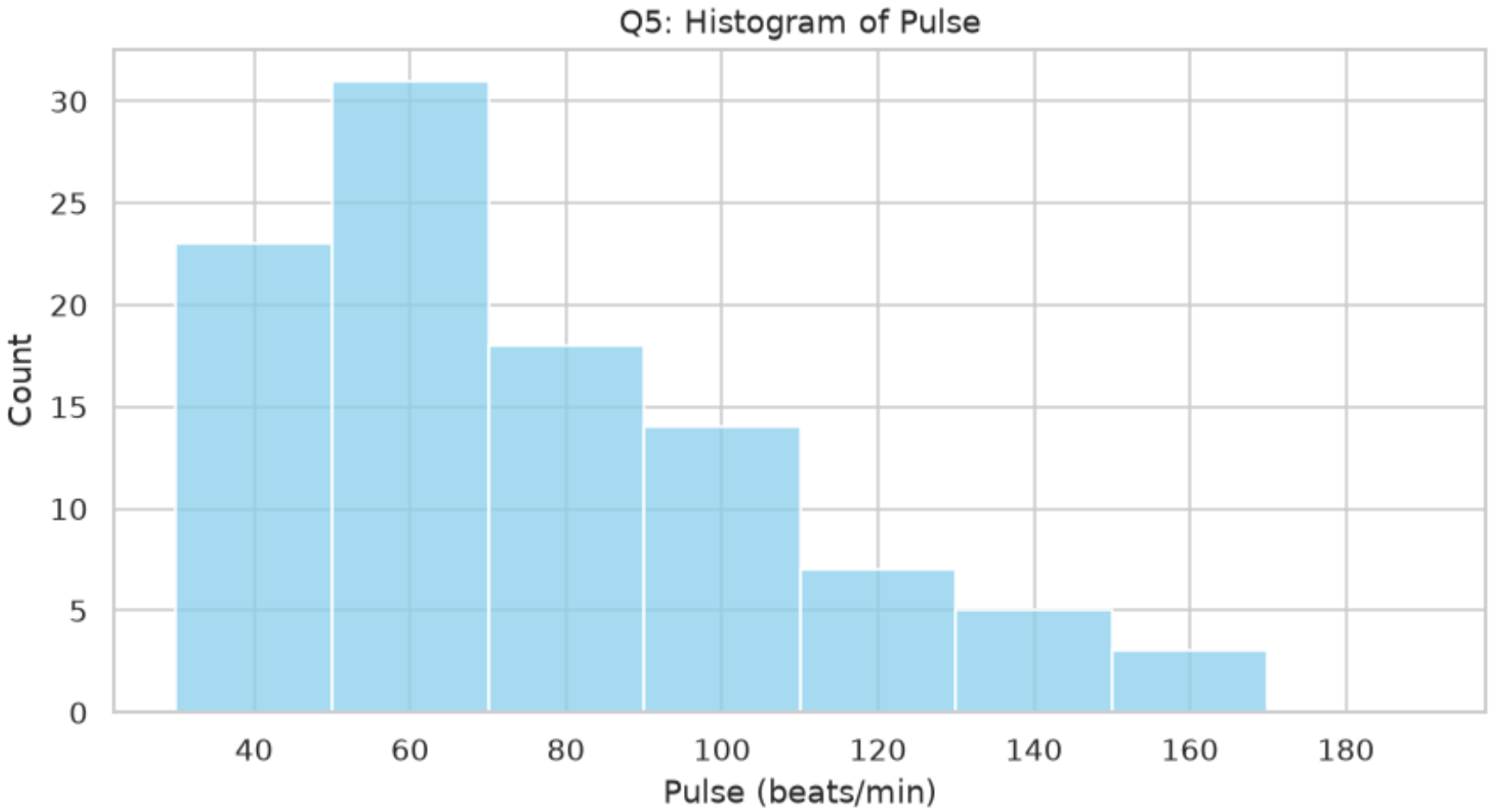

Q4: Distribution of Outcome
died
27.5%
58.7%
lived
13.8%
euthanized

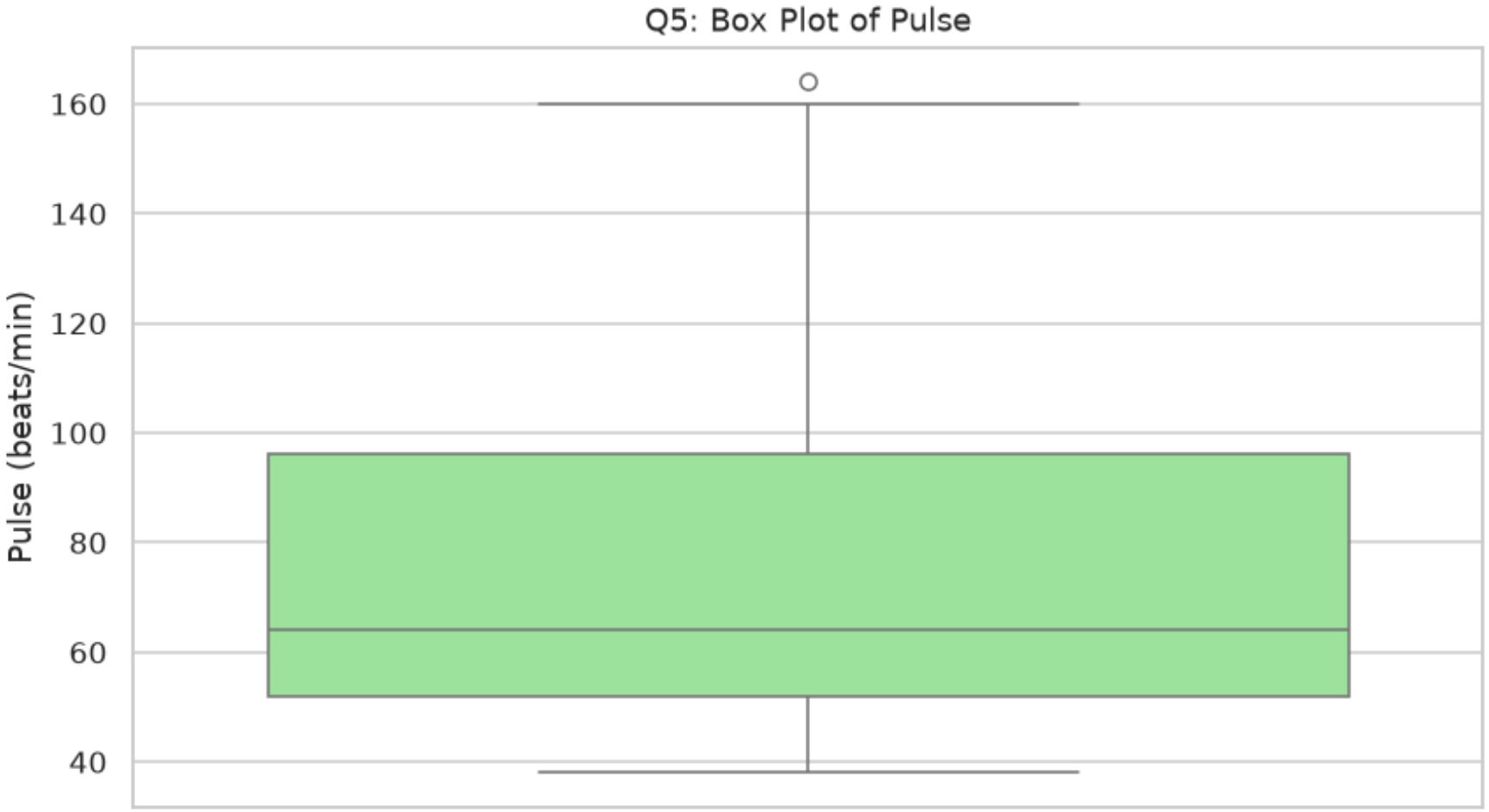



```
Q5: Pulse frequency by interval

pulse
[30, 50)      23
[50, 70)      31
[70, 90)      18
[90, 110)     14
[110, 130)     7
[130, 150)     5
[150, 170)     3
[170, 190)     0
Name: count, dtype: int64
```

```
Non-missing pulse values outside [30,190): 0
```





Q6-Q9 Cross-tabulations and grouped distributions

Python code

```
ctable = pd.crosstab(df["surgery"], df["outcome"])
show("Surgery × Outcome", ctable)
row_pct = ctable.div(ctable.sum(axis=1), axis=0) * 100
col_pct = ctable.div(ctable.sum(axis=0), axis=1) * 100
show("Row percentages", row_pct)
show("Column percentages", col_pct)
ctable.T.plot.bar(stacked=True, rot=0, color=["coral", "lightgreen"])
```

```
plt.legend(title="Surgery")
finish("Q7: Outcome by Surgery Status", "Outcome", "Count")
grouped_stats = df.groupby("outcome")["packed_cell_volume"].agg(Mean="mean", Median="median", SD="std")
show("Q8: Packed cell volume by outcome", grouped_stats)
sns.boxplot(data=df, x="outcome", y="packed_cell_volume", color="lightblue")
finish("Q9: Packed Cell Volume by Outcome", "Outcome", "Packed cell volume (%)")
sns.violinplot(data=df, x="outcome", y="packed_cell_volume", hue="outcome", cut=0, legend=False)
finish("Q9: Violin Plot of Packed Cell Volume", "Outcome", "Packed cell volume (%)")
```

Output

Surgery × Outcome

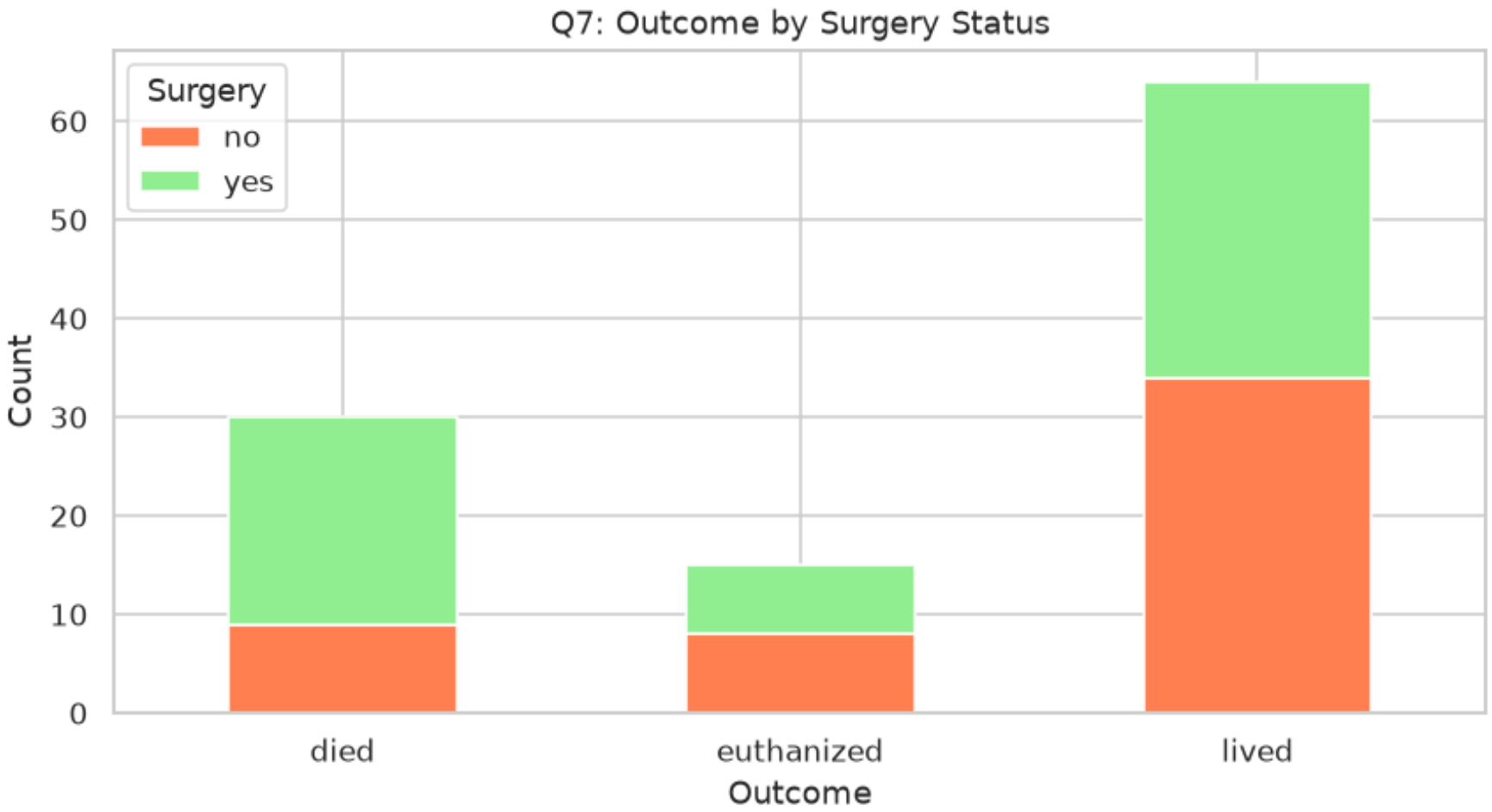
outcome	died	euthanized	lived
surgery			
no	9	8	34
yes	21	7	30

Row percentages

outcome	died	euthanized	lived
surgery			
no	17.647059	15.686275	66.666667
yes	36.206897	12.068966	51.724138

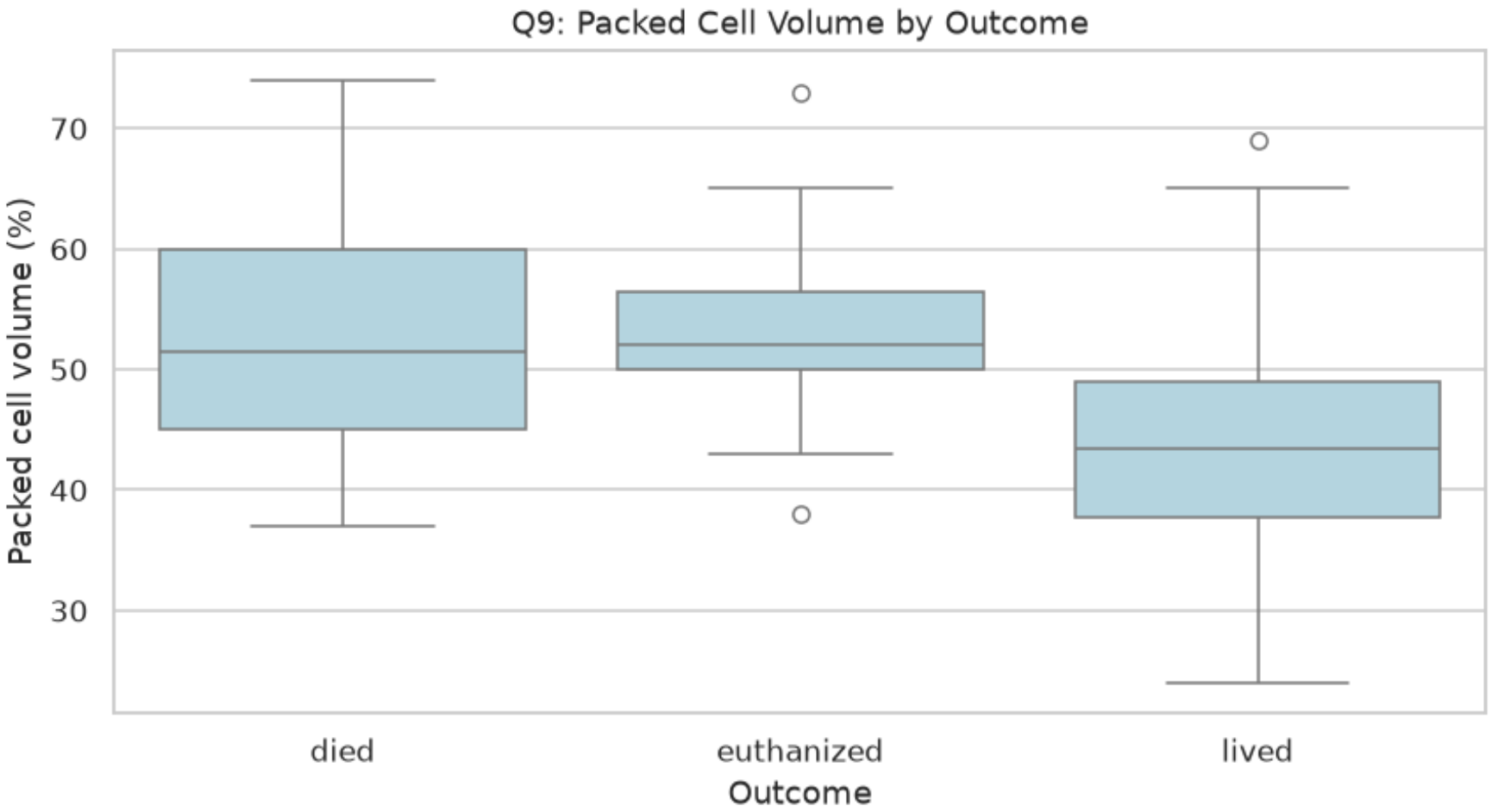
Column percentages

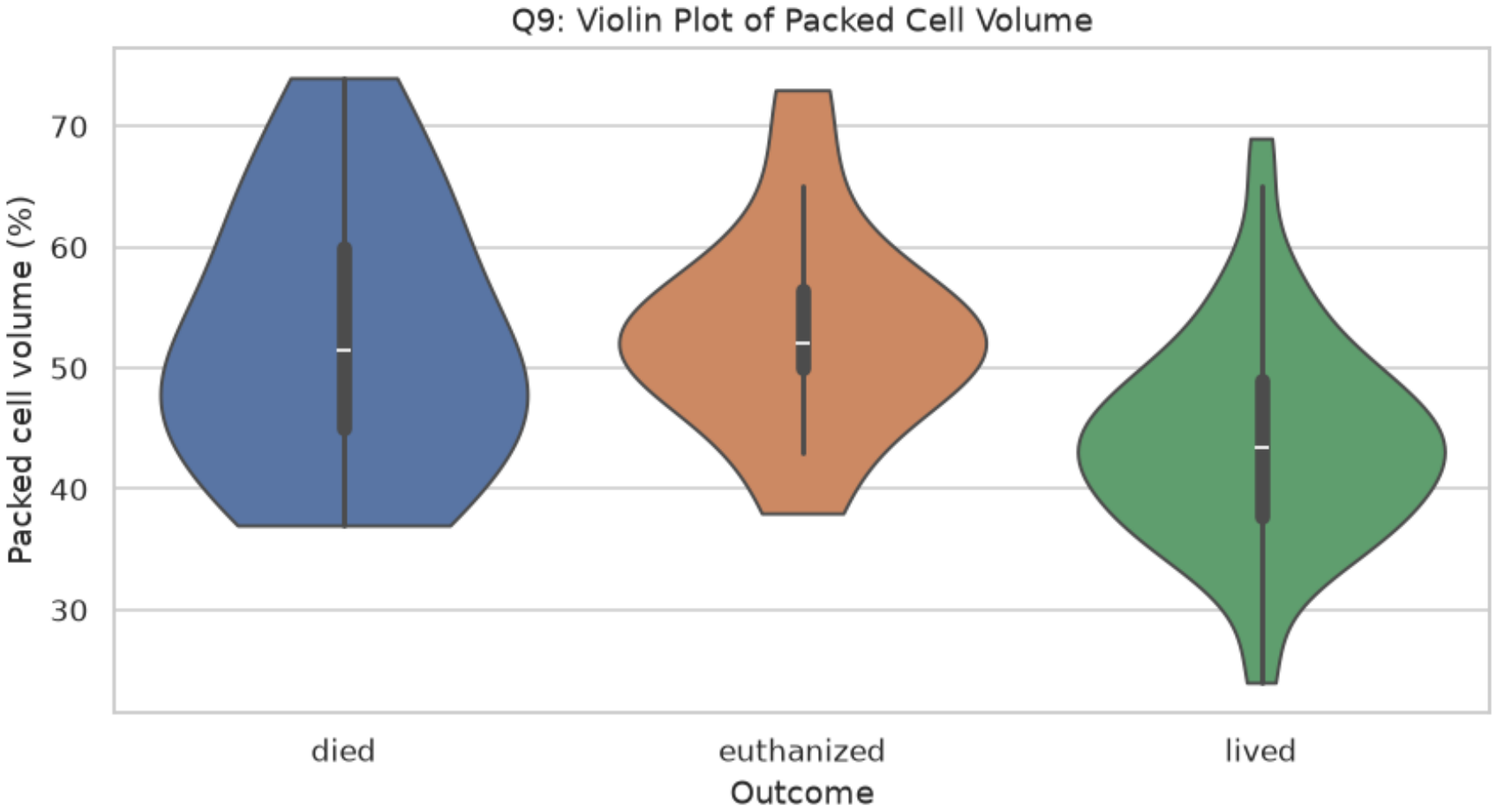
outcome	died	euthanized	lived
surgery			
no	30.0	53.333333	53.125
yes	70.0	46.666667	46.875



Q8: Packed cell volume by outcome

	Mean	Median	SD
outcome			
died	52.461538	51.5	10.955294
euthanized	53.142857	52.0	8.565199
lived	43.950000	43.5	8.387693


Q9: Packed Cell Volume by Outcome
Packed cell volume (%)
70
60
50
40
30
died
euthanized
lived
Outcome



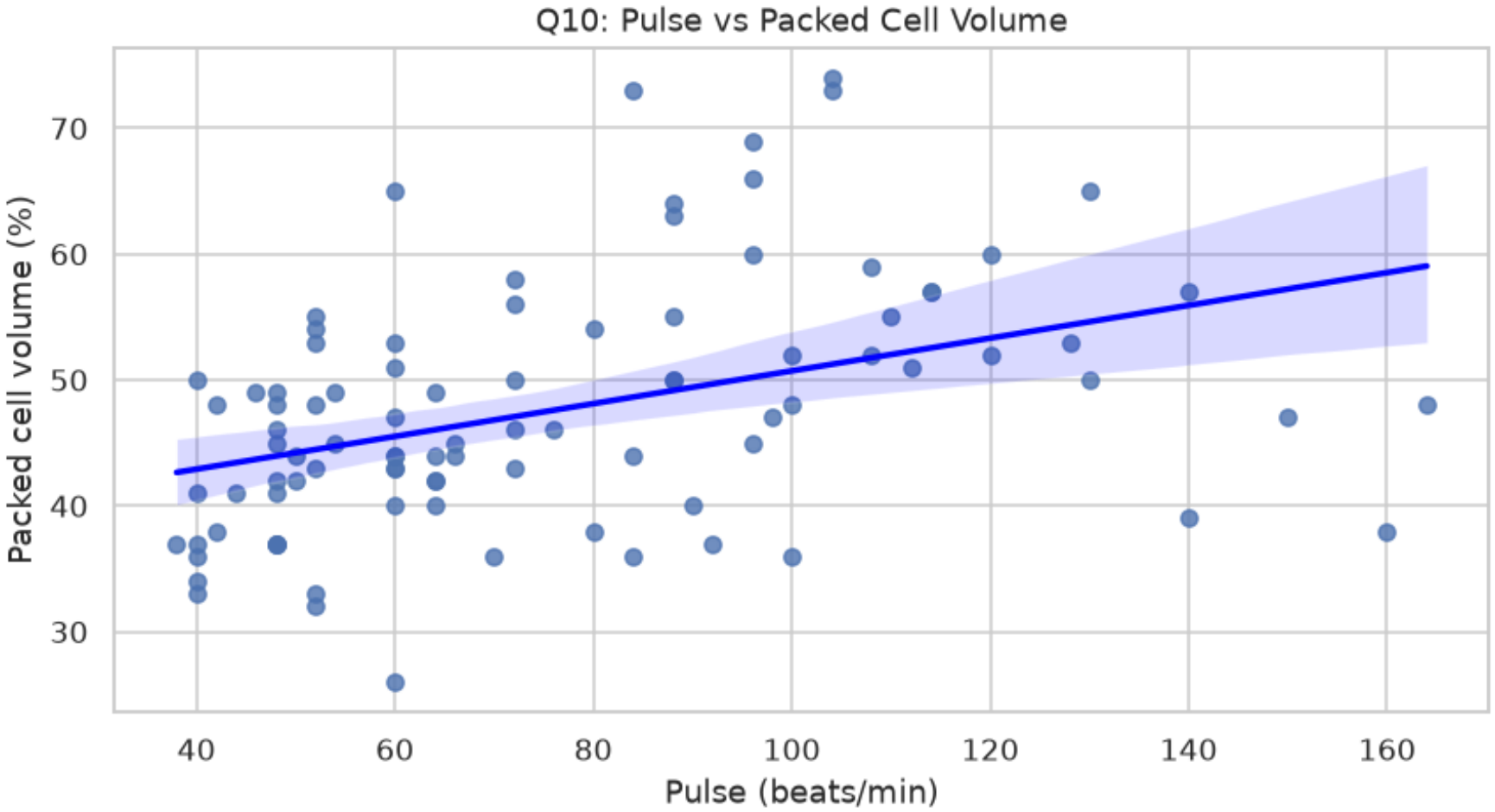
Q10 Correlation and fitted regression

Python code

```
num_pair = df[["pulse", "packed_cell_volume"]].dropna()
print("Pearson correlation:", num_pair.corr(method="pearson").iloc[0, 1])
print("Spearman correlation:", num_pair.corr(method="spearman").iloc[0, 1])
sns.regplot(data=num_pair, x="pulse", y="packed_cell_volume", line_kws={"color": "blue"})
finish("Q10: Pulse vs Packed Cell Volume", "Pulse (beats/min)", "Packed cell volume (%)")
```

Output

```
Pearson correlation: 0.4061701397352422
Spearman correlation: 0.4888465753691219
```



Q11 Two categorical variables and one numerical variable

Python code

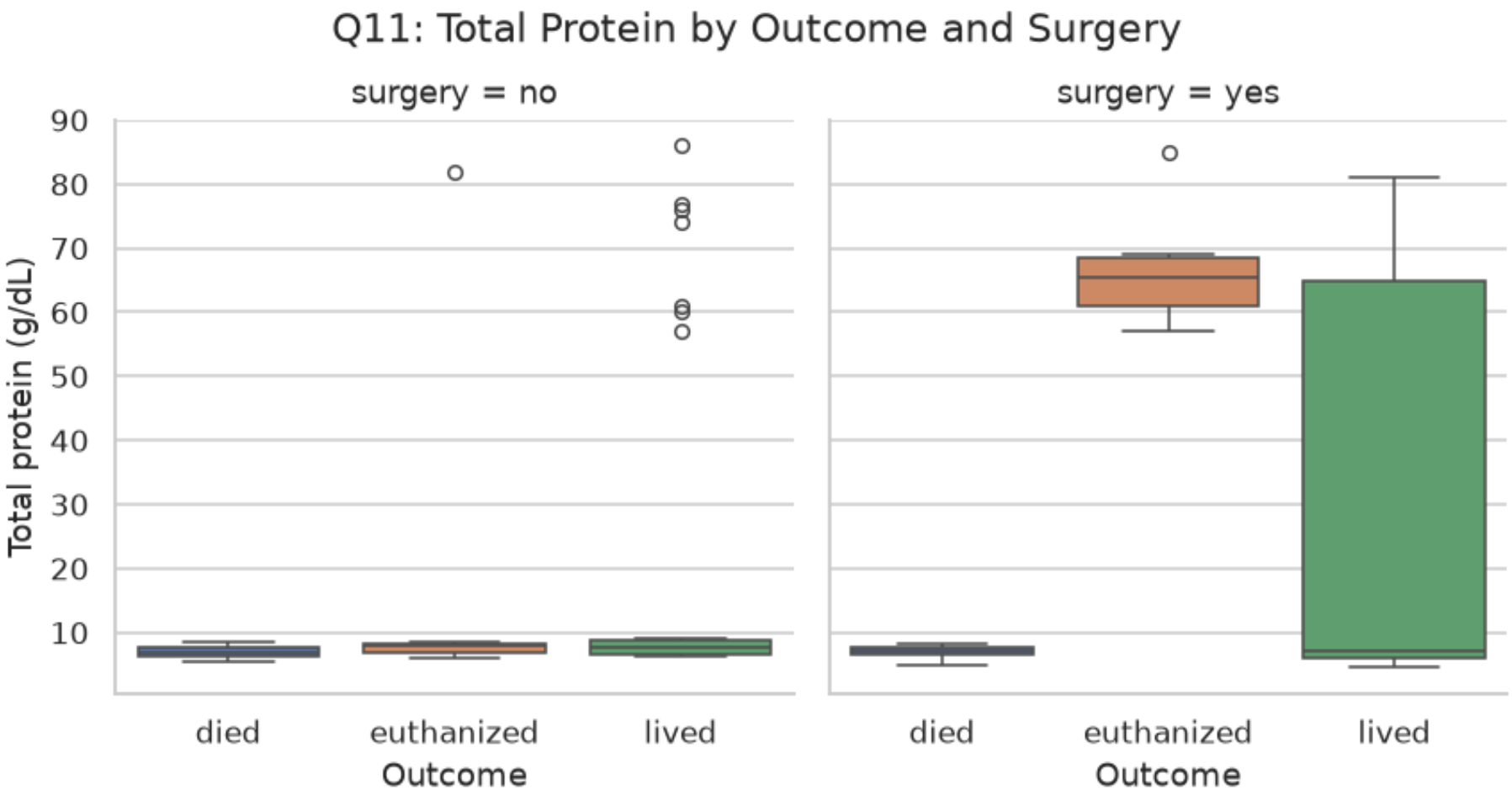
```
cat_stats_df = df.groupby(["surgery", "outcome"])["total_protein"].agg(Mean="mean", SD="std").reset_index()
show("Total protein by surgery and outcome", cat_stats_df)
g = sns.catplot(data=df.dropna(subset=["surgery", "outcome", "total_protein"]), x="outcome", y="total_protein", col="surgery", hue="outcome", kind="box",
    height=4)
```

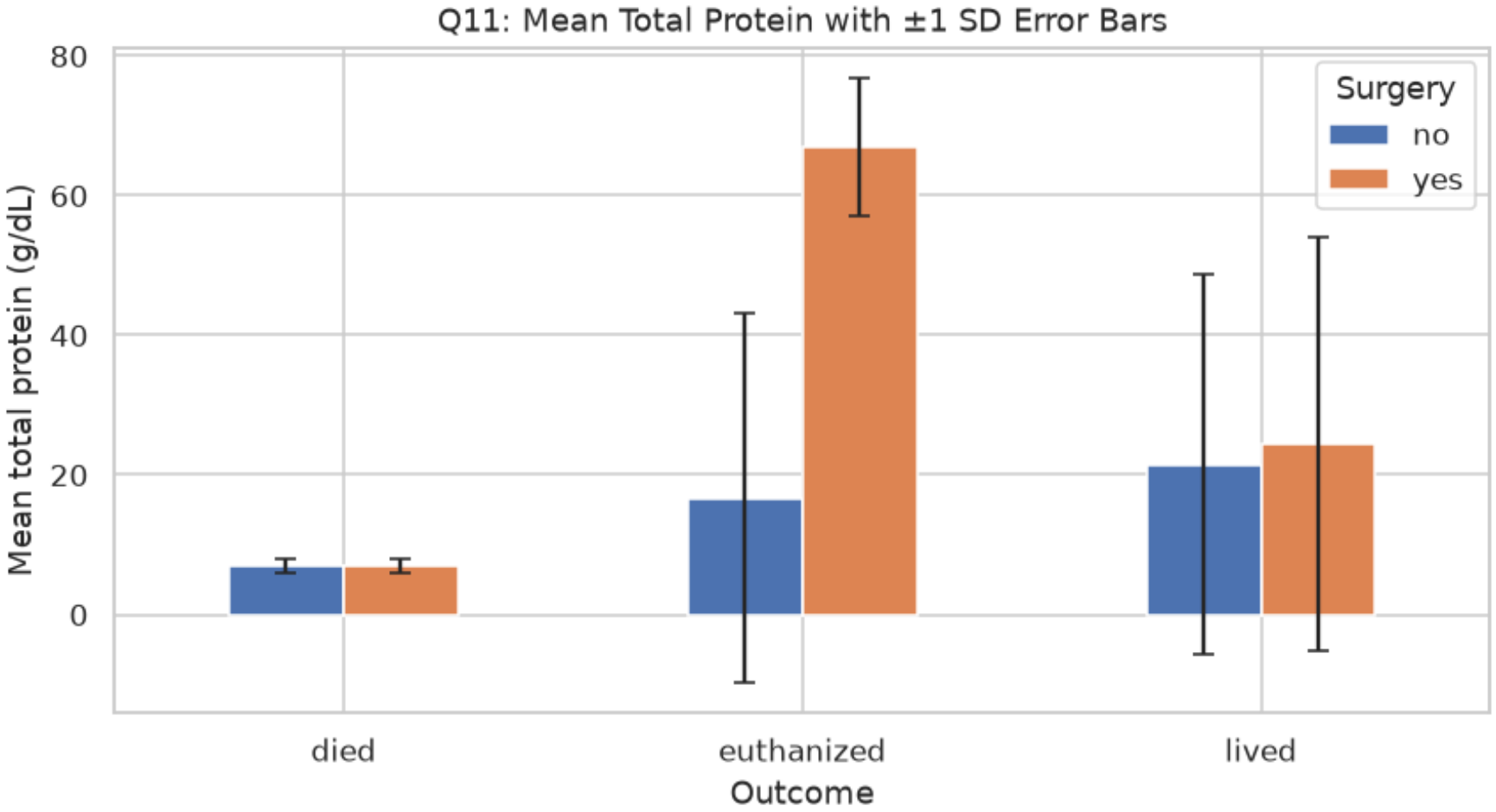
```
g.set_axis_labels("Outcome", "Total protein (g/dL)")
g.figure.suptitle("Q11: Total Protein by Outcome and Surgery", y=1.04)
plt.show()
means = cat_stats_df.pivot(index="outcome", columns="surgery", values="Mean")
errors = cat_stats_df.pivot(index="outcome", columns="surgery", values="SD").reindex(index=means.index, columns=means.columns)
means.plot.bar(yerr=errors, capsize=4, rot=0)
plt.legend(title="Surgery")
finish("Q11: Mean Total Protein with ±1 SD Error Bars", "Outcome", "Mean total protein (g/dL)")
```

Output

```
Total protein by surgery and outcome
```

	surgery	outcome	Mean	SD
0	no	died	7.000000	1.009950
1	no	euthanized	16.687500	26.405705
2	no	lived	21.487097	27.160360
3	yes	died	6.968750	0.935036
4	yes	euthanized	67.000000	9.859006
5	yes	lived	24.517241	29.601135


Q11: Total Protein by Outcome and Surgery
surgery = no
surgery = yes
Total protein (g/dL)
90
80
70
60
50
40
30
20
10
died
euthanized
lived
Outcome
died
euthanized
lived
Outcome



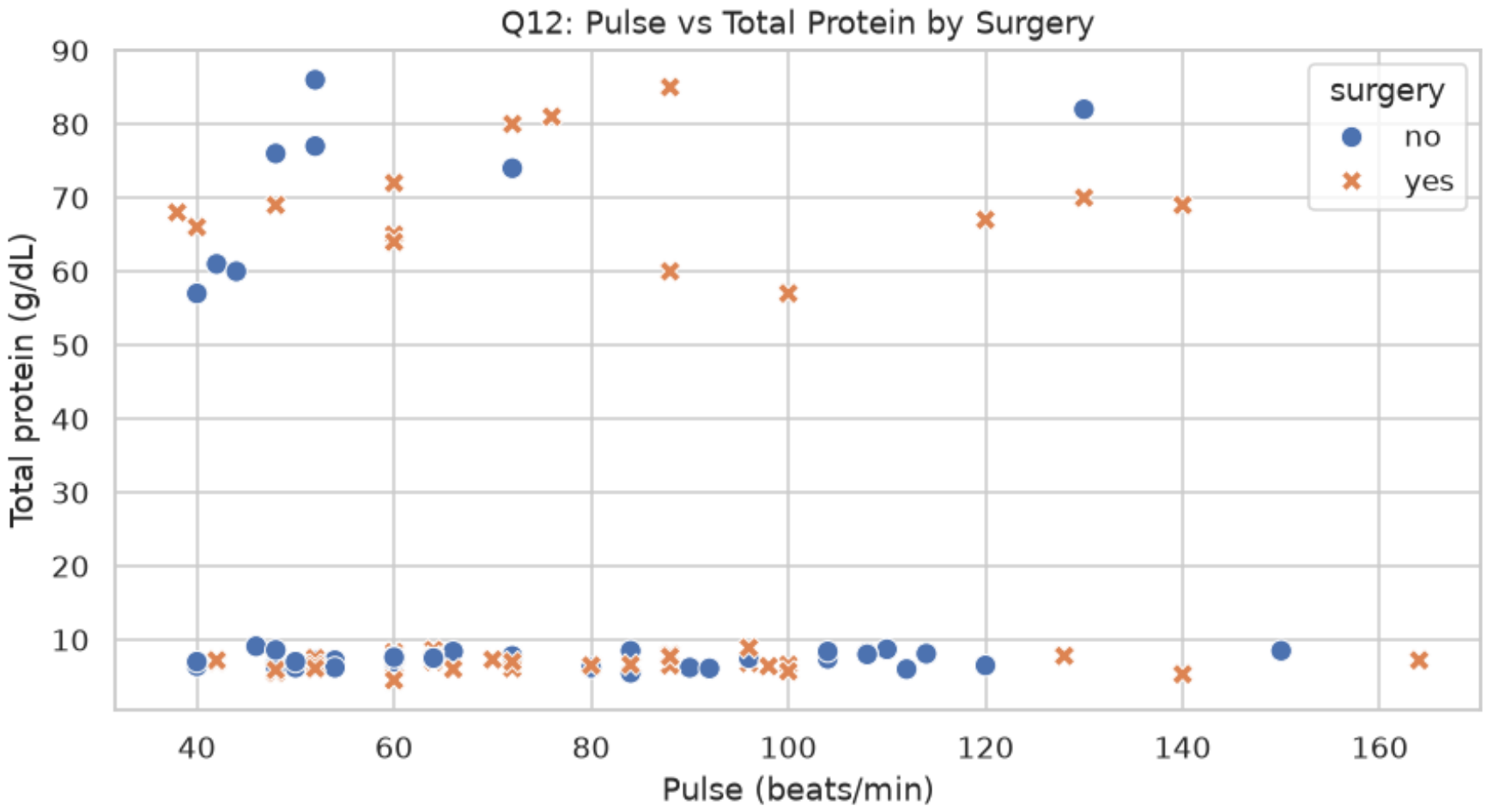
Q12-Q14 Group-specific correlations and multiple variables

Python code

```
df_sub12 = df[["surgery", "pulse", "total_protein"]].dropna()
sns.scatterplot(data=df_sub12, x="pulse", y="total_protein", hue="surgery", style="surgery", s=65)
finish("Q12: Pulse vs Total Protein by Surgery", "Pulse (beats/min)", "Total protein (g/dL)")
for surgery, group in df_sub12.groupby("surgery"):
    show(f"Q12: Correlation matrix for surgery={surgery}", group[["pulse", "total_protein"]].corr())
num_df = df[clinical_columns].dropna()
cor_mat = num_df.corr()
```

```
show("Q13: Full correlation matrix", cor_mat)
sns.heatmap(cor_mat, cmap="RdBu_r", vmin=-1, vmax=1, annot=True, fmt=".2f")
finish("Q13: Correlation Heatmap")
g = sns.pairplot(num_df, diag_kind="hist", plot_kws={"s": 25, "alpha": .6})
g.figure.suptitle("Q14: Scatterplot Matrix", y=1.01)
plt.show()
```

Output




```
Q12: Correlation matrix for surgery=no
```

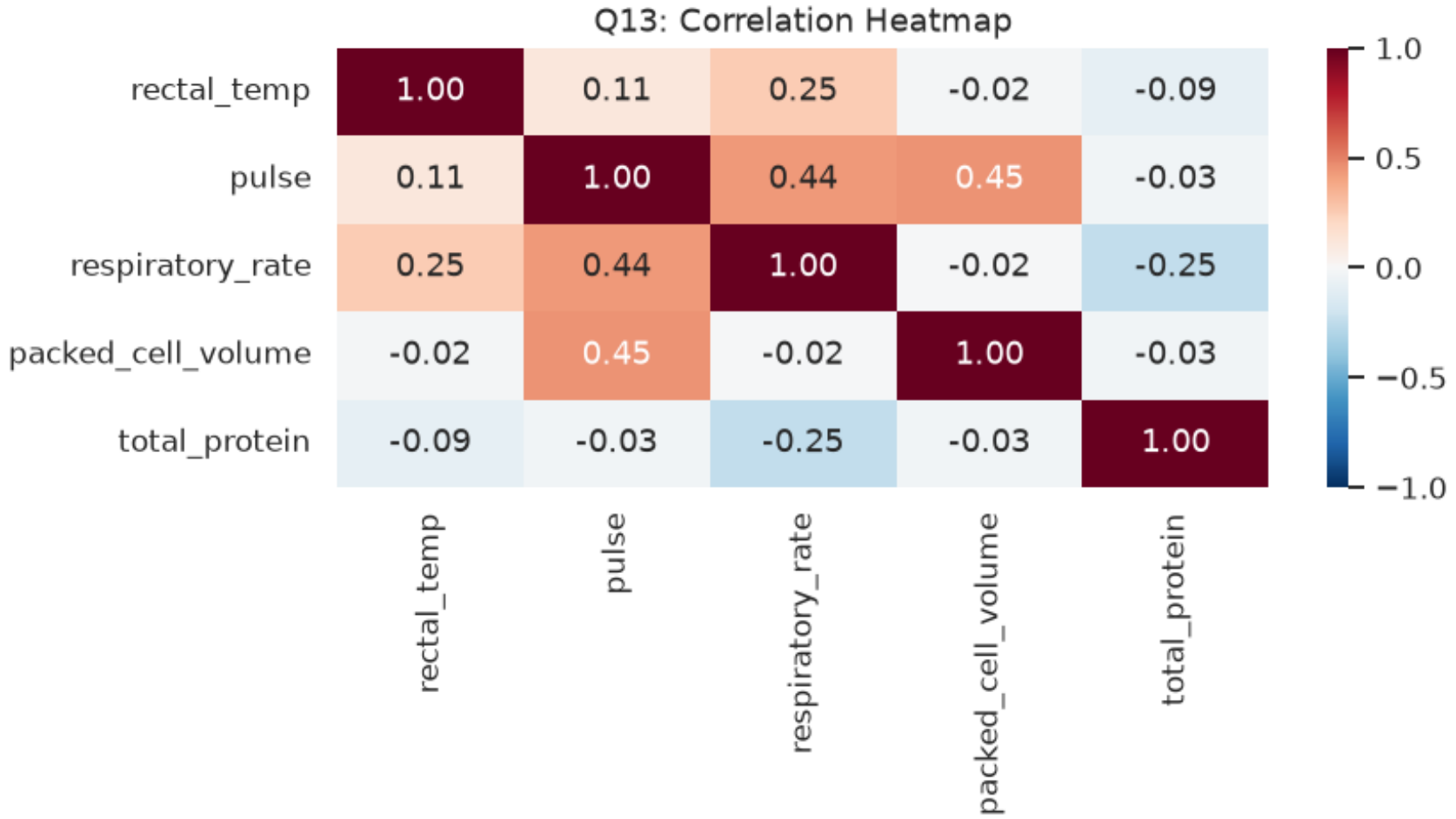
	pulse	total_protein
pulse	1.000000	-0.146503
total_protein	-0.146503	1.000000

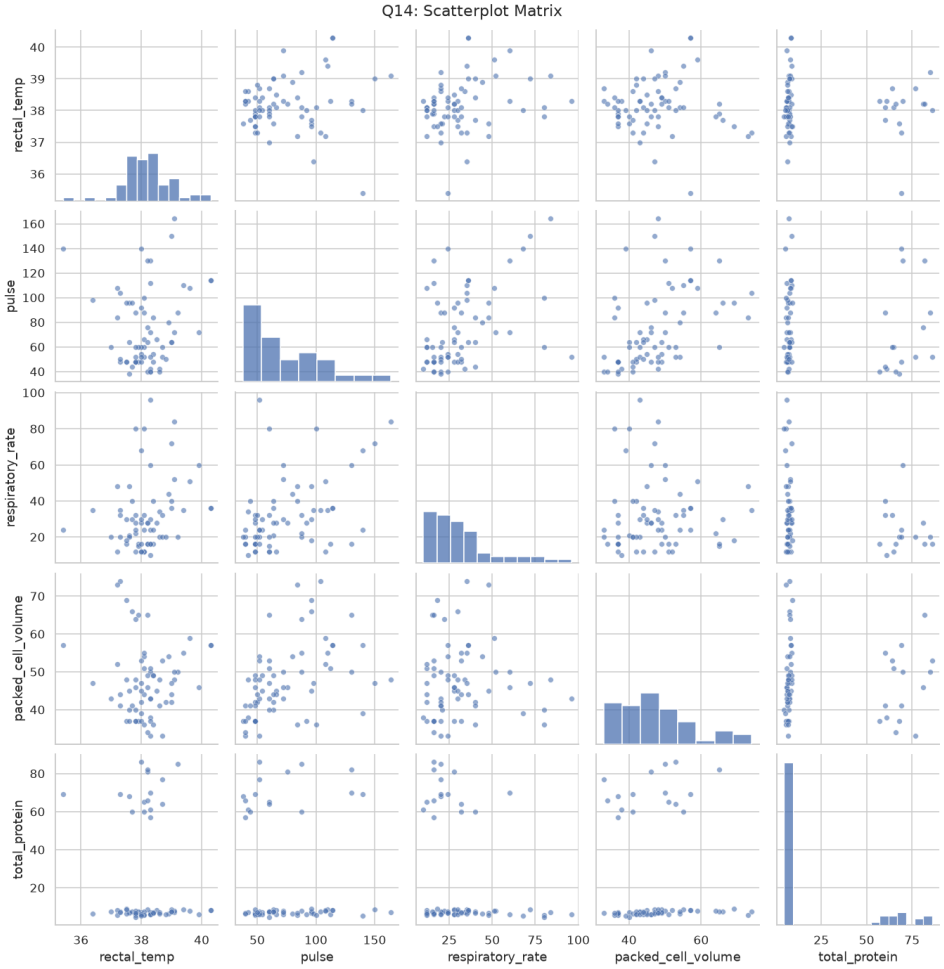
```
Q12: Correlation matrix for surgery=yes
```

	pulse	total_protein
pulse	1.000000	0.060133
total_protein	0.060133	1.000000

```
Q13: Full correlation matrix
```

	rectal_temp	pulse	respiratory_rate	packed_cell_volume	total_protein
rectal_temp	1.000000	0.110048	0.251167	-0.022367	-0.086338
pulse	0.110048	1.000000	0.436753	0.448625	-0.034349
respiratory_rate	0.251167	0.436753	1.000000	-0.015595	-0.246463
packed_cell_volume	-0.022367	0.448625	-0.015595	1.000000	-0.033432
total_protein	-0.086338	-0.034349	-0.246463	-0.033432	1.000000


Q13: Correlation Heatmap
rectal_temp
pulse
respiratory_rate
packed_cell_volume
total_protein
1.00
0.11
0.25
-0.02
-0.09
0.11
1.00
0.44
0.45
-0.03
0.25
0.44
1.00
-0.02
-0.25
-0.02
0.45
-0.02
1.00
-0.03
-0.09
-0.03
-0.25
-0.03
1.00
rectal_temp
pulse
respiratory_rate
packed_cell_volume
total_protein
1.0
0.5
0.0
−0.5
−1.0


Q14: Scatterplot Matrix
rectal_temp
pulse
respiratory_rate
packed_cell_volume
total_protein

Assignment 8 - K-Means and Hierarchical Clustering

This assignment prepares numeric data by removing the hospital identifier, imputing column means, and removing constant columns before standardization. It runs K-means, plots the WCSS elbow curve, and performs Euclidean Ward hierarchical clustering with three final clusters. The source solution's K=3 is retained as a choice to inspect against the elbow curve; outcome counts do not determine the unsupervised optimum.

Questions

K-Means Clustering

1. Load the dataset shared through e-mail.

2. View the dataset

3. Check missing values

4. Check for the dependent variable distribution. (table)

5. Select only numerical features for K-Means clustering (Others remove it).

6. Standardize the data. (scale)

7. Apply K-Means Clustering (kmeans) for center as 2.

8. Display K-Means results

9. Display cluster centers

10. Calculate WCSS for K = 1 to 10

11. Display the elbow plot.

12. Apply K-means clustering based on the elbow recommendation.

Hierarchical Clustering

1. Load the dataset shared through e-mail.

2. View the dataset

3. Check missing values

4. Select only numerical attributes

5. Standardize the data (scale)

6. Calculate Euclidean distance (dist)

7. Perform Hierarchical Clustering using Ward's method (hclust)

8. Plot the dendrogram. (plot)

9. Divide the data into 3 clusters and Display cluster assignments(cutree)

10. Display number of samples in each cluster. (table)

K-Means Q1-Q6 Load and prepare the data

Python code

```
data = load_horse()
show("Dataset preview", data.head(6))
data.info()
show("Missing values", data.isna().sum())
show("Outcome distribution", data["outcome"].value_counts())
num_data, scaled_data = prepare_clusters(data)
show("Numerical features used", num_data.head(6))
show("Standardized data (sample standard deviation, as in R scale)", scaled_data.head(6))
```

Output

```
Dataset preview
```

Table columns 1 of 3

	surgery	age	hospital_number	rectal_temp	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time	pain
0	no	adult	530101	38.5	66.0	28.0	cool	reduced	NaN	more_3_sec	extreme_pain
1	yes	adult	534817	39.2	88.0	20.0	NaN	NaN	pale_cyanotic	less_3_sec	mild_pain
2	no	adult	530334	38.3	40.0	24.0	normal	normal	pale_pink	less_3_sec	mild_pain
3	yes	young	5290409	39.1	164.0	84.0	cold	normal	dark_cyanotic	more_3_sec	depressed
4	no	adult	530255	37.3	104.0	35.0	NaN	NaN	dark_cyanotic	more_3_sec	NaN
5	no	adult	528355	NaN	NaN	NaN	warm	normal	pale_pink	less_3_sec	depressed

Table columns 2 of 3

	peristalsis	abdominal_distention	nasogastric_tube	nasogastric_reflux	nasogastric_reflux_ph	rectal_exam_feces	abdomen	packed_cell_volume	total_protein
0	absent	severe	NaN	NaN	NaN	decreased	distend_large	45.0	8.4

	peristalsis	abdominal_distention	nasogastric_tube	nasogastric_reflux	nasogastric_reflux_ph	rectal_exam_feces	abdomen	packed_cell_volume	total_protein
1	absent	slight	NaN	NaN	NaN	absent	other	50.0	85.0
2	hypomotile	none	NaN	NaN	NaN	normal	normal	33.0	6.7
3	absent	severe	none	less_1_liter	5.0	decreased	NaN	48.0	7.2
4	NaN	NaN	NaN	NaN	NaN	NaN	NaN	74.0	7.4
5	hypomotile	slight	slight	none	NaN	decreased	firm	NaN	NaN

Table columns 3 of 3

	abdomo_appearance	abdomo_protein	outcome	surgical_lesion	lesion_1	lesion_2	lesion_3	cp_data
0	NaN	NaN	died	no	11300.0	0.0	0.0	no
1	cloudy	2.0	euthanized	no	2208.0	0.0	0.0	no
2	NaN	NaN	lived	no	0.0	0.0	0.0	yes
3	serosanguious	5.3	died	yes	2208.0	0.0	0.0	yes
4	NaN	NaN	died	no	4300.0	0.0	0.0	no
5	NaN	NaN	lived	no	0.0	0.0	0.0	no

```
<class 'pandas.DataFrame'>
RangeIndex: 110 entries, 0 to 109
Data columns (total 28 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   surgery                110 non-null    str
 1   age                    110 non-null    str
 2   hospital_number        110 non-null    int64
 3   rectal_temp            85 non-null     float64
 4   pulse                  101 non-null    float64
 5   respiratory_rate       87 non-null     float64
 6   temp_of_extremities    90 non-null     str
 7   peripheral_pulse       83 non-null     str
 8   mucous_membrane        94 non-null     str
 9   capillary_refill_time  99 non-null     str
 10  pain                   90 non-null     str
 11  peristalsis            91 non-null     str
 12  abdominal_distention   86 non-null     str
 13  nasogastric_tube       73 non-null     str
 14  nasogastric_reflux     72 non-null     str
 15  nasogastric_reflux_ph  21 non-null     float64
 16  rectal_exam_feces      80 non-null     str
 17  abdomen                74 non-null     str
```

```
 18  packed_cell_volume     101 non-null    float64
 19  total_protein          99 non-null     float64
 20  abdomo_appearance      49 non-null     str
 21  abdomo_protein         38 non-null     float64
 22  outcome                109 non-null    str
 23  surgical_lesion        109 non-null    str
 24  lesion_1               109 non-null    float64
 25  lesion_2               109 non-null    float64
 26  lesion_3               109 non-null    float64
 27  cp_data                109 non-null    str
dtypes: float64(10), int64(1), str(17)
memory usage: 24.2 KB
Missing values

surgery                   0
age                       0
hospital_number           0
rectal_temp              25
pulse                     9
respiratory_rate         23
temp_of_extremities      20
peripheral_pulse         27
mucous_membrane          16
capillary_refill_time    11
pain                     20
peristalsis              19
abdominal_distention     24
nasogastric_tube         37
nasogastric_reflux       38
nasogastric_reflux_ph    89
rectal_exam_feces        30
abdomen                  36
packed_cell_volume        9
total_protein            11
abdomo_appearance        61
abdomo_protein           72
outcome                   1
surgical_lesion           1
lesion_1                  1
lesion_2                  1
lesion_3                  1
cp_data                   1
dtype: int64

Outcome distribution

outcome
lived         64
died          30
euthanized    15
Name: count, dtype: int64
```

Numerical features used

	rectal_temp	pulse	respiratory_rate	nasogastric_reflux_ph	packed_cell_volume	total_protein	abdomo_protein	lesion_1	lesion_2
0	38.500000	66.000000	28.000000	4.87619	45.000000	8.400000	3.184211	11300.0	0.0
1	39.200000	88.000000	20.000000	4.87619	50.000000	85.000000	2.000000	2208.0	0.0
2	38.300000	40.000000	24.000000	4.87619	33.000000	6.700000	3.184211	0.0	0.0
3	39.100000	164.000000	84.000000	5.00000	48.000000	7.200000	5.300000	2208.0	0.0
4	37.300000	104.000000	35.000000	4.87619	74.000000	7.400000	3.184211	4300.0	0.0
5	38.149412	75.544554	31.954023	4.87619	47.336634	21.077778	3.184211	0.0	0.0

Standardized data (sample standard deviation, as in R scale)

	rectal_temp	pulse	respiratory_rate	nasogastric_reflux_ph	packed_cell_volume	total_protein	abdomo_protein	lesion_1	lesion_2
0	5.098825e-01	-3.248796e-01	-2.362446e-01	3.067223e-15	-0.243023	-0.499604	1.279204e-15	2.345588	-0.096225
1	1.527936e+00	4.239611e-01	-7.142279e-01	3.067223e-15	0.277005	2.519038	-1.137044e+00	-0.259830	-0.096225
2	2.190099e-01	-1.209873e+00	-4.752362e-01	3.067223e-15	-1.491092	-0.566597	1.279204e-15	-0.892558	-0.096225
3	1.382500e+00	3.010866e+00	3.109638e+00	1.425207e-01	0.068994	-0.546894	2.031519e+00	-0.259830	-0.096225
4	-1.235353e+00	9.685726e-01	1.819908e-01	3.067223e-15	2.773143	-0.539012	1.279204e-15	0.339657	-0.096225
5	1.033387e-14	4.837121e-16	-2.122672e-16	3.067223e-15	0.000000	0.000000	1.279204e-15	-0.892558	-0.096225

K-Means Q7-Q9 Fit two clusters and inspect centers

Python code

```
kmeans_k2 = KMeans(n_clusters=2, n_init=25, random_state=42).fit(scaled_data)
print(kmeans_k2)
print("Iterations:", kmeans_k2.n_iter_)
print("Total within-cluster sum of squares:", kmeans_k2.inertia_)
show("Cluster assignments (1-based labels)", pd.Series(kmeans_k2.labels_ + 1, index=data.index, name="Cluster"))
show("Cluster sizes", pd.Series(kmeans_k2.labels_ + 1).value_counts().sort_index())
show("Standardized cluster centers", pd.DataFrame(kmeans_k2.cluster_centers_, columns=scaled_data.columns, index=["Cluster 1", "Cluster 2"]))
```

Output

```
KMeans(n_clusters=2, n_init=25, random_state=42)
Iterations: 7
Total within-cluster sum of squares: 844.450618140584
```

```
Cluster assignments (1-based labels)

0      2
1      1
2      2
3      2
4      2
      ..
105    2
106    2
107    2
108    1
109    2
Name: Cluster, Length: 110, dtype: int32

Cluster sizes

1    22
2    88
Name: count, dtype: int64

Standardized cluster centers
```

	rectal_temp	pulse	respiratory_rate	nasogastric_reflux_ph	packed_cell_volume	total_protein	abdomo_protein	lesion_1	lesion_2
Cluster 1	-0.104994	-0.095895	-0.340070	-0.706624	-0.072832	1.938668	-0.560482	-0.327979	0.380526
Cluster 2	0.026249	0.023974	0.085017	0.176656	0.018208	-0.484667	0.140121	0.081995	-0.095132

K-Means Q10-Q12 Elbow curve and three-cluster solution

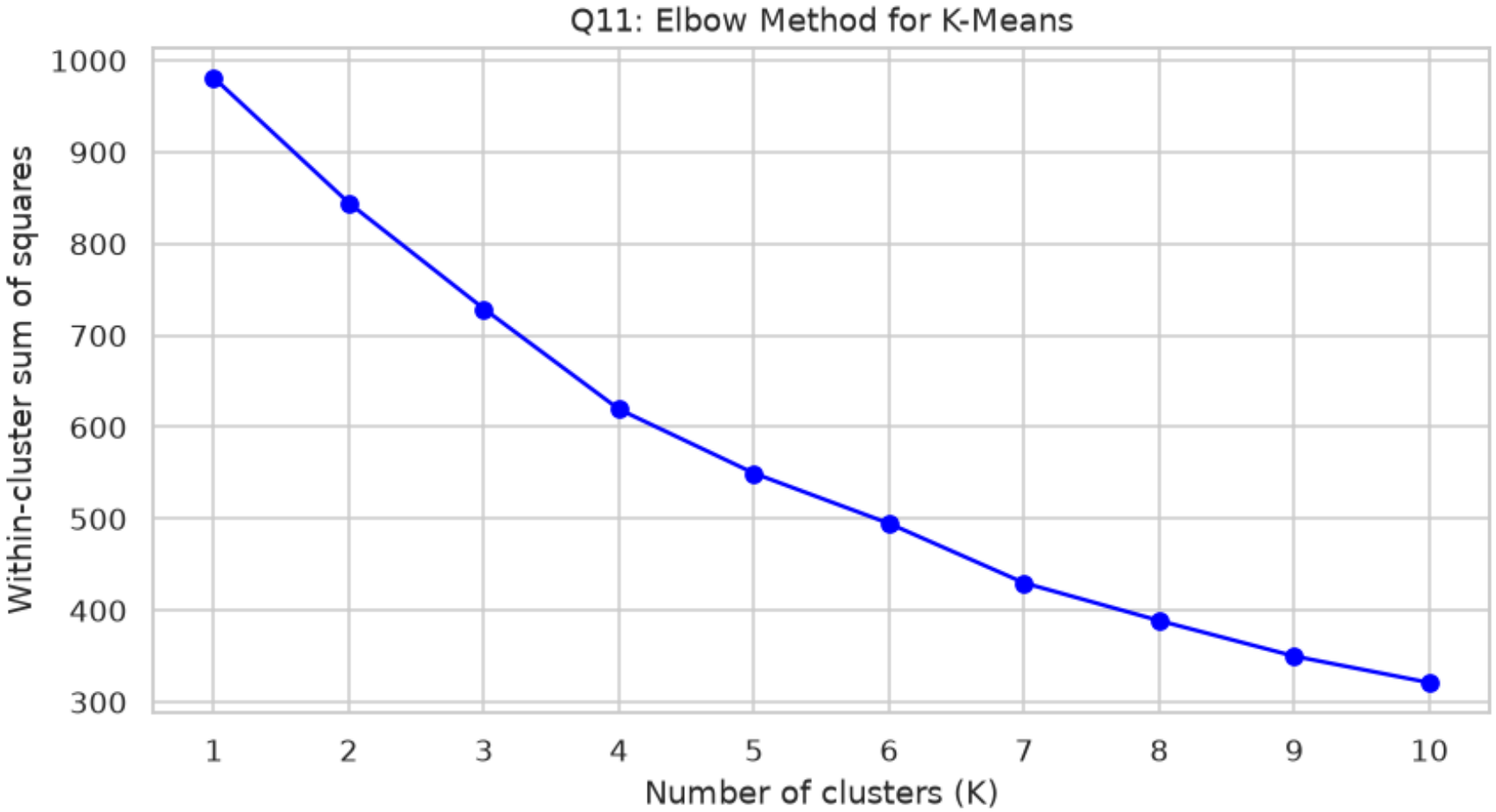
Python code

```
wcss = [KMeans(n_clusters=k, n_init=25, random_state=42).fit(scaled_data).inertia_ for k in range(1, 11)]
wcss_results = pd.DataFrame({"K": range(1, 11), "WCSS": wcss})
show("WCSS for K=1 to 10", wcss_results)
plt.plot(wcss_results["K"], wcss_results["WCSS"], "o-", color="blue")
plt.xticks(range(1, 11))
finish("Q11: Elbow Method for K-Means", "Number of clusters (K)", "Within-cluster sum of squares")
chosen_k = 3
kmeans_optimal = KMeans(n_clusters=chosen_k, n_init=25, random_state=42).fit(scaled_data)
print("Q12: K=3 retained from the submitted R solution; inspect the elbow plot to assess this choice.")
print(kmeans_optimal)
print("WCSS:", kmeans_optimal.inertia_)
show("Cluster sizes", pd.Series(kmeans_optimal.labels_ + 1).value_counts().sort_index())
show("Cluster centers", pd.DataFrame(kmeans_optimal.cluster_centers_, columns=scaled_data.columns, index=[f"Cluster {i}" for i in range(1, chosen_k + 1)]))
```

Output

WCSS for K=1 to 10

	K	WCSS
0	1	981.000000
1	2	844.450618
2	3	728.934083
3	4	619.333438
4	5	549.445005
5	6	494.692080
6	7	429.794520
7	8	388.657149
8	9	349.890482
9	10	321.053288




```
Q12: K=3 retained from the submitted R solution; inspect the elbow plot to assess this choice.
KMeans(n_clusters=3, n_init=25, random_state=42)
WCSS: 728.9340825083096
Cluster sizes

1    34
2    75
3     1
Name: count, dtype: int64

Cluster centers
```

	rectal_temp	pulse	respiratory_rate	nasogastric_reflux_ph	packed_cell_volume	total_protein	abdomo_protein	lesion_1	lesion_2
Cluster 1	0.422017	0.851986	0.763412	3.127717e-01	0.620642	-0.423184	0.445454	0.486613	-0.096225
Cluster 2	-0.192295	-0.410948	-0.333370	-1.417898e-01	-0.305852	0.159832	-0.186778	-0.217137	-0.094942
Cluster 3	0.073574	1.853566	-0.953220	3.067223e-15	1.837091	2.400814	-1.137044	-0.259543	10.392305

Hierarchical Clustering Q1-Q7 Prepare distances and apply Ward linkage

Python code

```
hc_data = load_horse()
show("Dataset preview", hc_data.head(6))
hc_data.info()
show("Missing values", hc_data.isna().sum())
num_hc, scaled_hc = prepare_clusters(hc_data)
show("Numerical attributes", num_hc.head(6))
show("Standardized attributes", scaled_hc.head(6))
dist_matrix = pdist(scaled_hc.to_numpy(), metric="euclidean")
print("Number of pairwise Euclidean distances:", len(dist_matrix))
show("First 10 pairwise distances", pd.Series(dist_matrix).head(10))
hc_ward = linkage(dist_matrix, method="ward")
show("Ward linkage matrix (first six merges)", pd.DataFrame(hc_ward, columns=["Left", "Right", "Distance", "Samples"]).head(6))
```

Output

```
Dataset preview
```

Table columns 1 of 3

	surgery	age	hospital_number	rectal_temp	pulse	respiratory_rate	temp_of_extremities	peripheral_pulse	mucous_membrane	capillary_refill_time	pain
0	no	adult	530101	38.5	66.0	28.0	cool	reduced	NaN	more_3_sec	extreme_pain
1	yes	adult	534817	39.2	88.0	20.0	NaN	NaN	pale_cyanotic	less_3_sec	mild_pain
2	no	adult	530334	38.3	40.0	24.0	normal	normal	pale_pink	less_3_sec	mild_pain
3	yes	young	5290409	39.1	164.0	84.0	cold	normal	dark_cyanotic	more_3_sec	depressed
4	no	adult	530255	37.3	104.0	35.0	NaN	NaN	dark_cyanotic	more_3_sec	NaN
5	no	adult	528355	NaN	NaN	NaN	warm	normal	pale_pink	less_3_sec	depressed

Table columns 2 of 3

	peristalsis	abdominal_distention	nasogastric_tube	nasogastric_reflux	nasogastric_reflux_ph	rectal_exam_feces	abdomen	packed_cell_volume	total_protein
0	absent	severe	NaN	NaN	NaN	decreased	distend_large	45.0	8.4
1	absent	slight	NaN	NaN	NaN	absent	other	50.0	85.0
2	hypomotile	none	NaN	NaN	NaN	normal	normal	33.0	6.7
3	absent	severe	none	less_1_liter	5.0	decreased	NaN	48.0	7.2
4	NaN	NaN	NaN	NaN	NaN	NaN	NaN	74.0	7.4
5	hypomotile	slight	slight	none	NaN	decreased	firm	NaN	NaN

Table columns 3 of 3

	abdomo_appearance	abdomo_protein	outcome	surgical_lesion	lesion_1	lesion_2	lesion_3	cp_data
0	NaN	NaN	died	no	11300.0	0.0	0.0	no
1	cloudy	2.0	euthanized	no	2208.0	0.0	0.0	no
2	NaN	NaN	lived	no	0.0	0.0	0.0	yes
3	serosanguious	5.3	died	yes	2208.0	0.0	0.0	yes
4	NaN	NaN	died	no	4300.0	0.0	0.0	no
5	NaN	NaN	lived	no	0.0	0.0	0.0	no

```
<class 'pandas.DataFrame'>
RangeIndex: 110 entries, 0 to 109
Data columns (total 28 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   surgery                110 non-null    str
 1   age                    110 non-null    str
 2   hospital_number        110 non-null    int64
 3   rectal_temp            85 non-null     float64
 4   pulse                  101 non-null    float64
 5   respiratory_rate       87 non-null     float64
 6   temp_of_extremities    90 non-null     str
 7   peripheral_pulse       83 non-null     str
 8   mucous_membrane        94 non-null     str
 9   capillary_refill_time  99 non-null     str
 10  pain                   90 non-null     str
 11  peristalsis            91 non-null     str
 12  abdominal_distention   86 non-null     str
 13  nasogastric_tube       73 non-null     str
 14  nasogastric_reflux     72 non-null     str
 15  nasogastric_reflux_ph  21 non-null     float64
```

```
 16  rectal_exam_feces      80 non-null     str
 17  abdomen                74 non-null     str
 18  packed_cell_volume     101 non-null    float64
 19  total_protein          99 non-null     float64
 20  abdomo_appearance      49 non-null     str
 21  abdomo_protein         38 non-null     float64
 22  outcome                109 non-null    str
 23  surgical_lesion        109 non-null    str
 24  lesion_1               109 non-null    float64
 25  lesion_2               109 non-null    float64
 26  lesion_3               109 non-null    float64
 27  cp_data                109 non-null    str
dtypes: float64(10), int64(1), str(17)
memory usage: 24.2 KB
Missing values

surgery                   0
age                       0
hospital_number           0
rectal_temp              25
pulse                     9
respiratory_rate         23
temp_of_extremities      20
peripheral_pulse         27
mucous_membrane          16
capillary_refill_time    11
pain                     20
peristalsis              19
abdominal_distention     24
nasogastric_tube         37
nasogastric_reflux       38
nasogastric_reflux_ph    89
rectal_exam_feces        30
abdomen                  36
packed_cell_volume        9
total_protein            11
abdomo_appearance        61
abdomo_protein           72
outcome                   1
surgical_lesion           1
lesion_1                  1
lesion_2                  1
lesion_3                  1
cp_data                   1
dtype: int64

Numerical attributes
```

	rectal_temp	pulse	respiratory_rate	nasogastric_reflux_ph	packed_cell_volume	total_protein	abdomo_protein	lesion_1	lesion_2
0	38.500000	66.000000	28.000000	4.87619	45.000000	8.400000	3.184211	11300.0	0.0
1	39.200000	88.000000	20.000000	4.87619	50.000000	85.000000	2.000000	2208.0	0.0
2	38.300000	40.000000	24.000000	4.87619	33.000000	6.700000	3.184211	0.0	0.0
3	39.100000	164.000000	84.000000	5.00000	48.000000	7.200000	5.300000	2208.0	0.0
4	37.300000	104.000000	35.000000	4.87619	74.000000	7.400000	3.184211	4300.0	0.0
5	38.149412	75.544554	31.954023	4.87619	47.336634	21.077778	3.184211	0.0	0.0

Standardized attributes

	rectal_temp	pulse	respiratory_rate	nasogastric_reflux_ph	packed_cell_volume	total_protein	abdomo_protein	lesion_1	lesion_2
0	5.098825e-01	-3.248796e-01	-2.362446e-01	3.067223e-15	-0.243023	-0.499604	1.279204e-15	2.345588	-0.096225
1	1.527936e+00	4.239611e-01	-7.142279e-01	3.067223e-15	0.277005	2.519038	-1.137044e+00	-0.259830	-0.096225
2	2.190099e-01	-1.209873e+00	-4.752362e-01	3.067223e-15	-1.491092	-0.566597	1.279204e-15	-0.892558	-0.096225
3	1.382500e+00	3.010866e+00	3.109638e+00	1.425207e-01	0.068994	-0.546894	2.031519e+00	-0.259830	-0.096225
4	-1.235353e+00	9.685726e-01	1.819908e-01	3.067223e-15	2.773143	-0.539012	1.279204e-15	0.339657	-0.096225
5	1.033387e-14	4.837121e-16	-2.122672e-16	3.067223e-15	0.000000	0.000000	1.279204e-15	-0.892558	-0.096225

```
Number of pairwise Euclidean distances: 5995
First 10 pairwise distances

0    4.391966
1    3.601763
2    5.841127
3    4.244568
4    3.348969
5    2.799206
6    2.675171
7    2.575690
8    3.530489
9    3.606955
dtype: float64
```

Ward linkage matrix (first six merges)

	Left	Right	Distance	Samples
0	8.0	93.0	0.285616	2.0

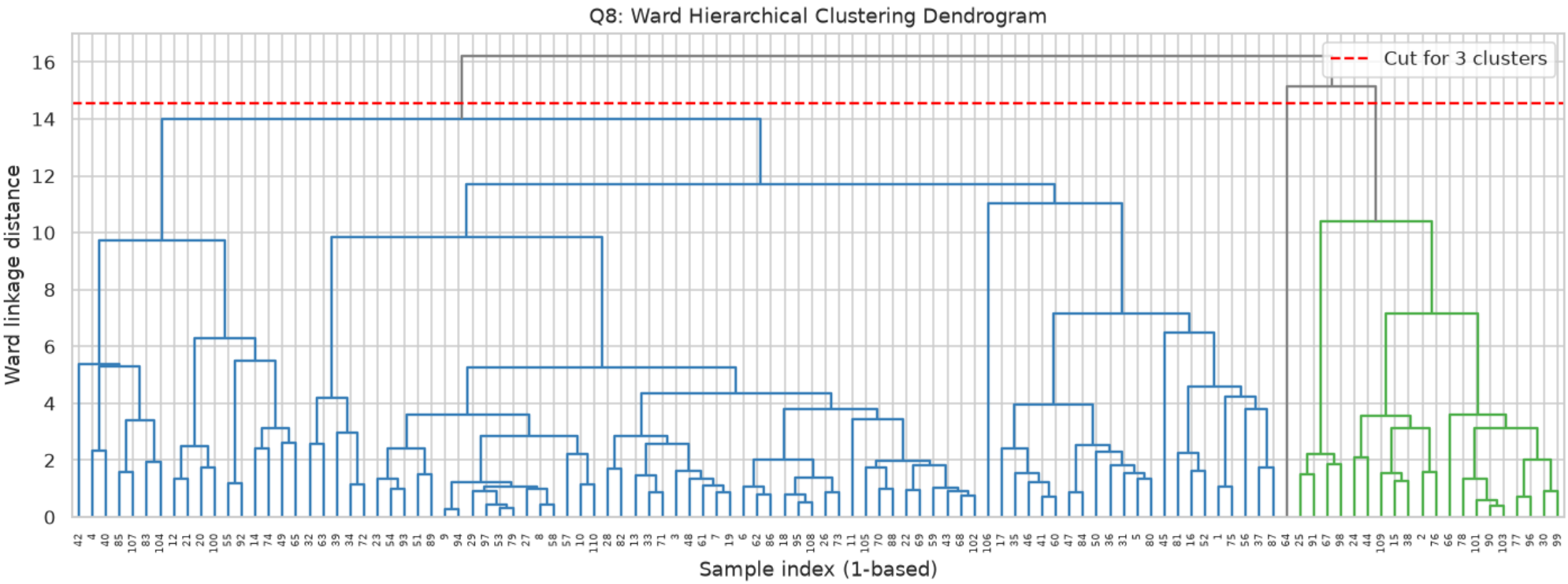
	Left	Right	Distance	Samples
1	52.0	78.0	0.320926	2.0
2	89.0	102.0	0.410868	2.0
3	96.0	111.0	0.445145	3.0
4	7.0	57.0	0.450649	2.0
5	94.0	107.0	0.529690	2.0

Hierarchical Clustering Q8-Q10 Dendrogram and cluster membership

Python code

```
threshold = (hc_ward[-3, 2] + hc_ward[-2, 2]) / 2
plt.figure(figsize=(13, 5))
set_link_color_palette(["#377eb8", "#4daf4a", "#e41a1c"])
dendrogram(hc_ward, labels=(hc_data.index + 1).astype(str).tolist(), color_threshold=threshold, leaf_rotation=90, leaf_font_size=6,
    above_threshold_color="gray")
plt.axhline(threshold, color="red", linestyle="--", label="Cut for 3 clusters")
plt.legend()
finish("Q8: Ward Hierarchical Clustering Dendrogram", "Sample index (1-based)", "Ward linkage distance")
cluster_assignments = pd.Series(cut_tree(hc_ward, n_clusters=3).ravel() + 1, index=hc_data.index, name="Cluster")
print("Q9: All cluster assignments:")
print(cluster_assignments.to_frame().to_string())
show("Q10: Samples per cluster", cluster_assignments.value_counts().sort_index())
```

Output



Q9: All cluster assignments (sample index is 0-based)

Sample	Cluster	Sample	Cluster	Sample	Cluster
0	1	37	2	74	1
1	2	38	1	75	2
2	1	39	1	76	2
3	1	40	1	77	2
4	1	41	1	78	1
5	1	42	1	79	1
6	1	43	2	80	1
7	1	44	1	81	1
8	1	45	1	82	1

Sample	Cluster	Sample	Cluster	Sample	Cluster
9	1	46	1	83	1
10	1	47	1	84	1
11	1	48	1	85	1
12	1	49	1	86	1
13	1	50	1	87	1
14	2	51	1	88	1
15	1	52	1	89	2
16	1	53	1	90	2
17	1	54	1	91	1
18	1	55	1	92	1
19	1	56	1	93	1
20	1	57	1	94	1
21	1	58	1	95	2
22	1	59	1	96	1
23	2	60	1	97	2
24	2	61	1	98	2
25	1	62	1	99	1
26	1	63	3	100	2
27	1	64	1	101	1
28	1	65	2	102	2
29	2	66	2	103	1
30	1	67	1	104	1
31	1	68	1	105	1
32	1	69	1	106	1
33	1	70	1	107	1
34	1	71	1	108	2
35	1	72	1	109	1

Sample	Cluster	Sample	Cluster	Sample	Cluster
36	1	73	1		

```
Q10: Samples per cluster
```

```
Cluster
1    89
2    20
3     1
Name: count, dtype: int64
```

Assignment 9 - Principal Component Analysis

This assignment selects numeric columns, removes incomplete records and constant columns, and standardizes each feature before PCA. It reports explained variance, creates a labeled scree plot and biplot, and displays component scores with the registration number in the final plot title.

Questions

Principal Component Analysis

1. Load the dataset sent to your email into R, and verify that the packages ggplot2, ggfortify, FactoMineR, and factoextra are loaded.

2. Display the total number of rows and columns in the dataset.

3. Extract only the numeric columns, clean the data, and normalize it using the scale() function into an object named data_scaled.

4. Perform Principal Component Analysis on data_scaled using either the PCA() or prcomp() function, and print its summary.

5. Plot a scree plot showing the percentage of variance for each component with data labels, setting the y-axis limit from 0 to 60.

6. Create a PCA biplot with non-overlapping labels and a proper title.

7. Extract the PCA scores into a data frame named pca_scores and display the first six rows.

8. Draw a scatter plot of the first two principal components (PC1 vs. PC2) with blue circular points and your registration number in the title

Q1-Q3 Load, clean, and standardize

Python code

```
data = load_horse()
print("Python equivalents loaded: pandas, matplotlib, scikit-learn PCA, and adjustText.")
print("Rows:", data.shape[0], "Columns:", data.shape[1])
numeric_data = data.select_dtypes(include="number")
numeric_clean = numeric_data.dropna().copy()
numeric_clean = numeric_clean.loc[:, numeric_clean.std(ddof=1).gt(0)]
data_scaled = (numeric_clean - numeric_clean.mean()) / numeric_clean.std(ddof=1)
print("Complete numeric records:", len(numeric_clean))
print("Retained numeric features:", numeric_clean.columns.tolist())
show("Standardized data", data_scaled.head(6))
```

Output

```
Python equivalents loaded: pandas, matplotlib, scikit-learn PCA, and adjustText.
Rows: 110 Columns: 28
Complete numeric records: 8
Retained numeric features: ['hospital_number', 'rectal_temp', 'pulse', 'respiratory_rate', 'nasogastric_reflux_ph', 'packed_cell_volume', 'total_protein',
    'abdomo_protein', 'lesion_1']
```

Standardized data

	hospital_number	rectal_temp	pulse	respiratory_rate	nasogastric_reflux_ph	packed_cell_volume	total_protein	abdomo_protein	lesion_1
3	2.474873	0.317259	1.817030	2.226732	-0.678684	-0.472456	-0.255774	0.506981	-0.571081
10	-0.353857	-0.557938	-0.647592	-0.916889	-2.088937	-1.228385	-1.790421	-0.681802	-0.613230
31	-0.352850	-1.345615	0.408675	-0.916889	0.026442	0.283473	1.023098	1.975476	-0.571583
54	-0.353314	1.367495	0.559570	0.130984	0.731569	1.228385	0.895211	-0.052446	2.034128
64	-0.353192	0.229739	-0.697890	0.130984	0.731569	-1.228385	0.127887	0.297196	-0.618750
73	-0.353904	-0.557938	-0.999681	-0.392953	0.731569	0.661438	0.127887	-1.381085	-0.619251

Q4 PCA and summary

Python code

```
pca_res = PCA(svd_solver="full")
scores_array = pca_res.fit_transform(data_scaled)
pc_names = [f"PC{i}" for i in range(1, len(pca_res.explained_variance_) + 1)]
pca_var = pca_res.explained_variance_
pca_var_pct = pca_res.explained_variance_ratio_ * 100
pca_summary = pd.DataFrame({"Standard deviation": np.sqrt(pca_var),
                            "Proportion of variance": pca_res.explained_variance_ratio_,
                            "Cumulative proportion": np.cumsum(pca_res.explained_variance_ratio_)}, index=pc_names)
show("PCA summary", pca_summary)
```

Output

PCA summary

	Standard deviation	Proportion of variance	Cumulative proportion
PC1	1.822424e+00	3.690255e-01	0.369026
PC2	1.633711e+00	2.965568e-01	0.665582
PC3	1.203072e+00	1.608203e-01	0.826403
PC4	8.626836e-01	8.269143e-02	0.909094
PC5	6.559803e-01	4.781224e-02	0.956906
PC6	6.151883e-01	4.205073e-02	0.998957
PC7	9.688417e-02	1.042949e-03	1.000000

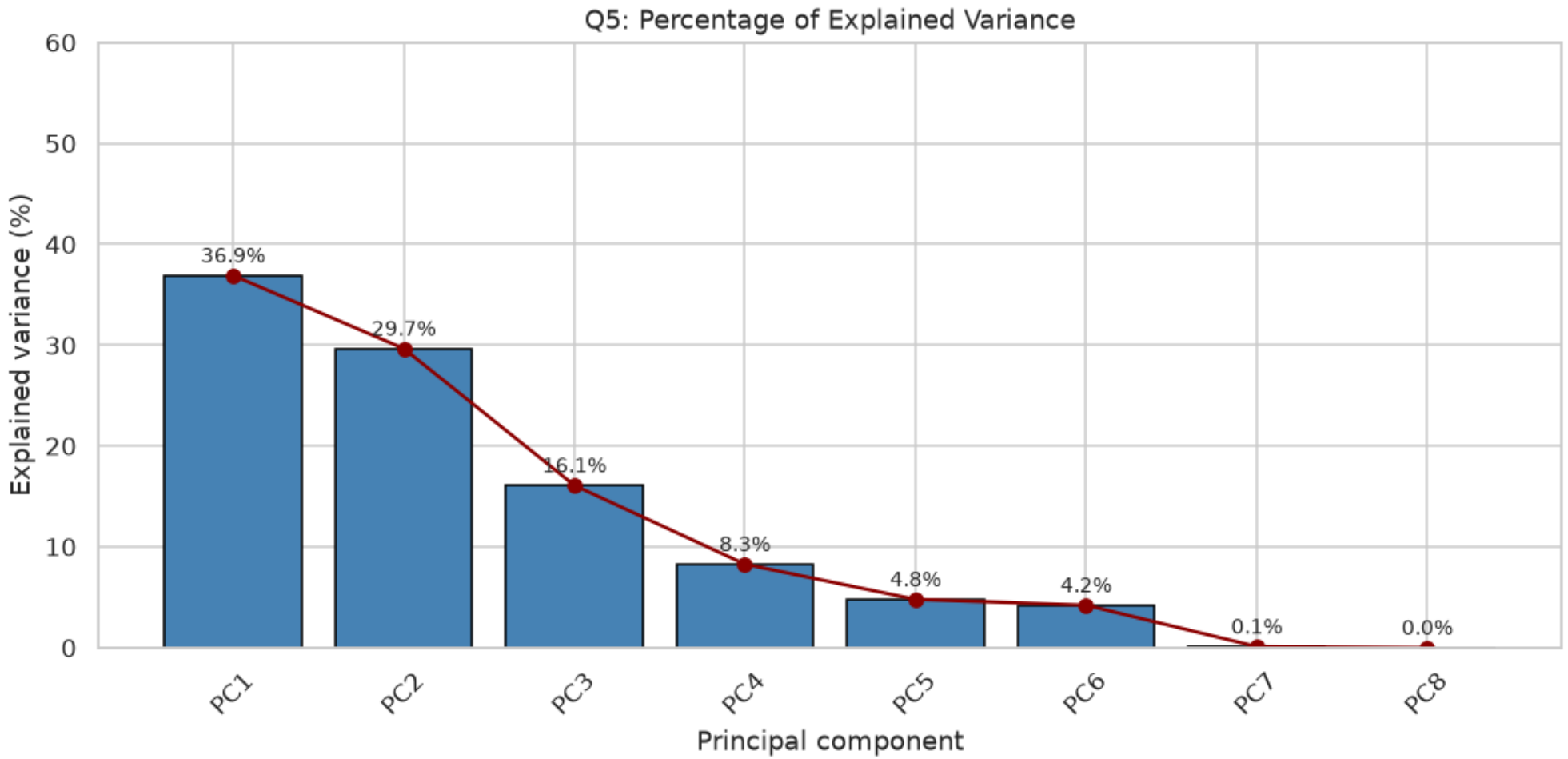
	Standard deviation	Proportion of variance	Cumulative proportion
PC8	5.046324e-17	2.829488e-34	1.000000

Q5 Scree plot

Python code

```
x = np.arange(len(pc_names))
fig, ax = plt.subplots(figsize=(10, 5))
ax.bar(x, pca_var_pct, color="steelblue", edgecolor="black")
ax.plot(x, pca_var_pct, "o-", color="darkred")
for pos, value in zip(x, pca_var_pct):
    ax.text(pos, min(value + .8, 58), f"{value:.1f}%", ha="center", va="bottom", fontsize=9)
ax.set_xticks(x, pc_names, rotation=45)
ax.set_ylim(0, 60)
finish("Q5: Percentage of Explained Variance", "Principal component", "Explained variance (%)")
```

Output



The biplot uses scores divided by the component singular values and loading vectors multiplied by those singular values. This gives both elements a shared PCA biplot scaling; arrow directions and relative lengths describe the features. Component signs may be reversed relative to R without changing the PCA result.

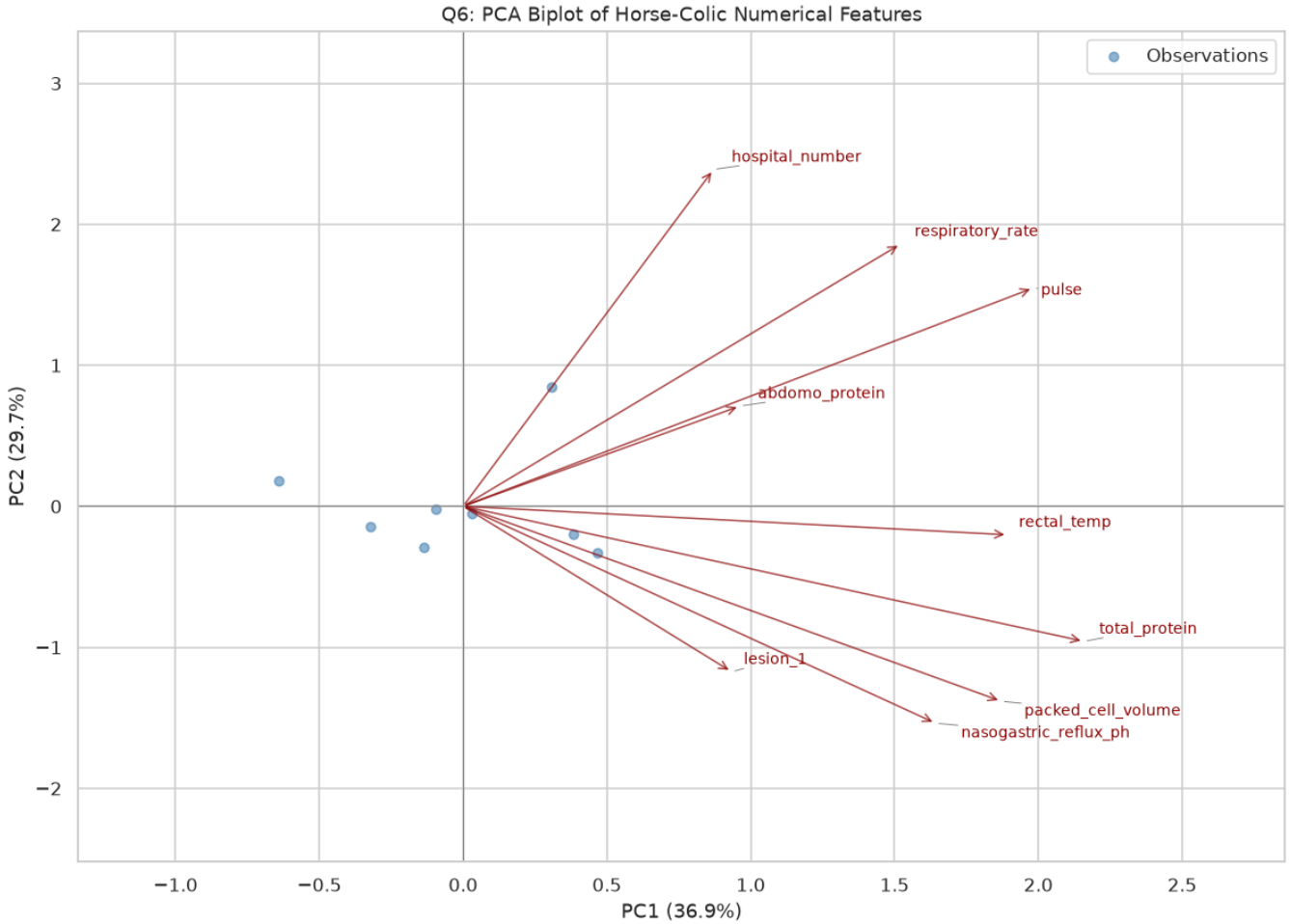
Q6 PCA biplot with non-overlapping feature labels

Python code

```
score_plot = scores_array[:, :2] / pca_res.singular_values_[:2]
loading_plot = pca_res.components_[:2].T * pca_res.singular_values_[:2]
fig, ax = plt.subplots(figsize=(11, 8))
ax.scatter(score_plot[:, 0], score_plot[:, 1], color="steelblue", s=30, alpha=.6, label="Observations")
texts = []
```

```
for label, (vx, vy) in zip(numeric_clean.columns, loading_plot):
    ax.annotate("", xy=(vx, vy), xytext=(0, 0), arrowprops={"arrowstyle": "->", "color": "darkred", "alpha": .7})
    texts.append(ax.text(vx, vy, label, color="darkred", fontsize=9))
all_x = np.r_[score_plot[:, 0], loading_plot[:, 0]]
all_y = np.r_[score_plot[:, 1], loading_plot[:, 1]]
margin_x = max(np.ptp(all_x) * .25, .2)
margin_y = max(np.ptp(all_y) * .25, .2)
ax.set_xlim(all_x.min() - margin_x, all_x.max() + margin_x)
ax.set_ylim(all_y.min() - margin_y, all_y.max() + margin_y)
adjust_text(texts, ax=ax, expand=(1.15, 1.25), force_text=(.7, .9), arrowprops={"arrowstyle": "-", "color": "gray", "lw": .6})
ax.axhline(0, color="gray", lw=.7)
ax.axvline(0, color="gray", lw=.7)
ax.legend()
finish("Q6: PCA Biplot of Horse-Colic Numerical Features", f"PC1 ({pca_var_pct[0]:.1f}%)", f"PC2 ({pca_var_pct[1]:.1f}%)")
```

Output



Q7-Q8 Component scores and PC1 vs PC2

Python code

```
pca_scores = pd.DataFrame(scores_array, columns=pc_names, index=numeric_clean.index)
show("First six PCA score rows", pca_scores.head(6))
plt.scatter(pca_scores["PC1"], pca_scores["PC2"], color="blue", marker="o", s=45, alpha=.8)
finish("Q8: PC1 vs PC2 - 23BDS0065", "Principal component 1 (PC1)", "Principal component 2 (PC2)")
```

Output

```
First six PCA score rows
```

	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8
3	1.482824	3.649595	-0.381759	-0.168099	0.057109	-0.349848	0.005201	4.720404e-17
10	-3.083504	0.792074	-0.522422	0.610685	0.541179	0.734424	0.046631	4.720404e-17
31	0.147188	-0.226930	2.785832	0.613142	0.191490	-0.230632	0.007676	4.720404e-17
54	2.254878	-1.415369	-1.024308	1.060463	-0.186660	0.020045	0.119664	4.720404e-17
64	-0.449750	-0.078447	0.413217	-0.938836	-1.279174	0.561559	0.034069	4.720404e-17
73	-0.650567	-1.255936	-0.298436	-1.380162	0.655935	-0.663024	0.076539	4.720404e-17

